# MASSA 2026

**Congress on Mass Spectrometry**

Chieti, Italy
22–25 June 2026

# Open Tools, Broad Knowledge: Empowering Researchers for Contextualized Metabolite Annotation

Adriano Rutz · ETH Zürich / Institute of Molecular Systems Biology

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Società Chimica Italiana
Divisione di Spettrometria di Massa

# Agenda

01

## Context

This talk

02

## Methods

Will be about

03

## Results

Open Source,
Open Data,
Open Knowledge

04

## Discussion

Mass Spectrometry
...

# Agenda

**01**

## Context

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## Results

Open Source,
Open Data,
Open Knowledge

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## Discussion

Mass Spectrometry
...

**05**

## Take home

… and a touch of bitterness!

# Agenda

01
**Taxonomically Informed Metabolite Annotation**

https://doi.org/10.3389/fpls.2019.01329

02
**LOTUS**

https://doi.org/10.7554/eLife.70780

03
**CASCADE**

https://doi.org/10.1021/acs.jafc.3c03099

04
**SAPID**

https://doi.org/10.1016/j.crfs.2025.101043

05
**The future**

MASSA 22-25 Giugno 2026

Metabolite Annotation

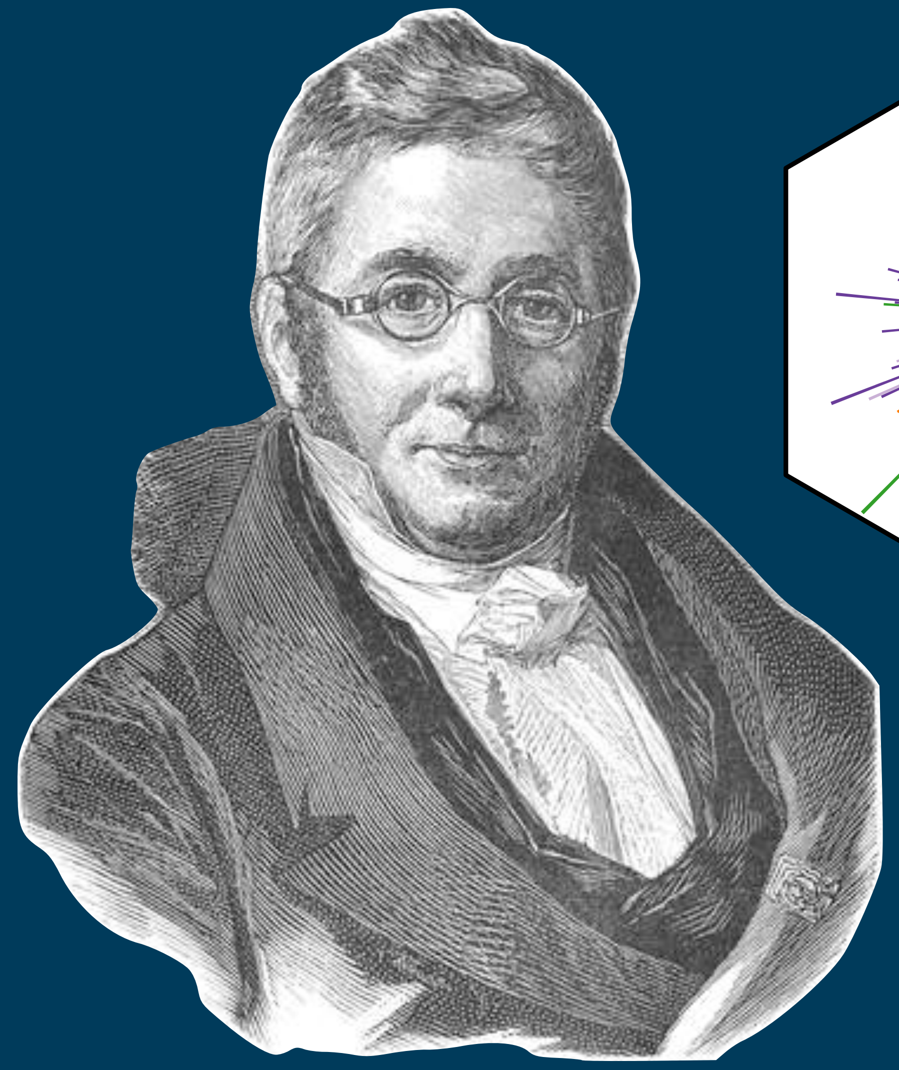
# Taxonomy as prior

Because spectra are not enough

MASSA
22-25 Giugno 2026

Metabolite Annotation

# Taxonomy as prior



tima

*“Plant taxonomy would be the most useful guide to man in his search for new industrial and medicinal plants”* (de Candolle, 1804)

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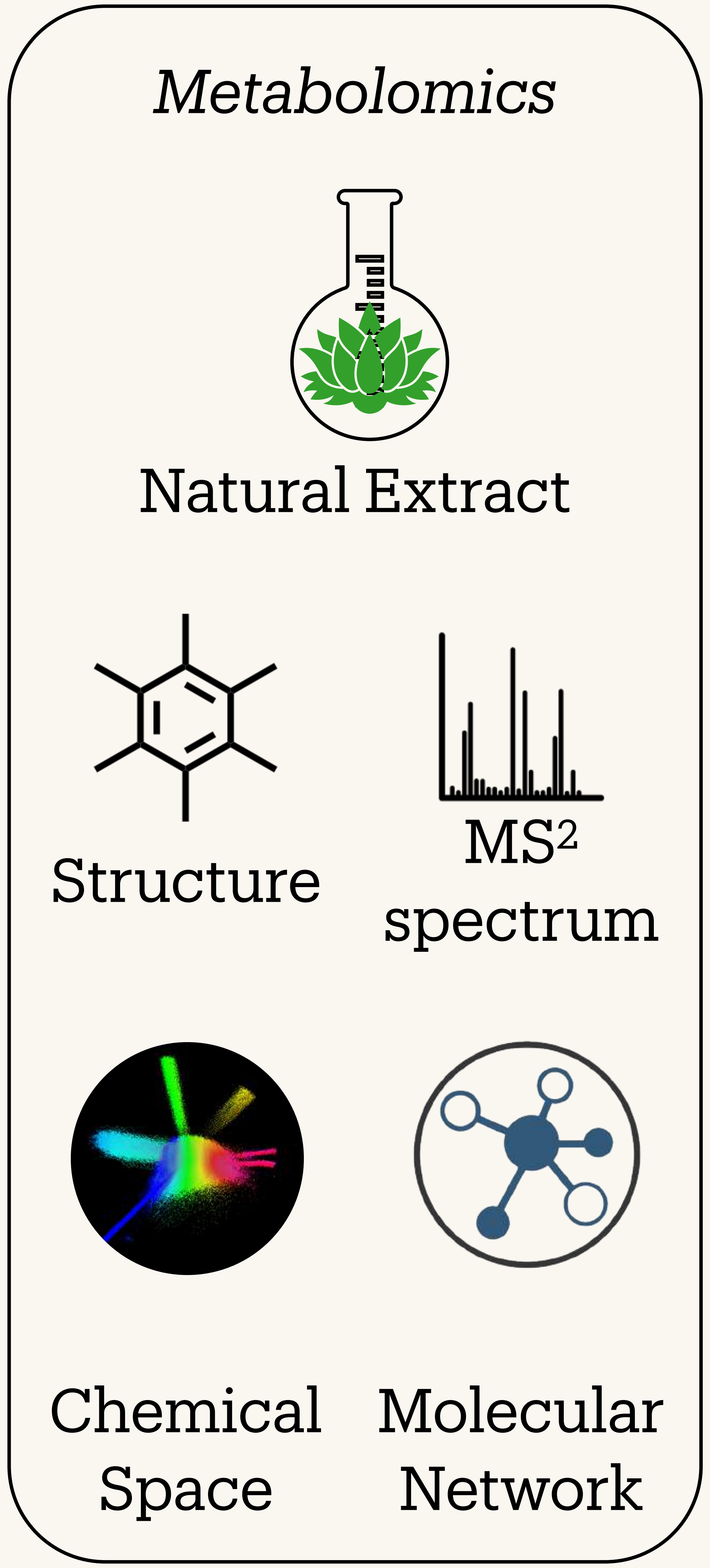
# Taxonomically Informed Metabolite Annotation

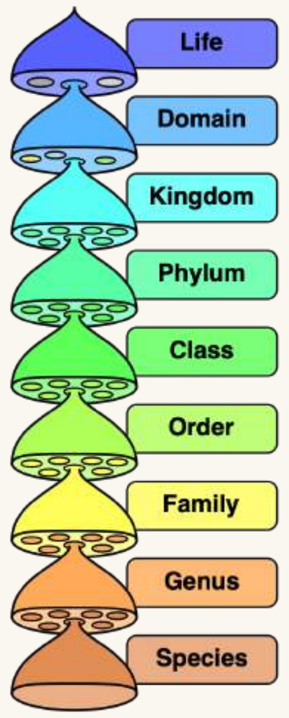


# Taxonomically Informed Metabolite Annotation

*Taxonomy*

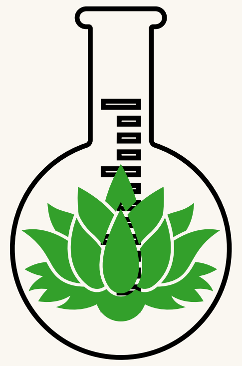
Biological organism

Morphological traits

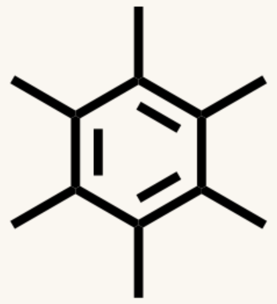



Taxonomy

*Metabolomics*



Natural Extract



Structure

$MS^2$ spectrum

Chemical Space

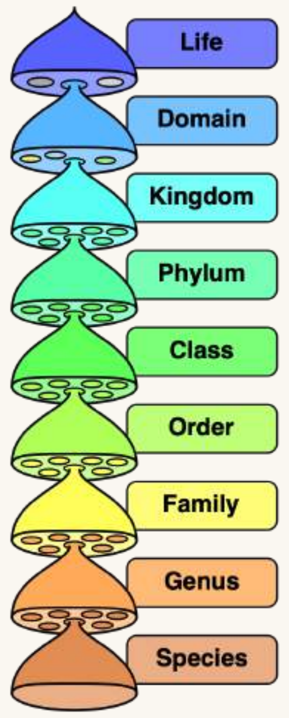
Molecular Network

# Taxonomically Informed Metabolite Annotation

*Taxonomy*

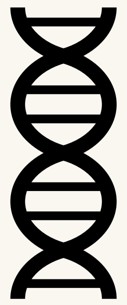
Biological organism

Morphological traits




Taxonomy

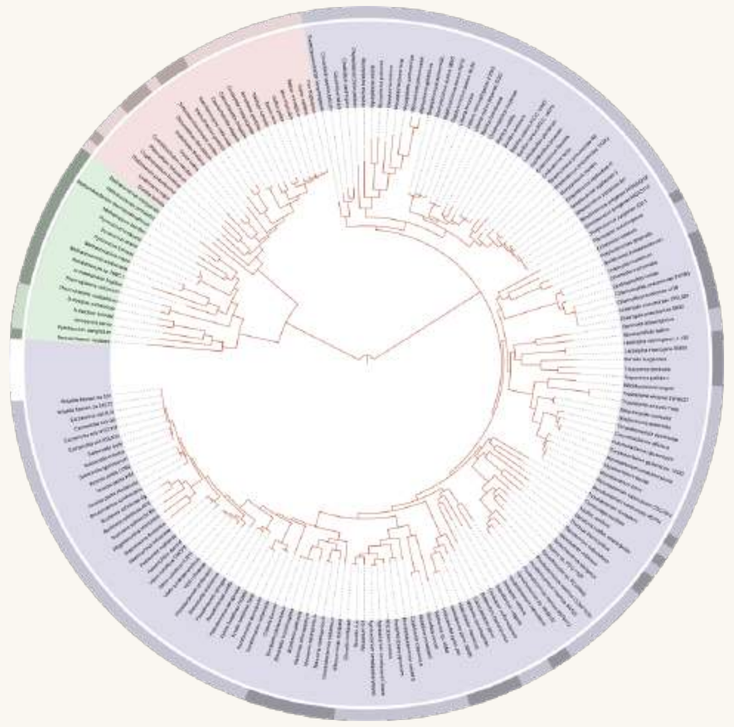
*Genomics*



Genetic material

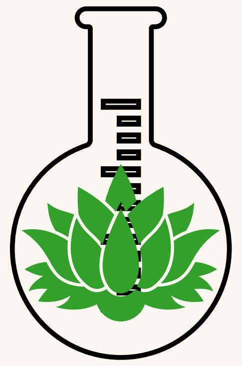
A T G C A A G C T T A
C G T A C C T A G G C

DNA sequence

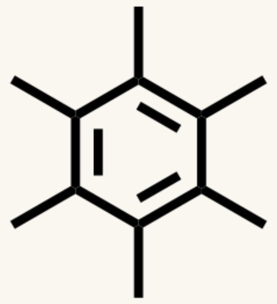


Phylogeny

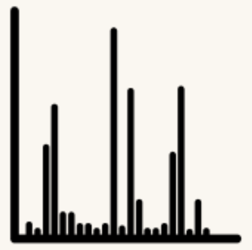
*Metabolomics*



Natural Extract



Structure



$MS^2$ spectrum

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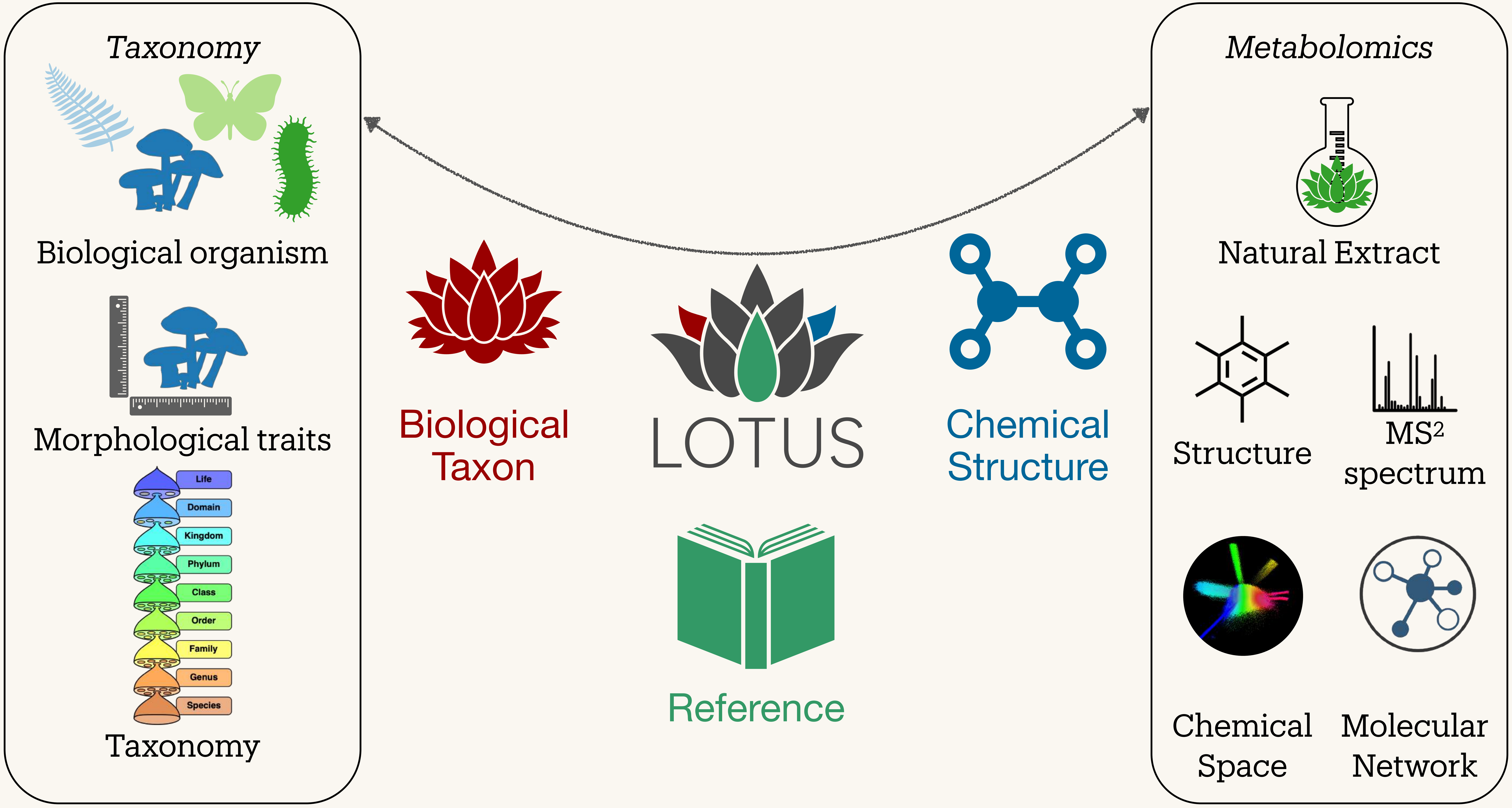
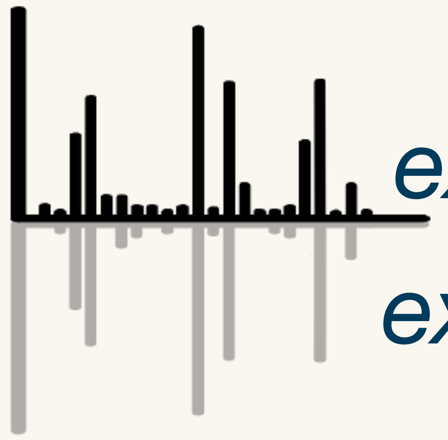
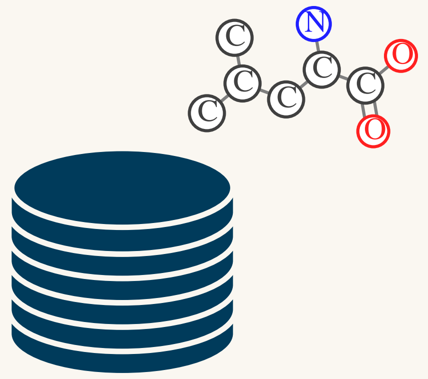


# Taxonomically Informed Metabolite Annotation

$$S_{tot} = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4$$

# Taxonomically Informed Metabolite Annotation

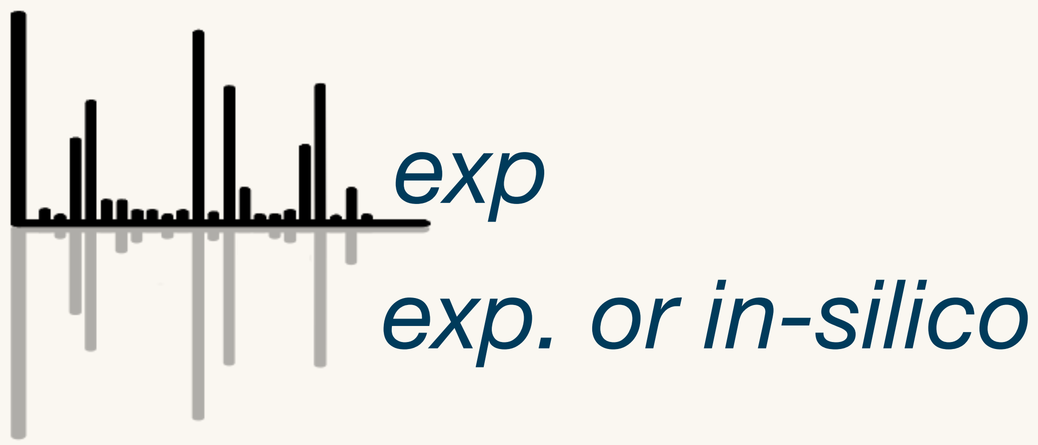


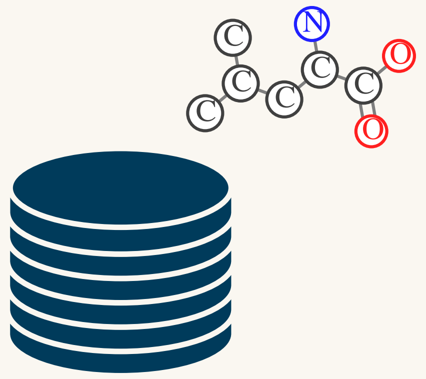



Spectral similarity

$$S_{tot} = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4$$

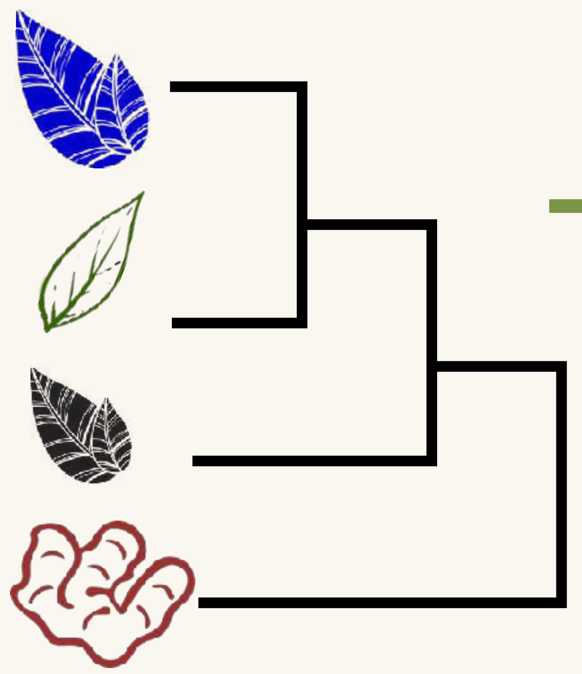
# Taxonomically Informed Metabolite Annotation



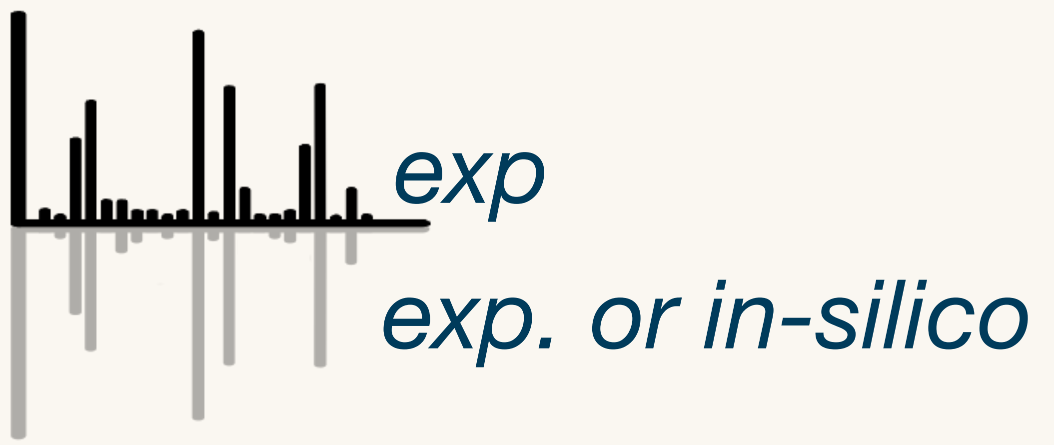
Spectral similarity

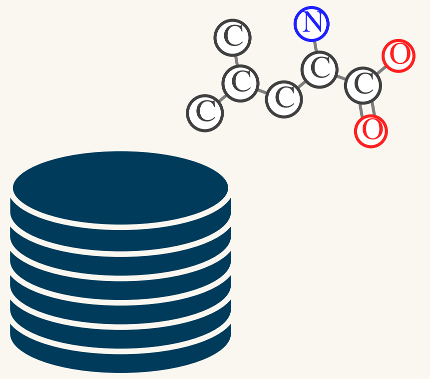
$$S_{tot} = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4$$



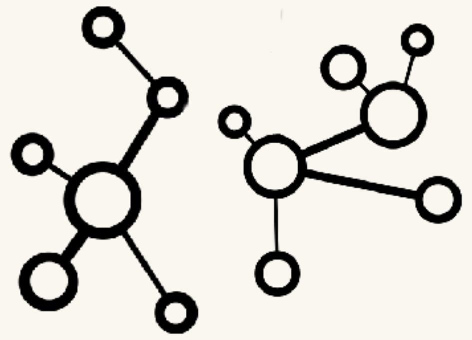
Taxonomic distance

## Taxonomically Informed Metabolite Annotation



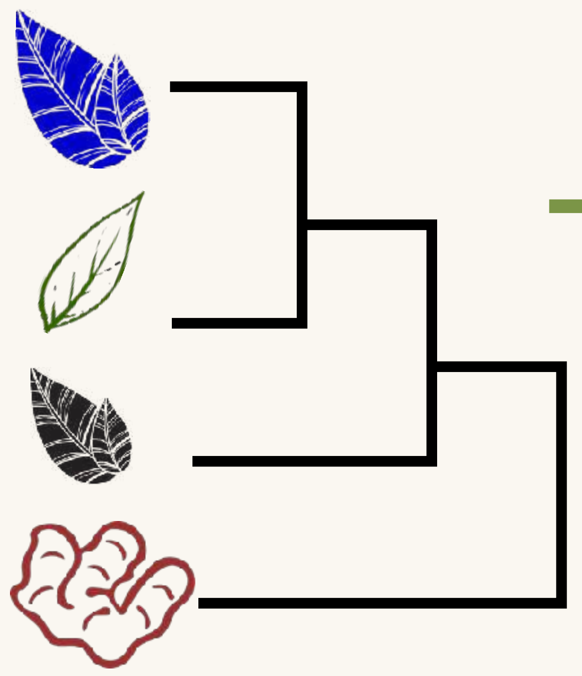



Spectral similarity



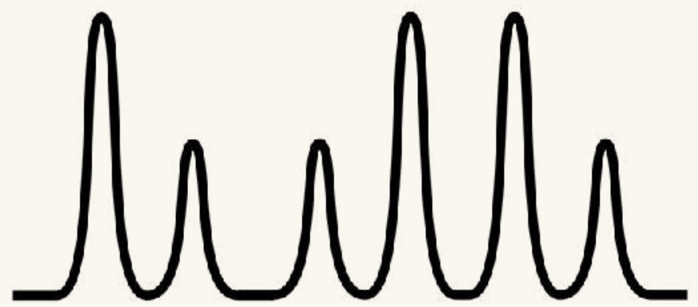
Structural consistency

$$S_{tot} = w_1 S_1 + w_2 S_2 + w_3 S_3 + w_4 S_4$$



Taxonomic distance

Physico-chemical consistency



# Taxonomically Informed Metabolite Annotation


**MS²-based annotation**

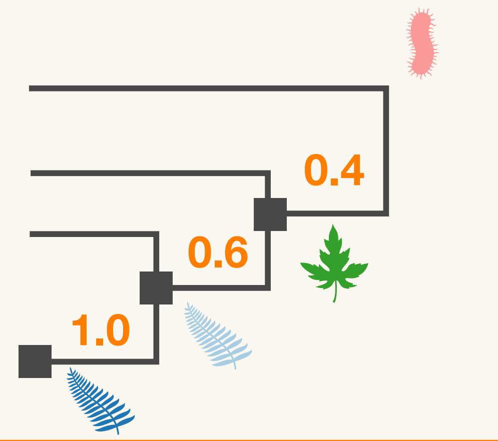
**MS¹-based annotation**

LOTUS

| Feature ID | Spectrum | Biological source | Candidate structure | Score $S_1$ | Initial rank |
|---|---|---|---|---|---|
| 1 | | | structure-1 | **0.55** | **1** |
| | | | structure-2 | 0.53 | 2 |
| | | | structure-3 | 0.50 | 3 |
| | | | structure-4 | 0.45 | 4 |
| | | | structure-5 | 0.00 | 9 |

re-ranking

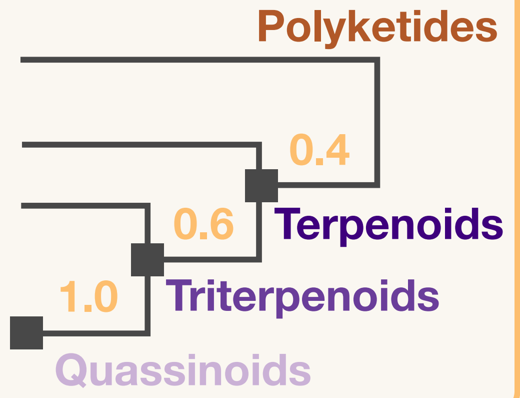
**Biological taxonomy weighting**




| Feature ID | Spectrum | Biological source | Candidate structure | Score $S_1$ | Initial rank | Candidate biological source | Score $S_2$ | Combined score | Final rank |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | structure-4 | **0.45** | **4** | | **1.00** | **0.78** | **1** |
| | | | structure-1 | 0.55 | 1 | | 0.60 | 0.58 | 2 |
| | | | structure-2 | 0.53 | 2 | | 0.40 | 0.47 | 3 |
| | | | structure-5 | 0.00 | 9 | | 0.60 | 0.30 | 4 |
| | | | structure-3 | 0.50 | 3 | | 0.00 | 0.25 | 5 |

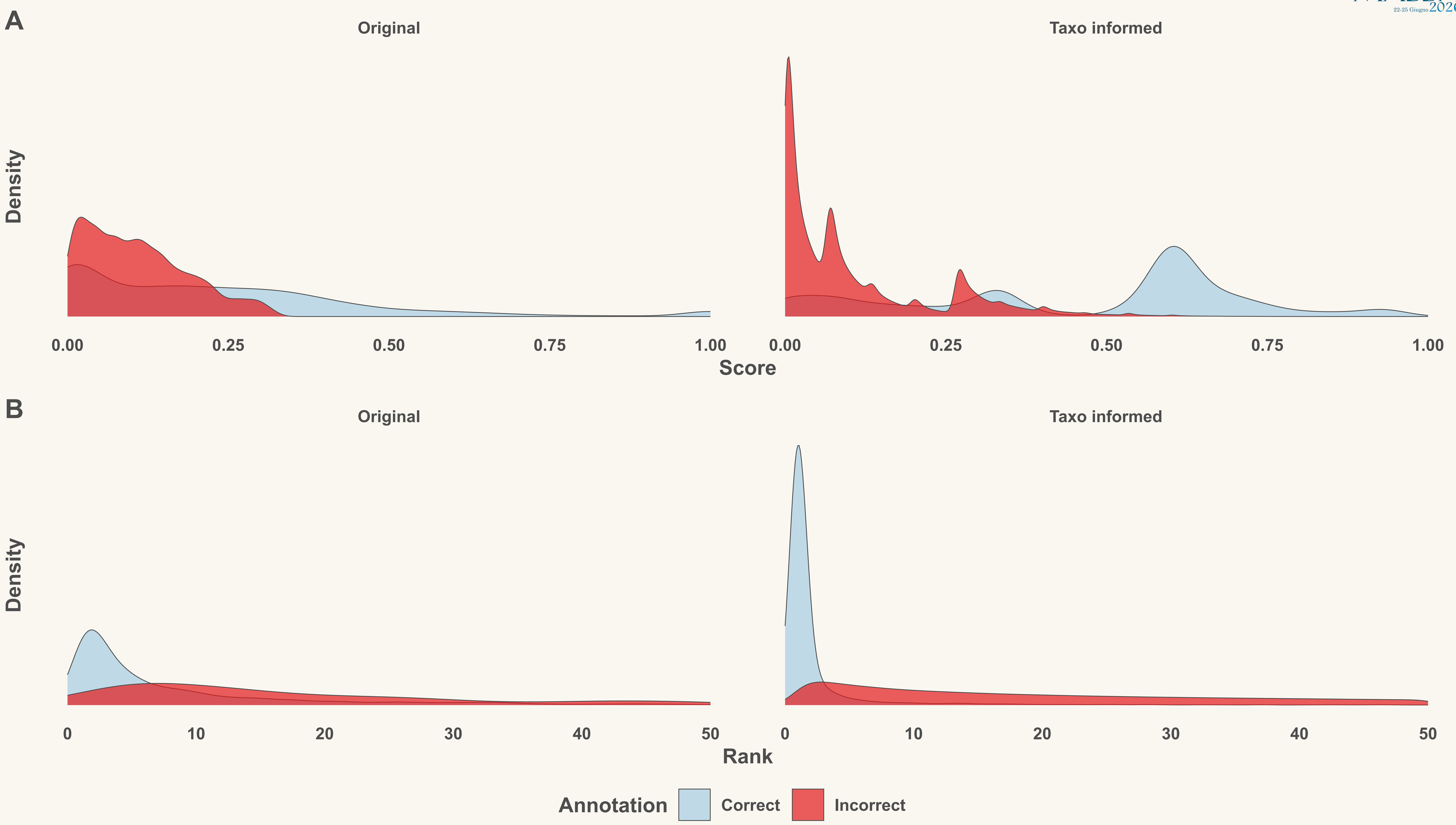
re-ranking

**Chemical taxonomy weighting**



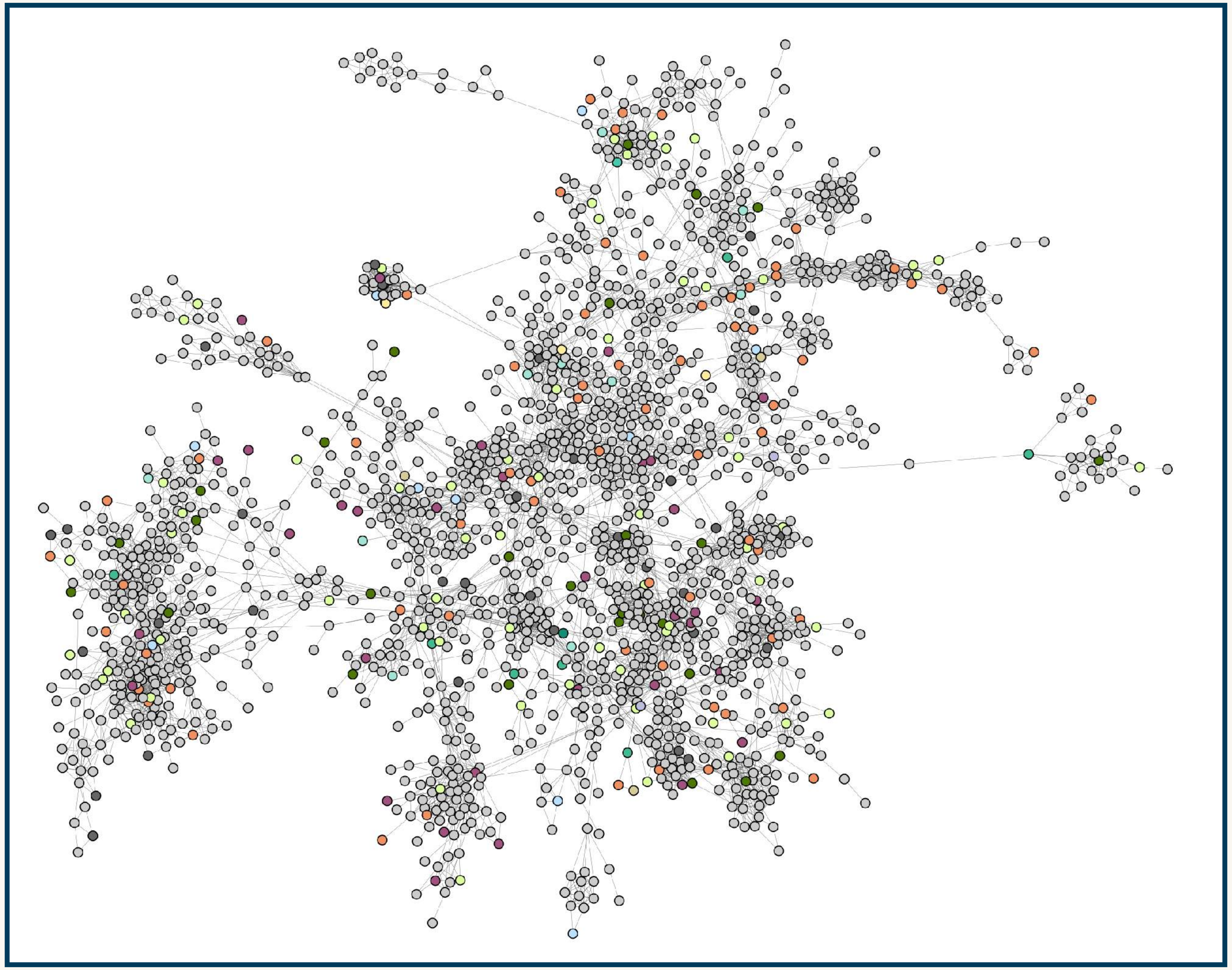

| Feature ID | Spectrum | Biological source | Candidate structure | Score $S_1$ | Initial rank | Candidate biological source | Score $S_2$ | Combined score | 2nd rank | Attributed chemical class | Candidate chemical class | Score $S_3$ | Combined score | Final rank |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | structure-4 | **0.45** | **4** | | **1.00** | **0.78** | **1** | Quassinoid | Quassinoid | 1.00 | **0.82** | **1** |
| | | | structure-1 | 0.55 | 1 | | 0.60 | 0.58 | 2 | | Triterpenoids | 0.60 | 0.58 | 2 |
| | | | structure-2 | 0.53 | 2 | | 0.40 | 0.47 | 3 | | Terpenoids | 0.40 | 0.45 | 3 |
| | | | structure-5 | 0.00 | 9 | | 0.60 | 0.30 | 4 | | Terpenoids | 0.40 | 0.32 | 4 |
| | | | structure-3 | 0.50 | 3 | | 0.00 | 0.25 | 5 | | Polyketides | 0.00 | 0.17 | 5 |

# Taxonomically Informed Metabolite Annotation



# Taxonomically Informed Metabolite Annotation

Feature:
RT @ *m/z* pointer

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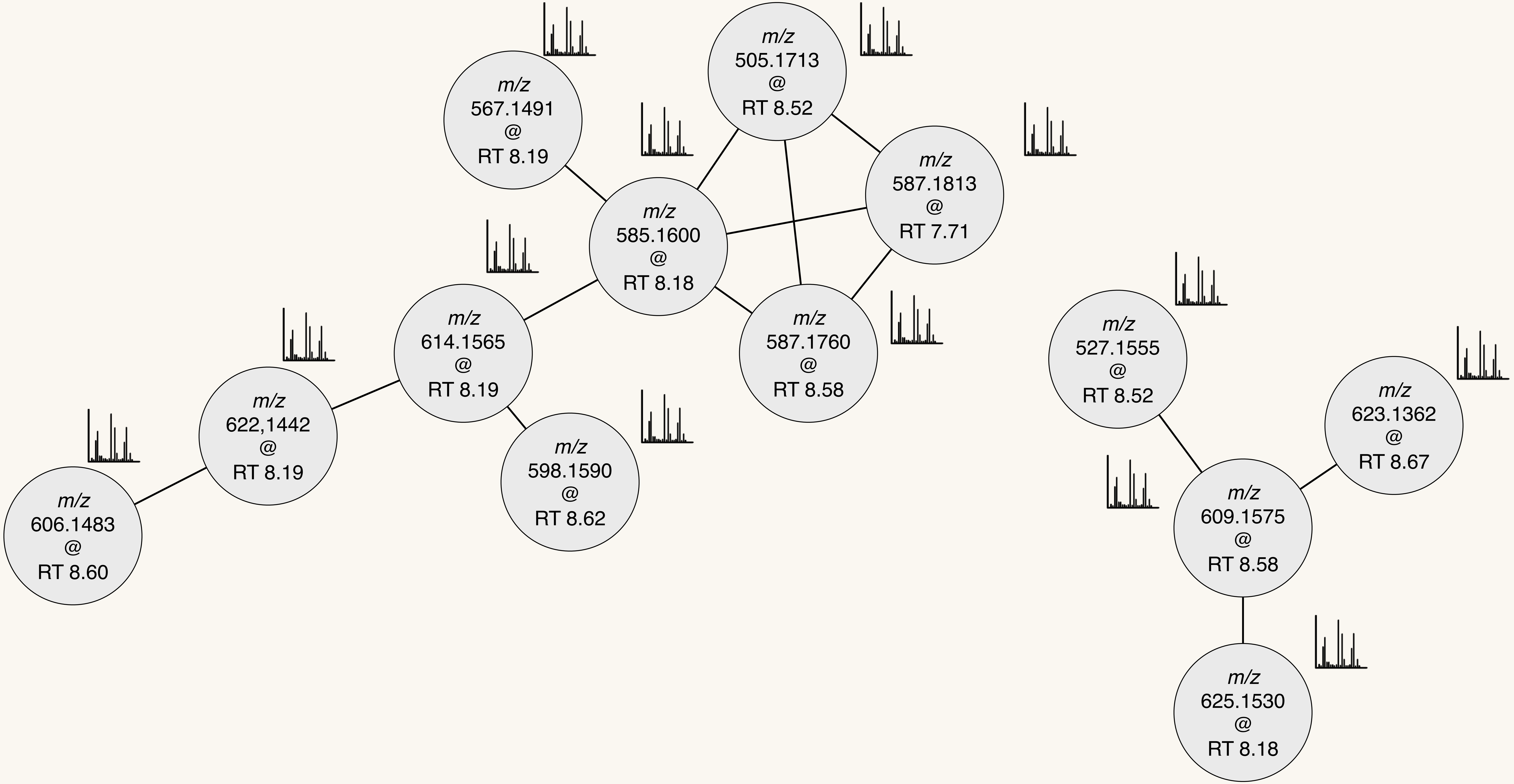

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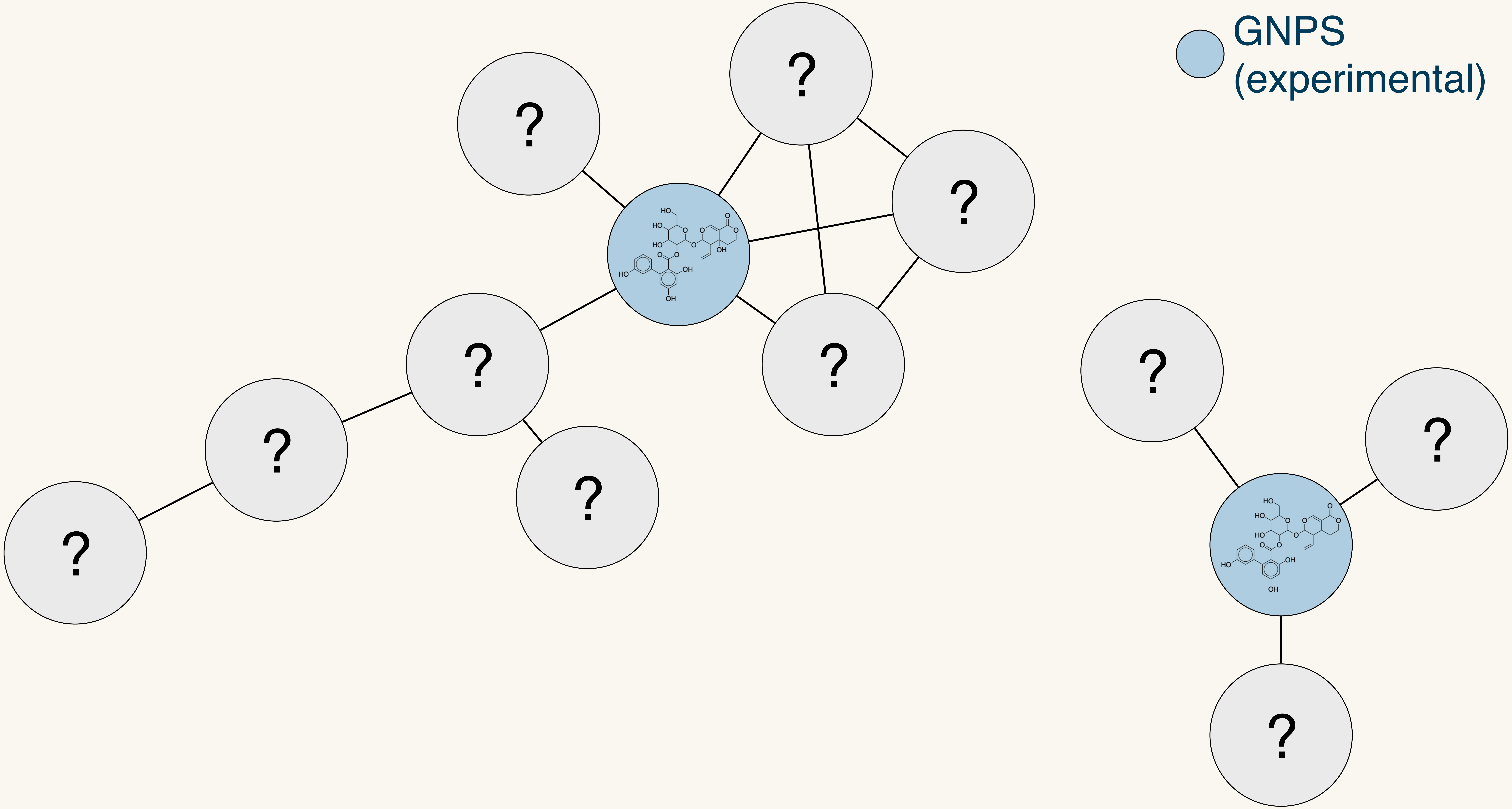

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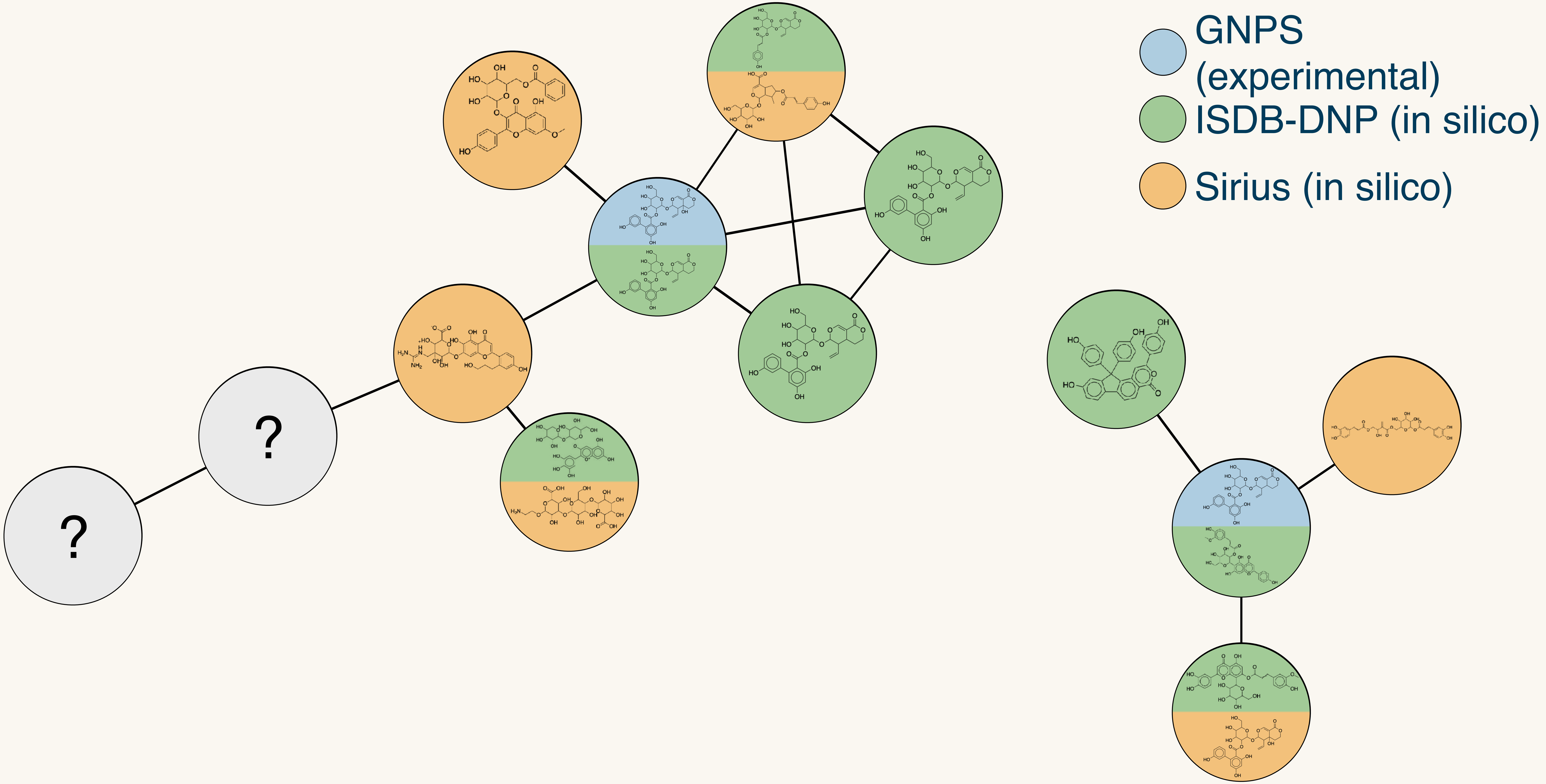

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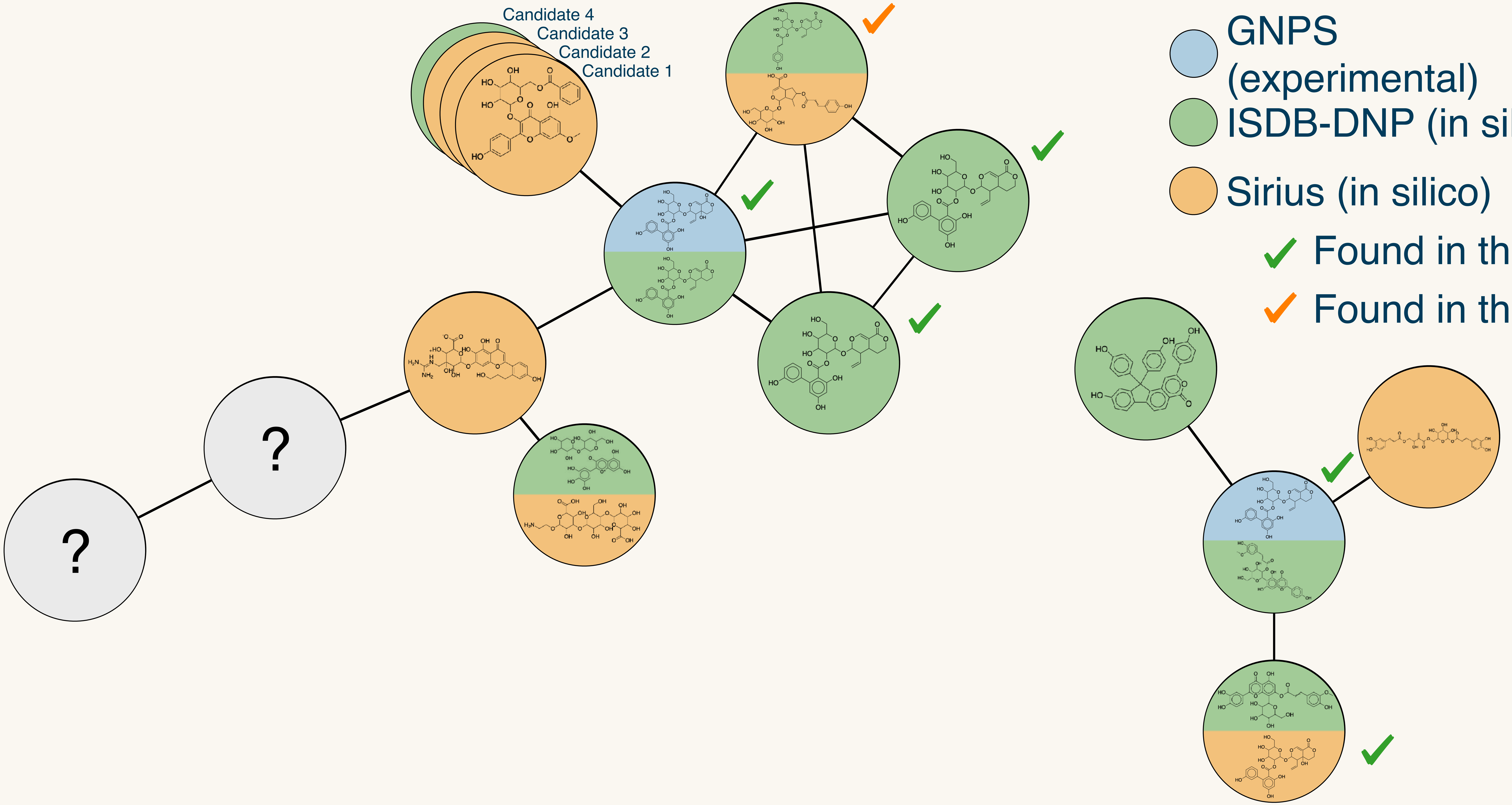
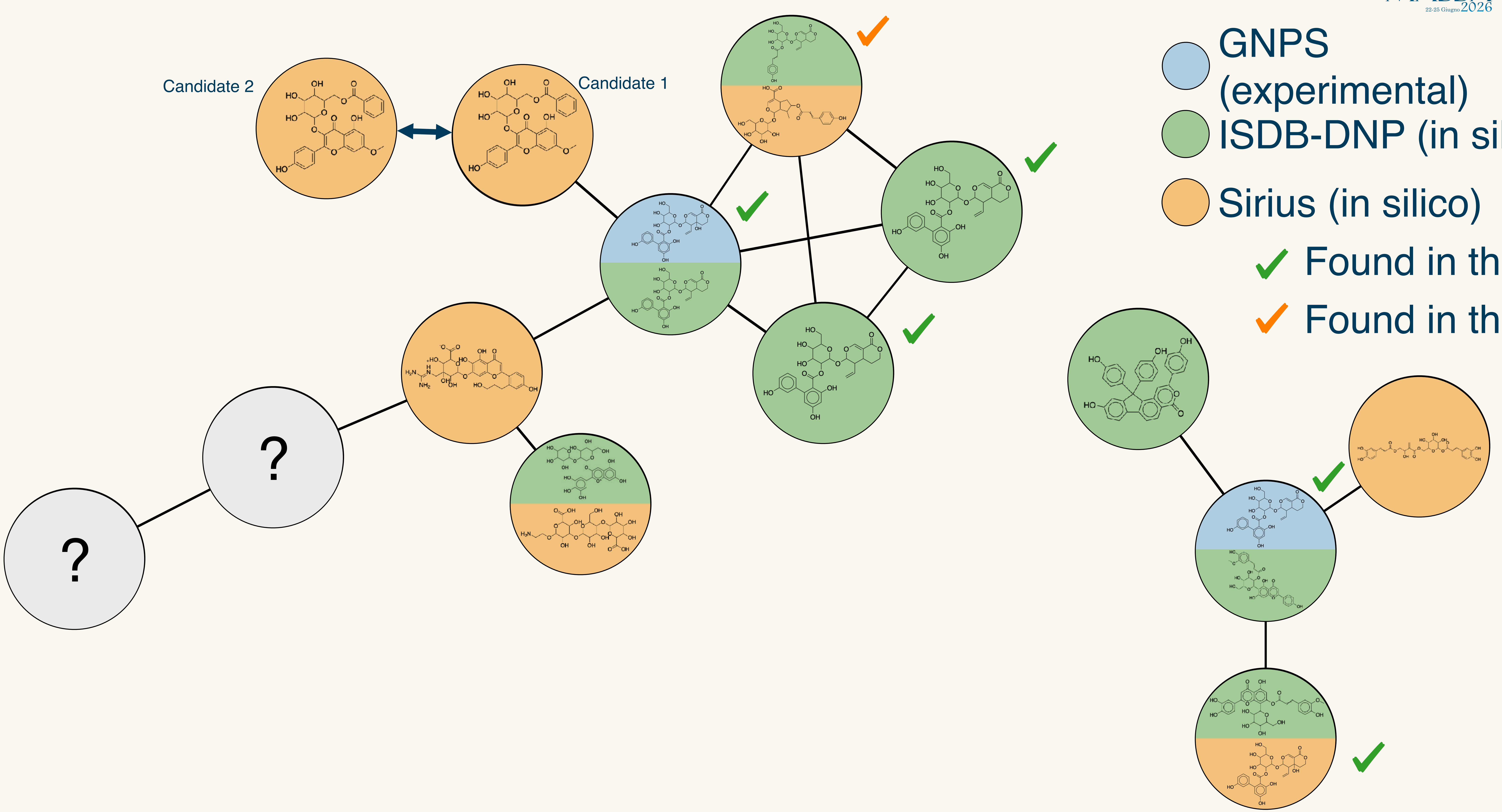



GNPS (experimental)
ISDB-DNP (in silico)
Sirius (in silico)
✓ Found in the genus
✓ Found in the family

# Taxonomically Informed Metabolite Annotation

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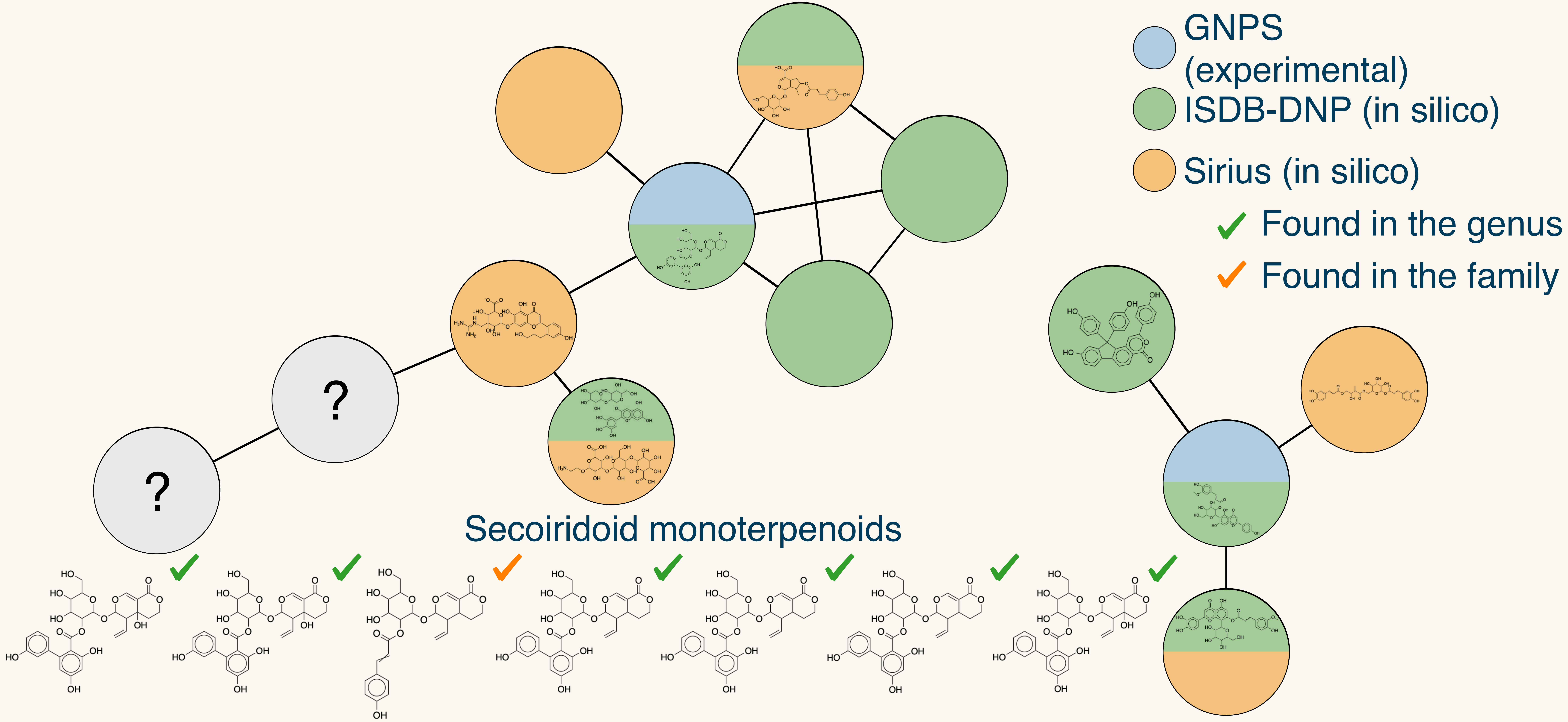

# Taxonomically Informed Metabolite Annotation

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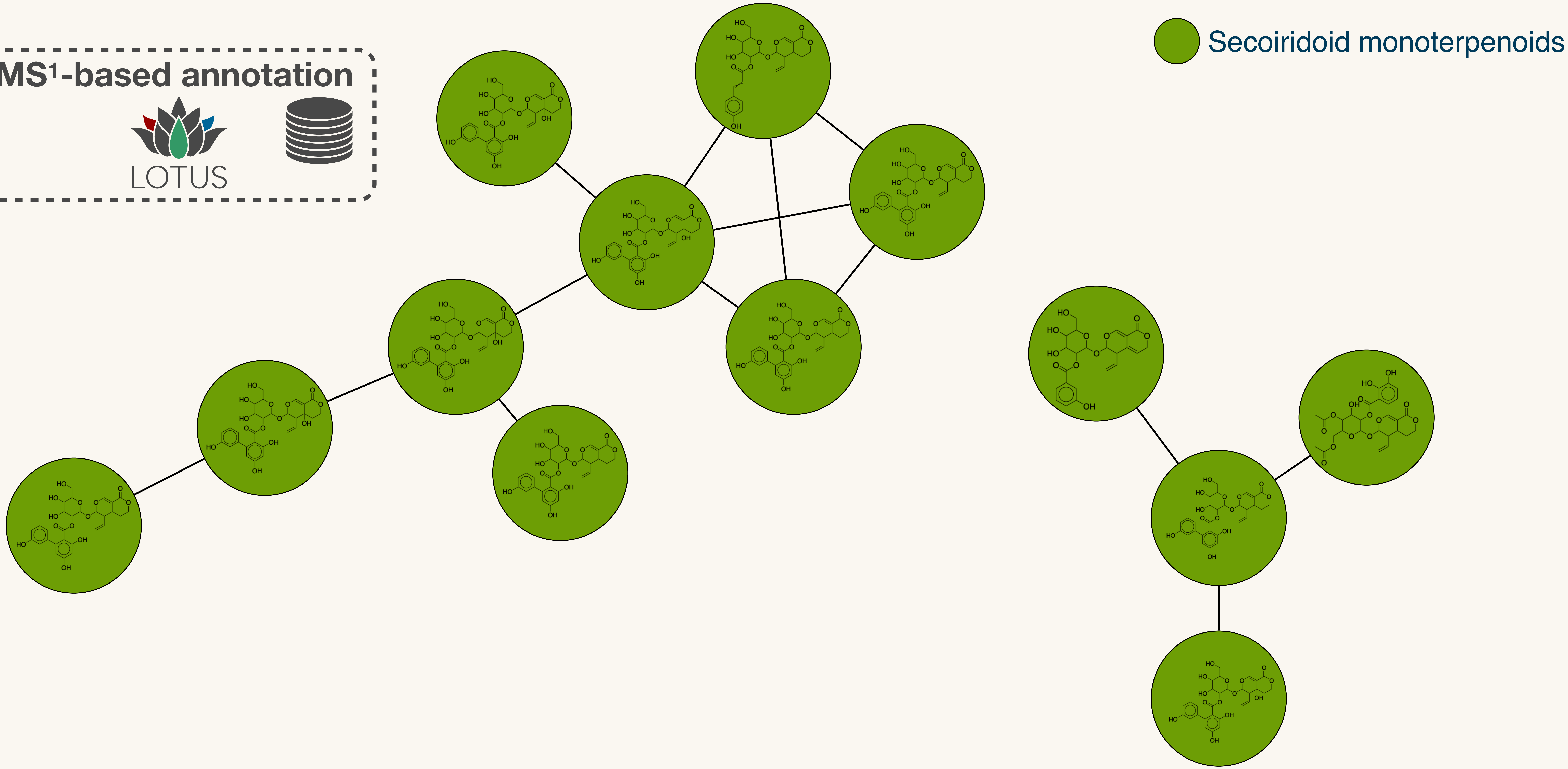
**MS¹-based annotation**

LOTUS



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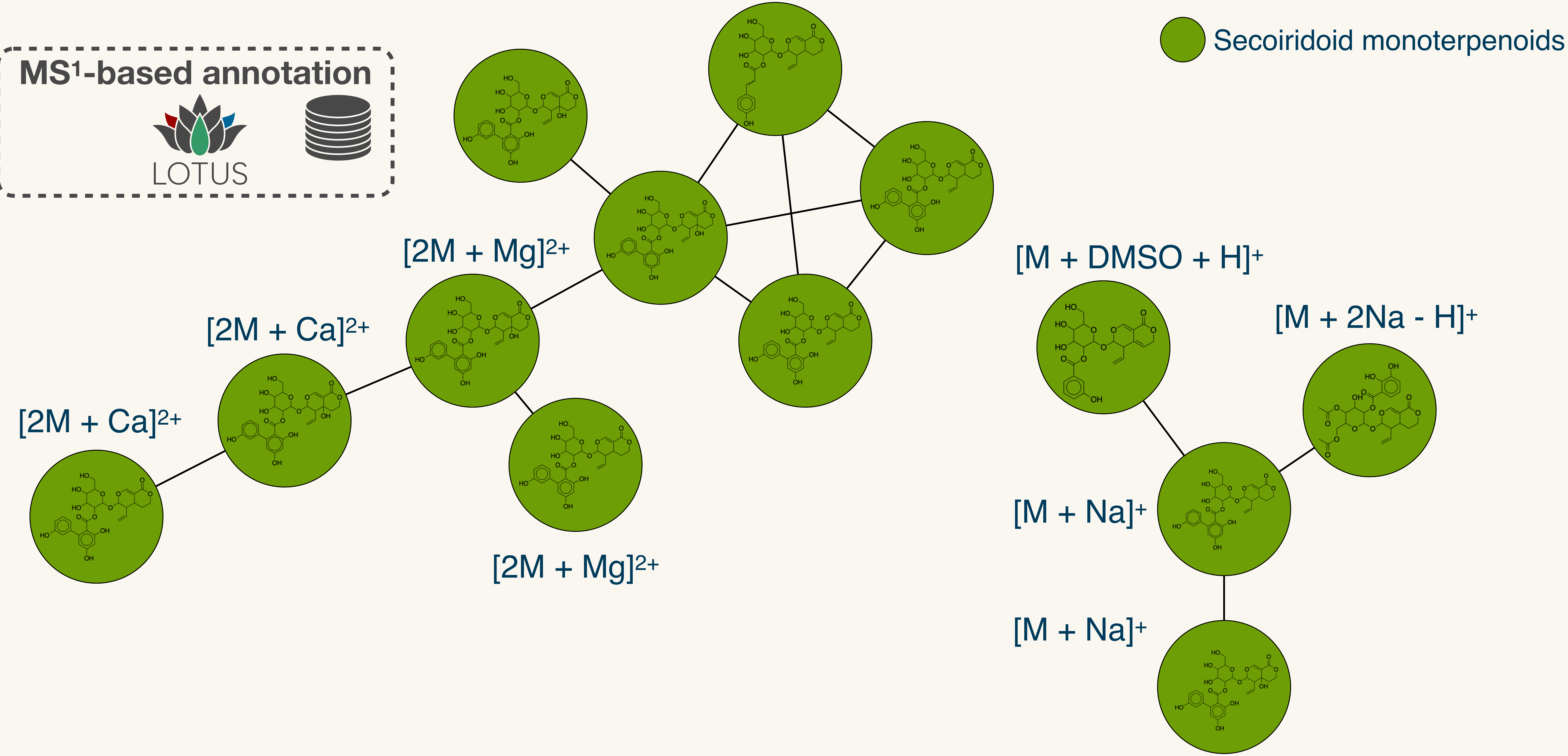
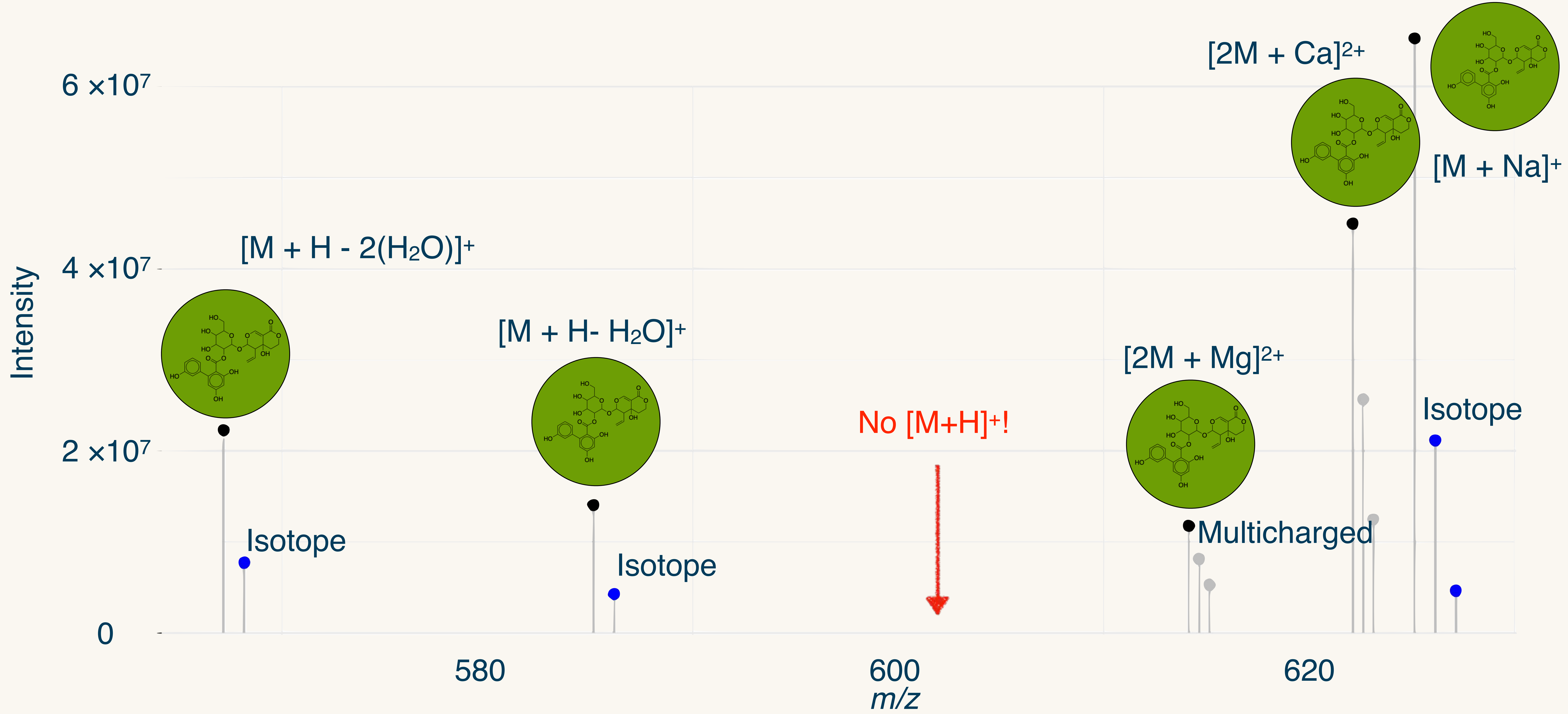


# Taxonomically Informed Metabolite Annotation



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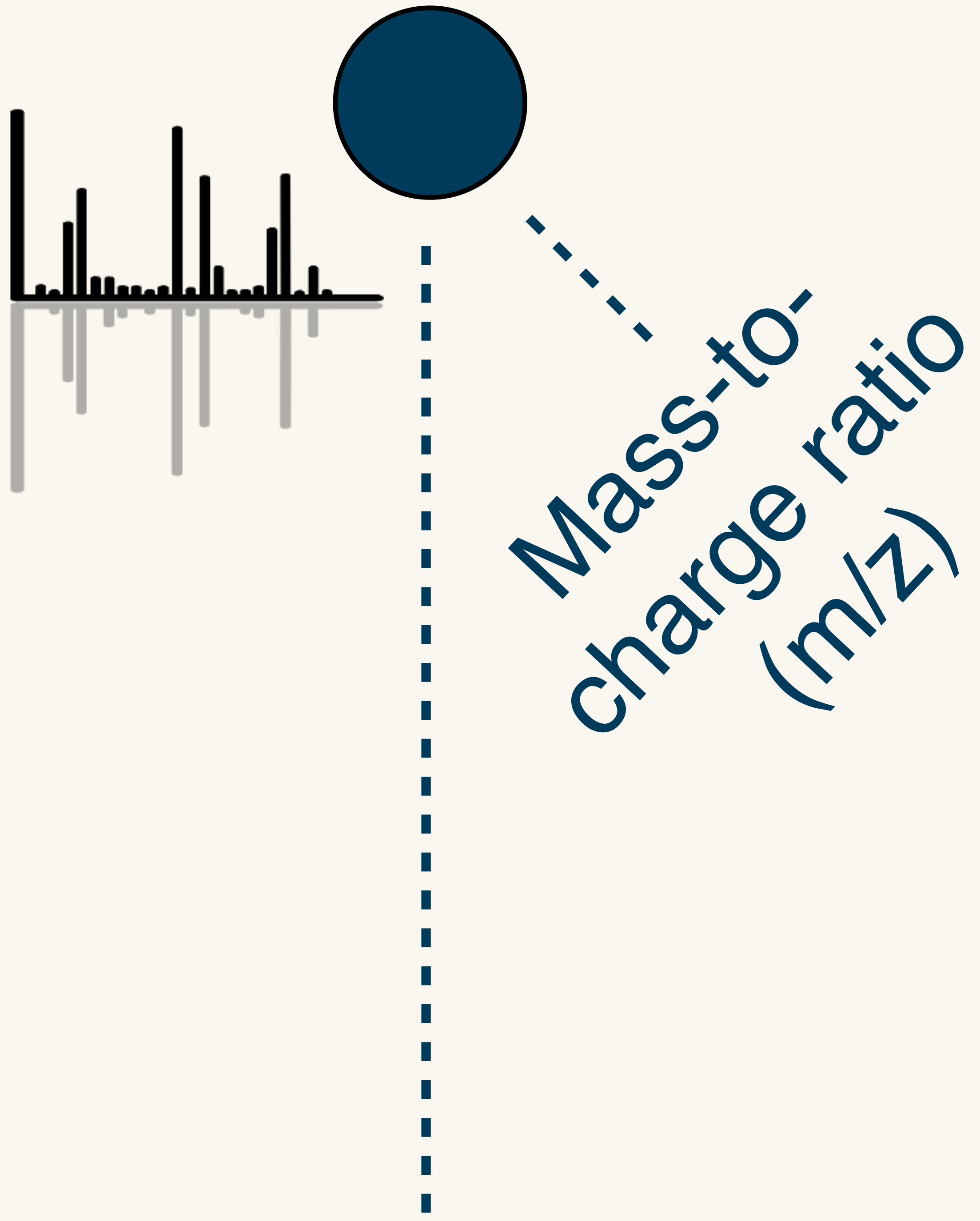
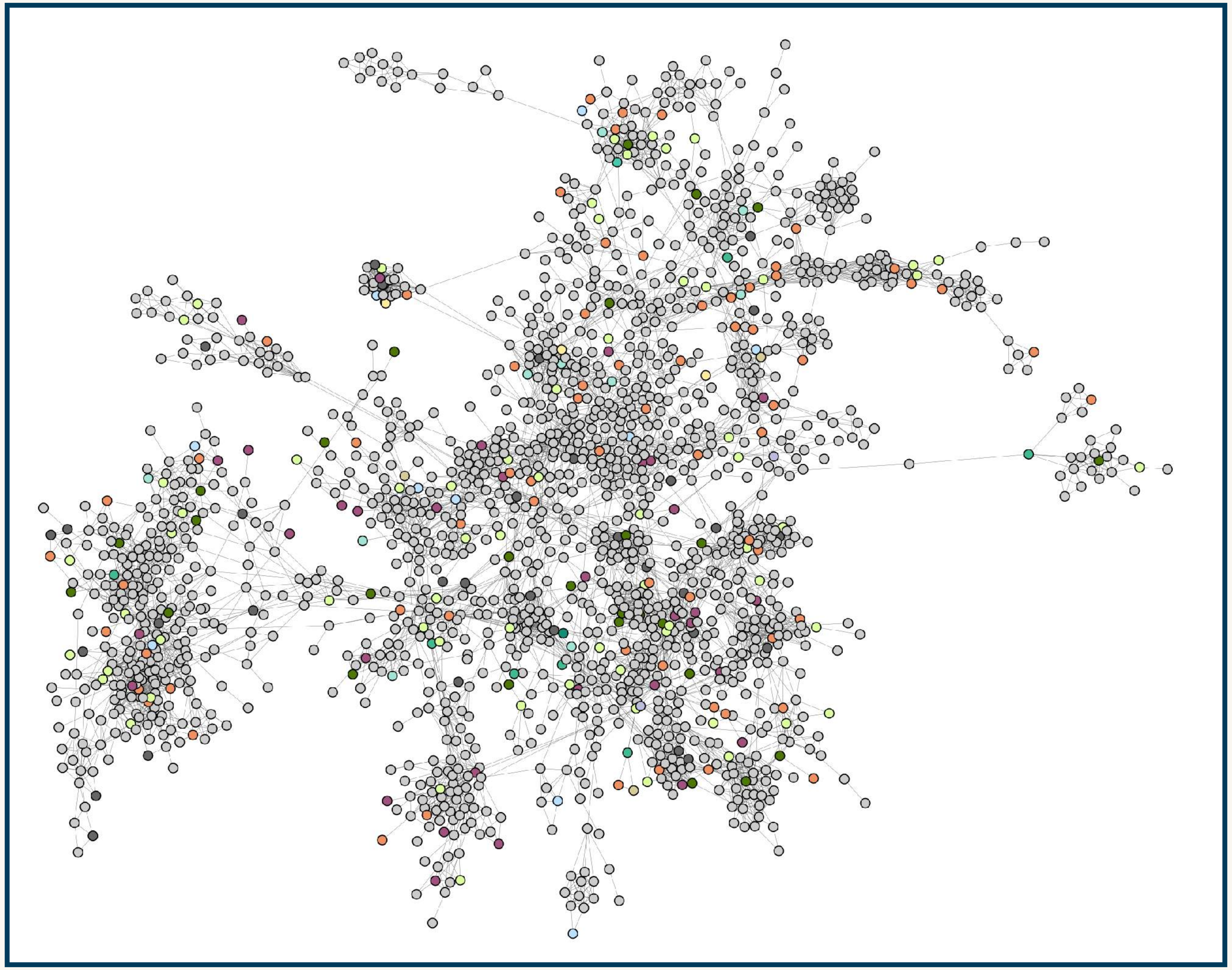



Retention Time (RT)



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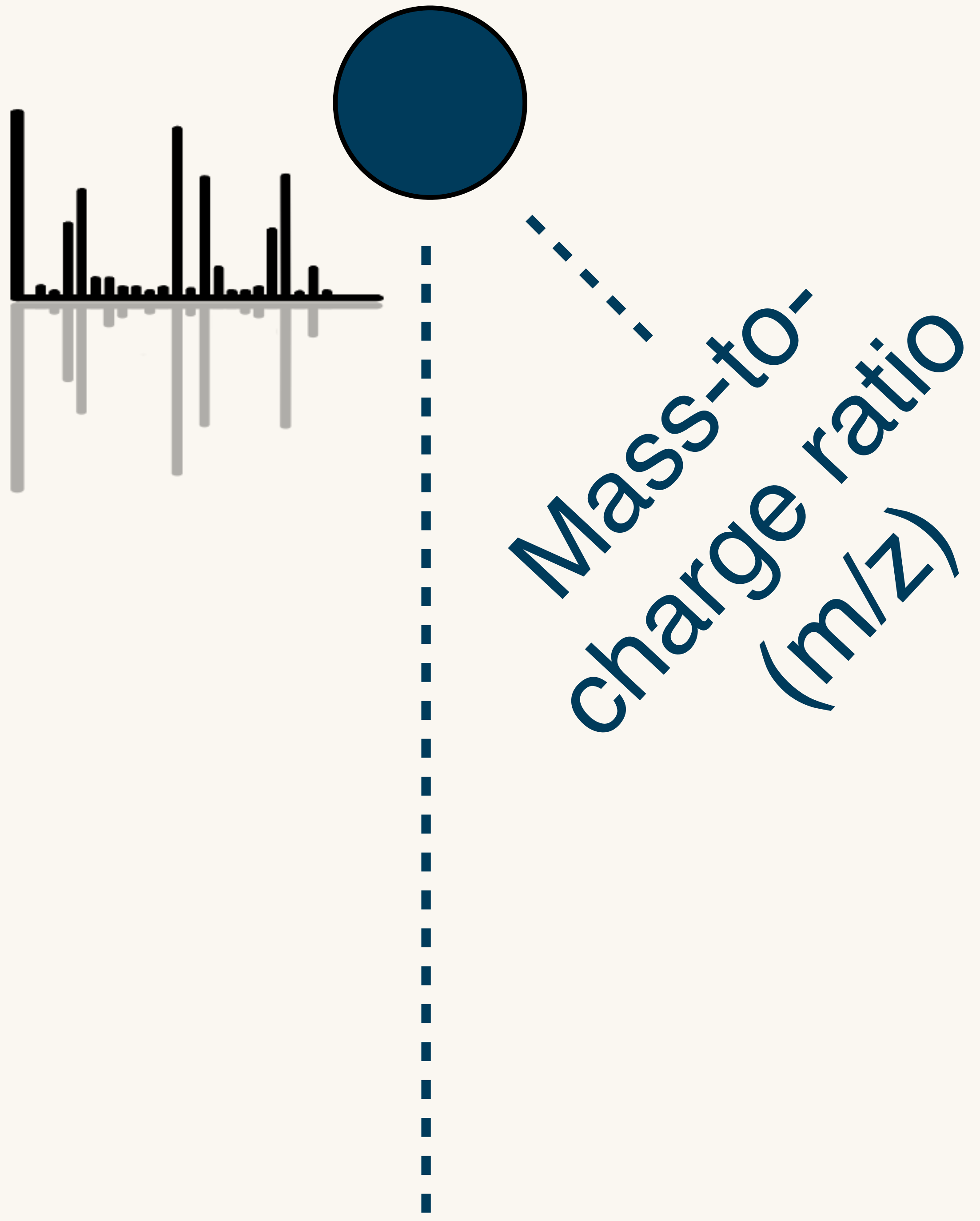
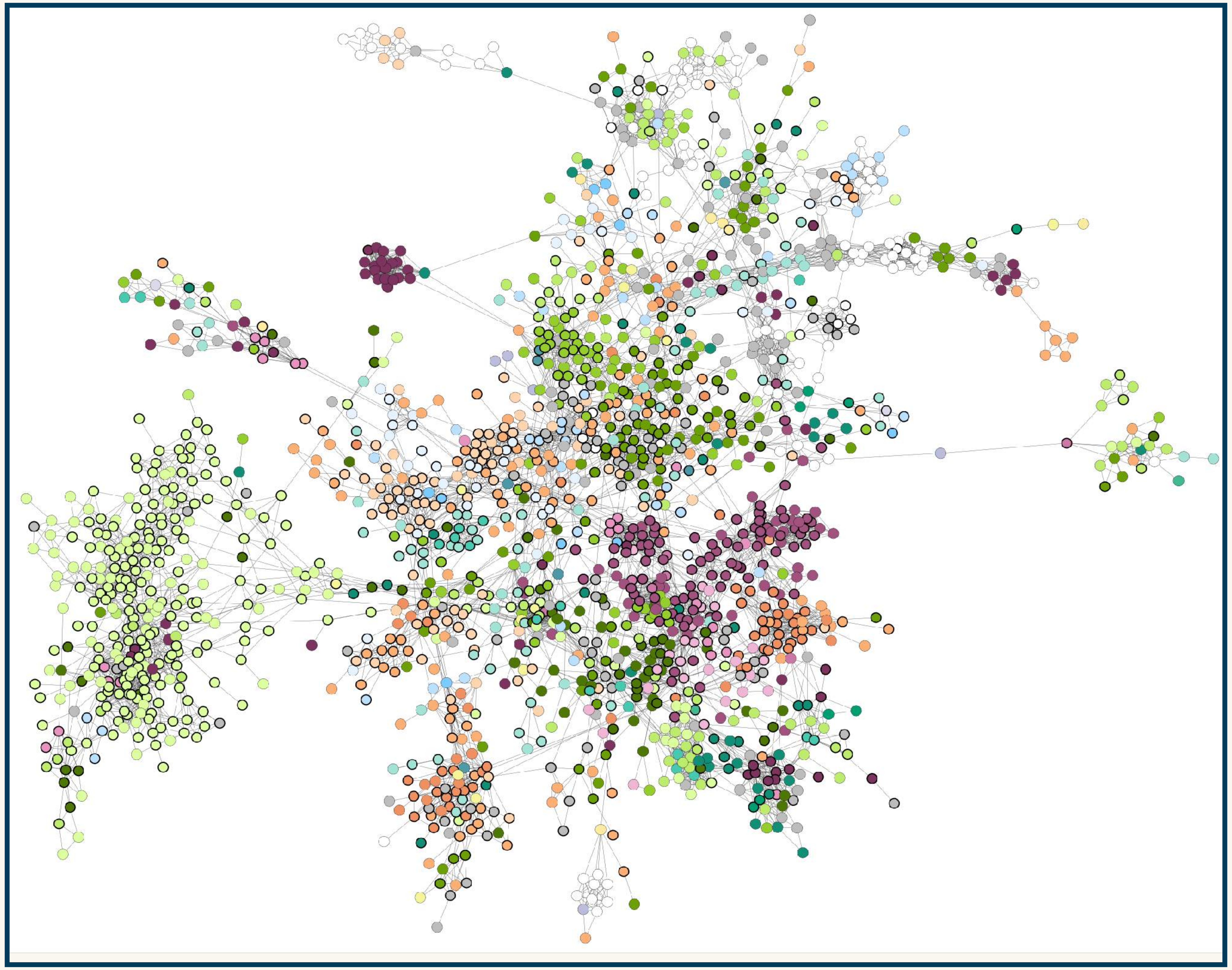
RT @ *m/z* pointer




Retention Time
(RT)



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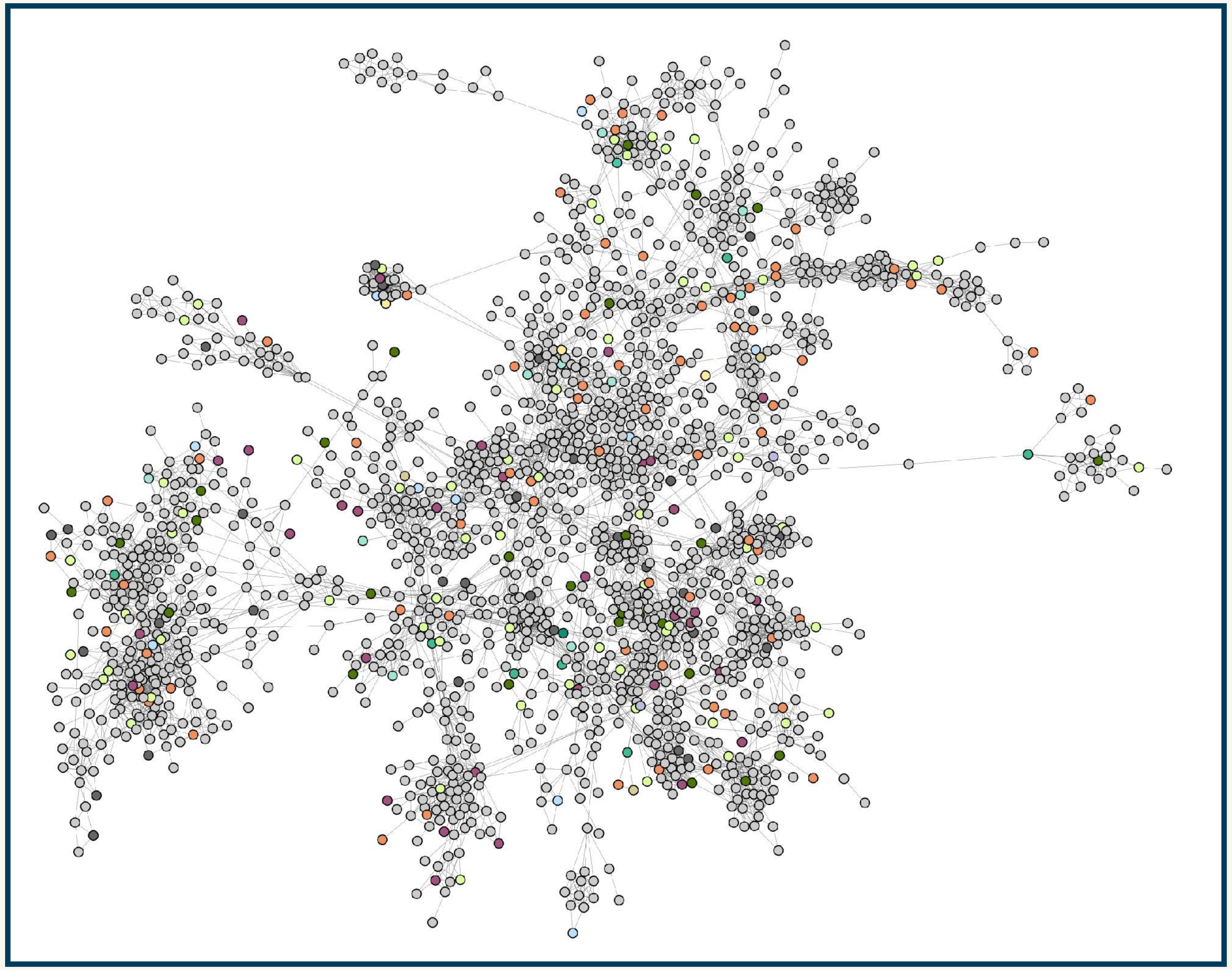

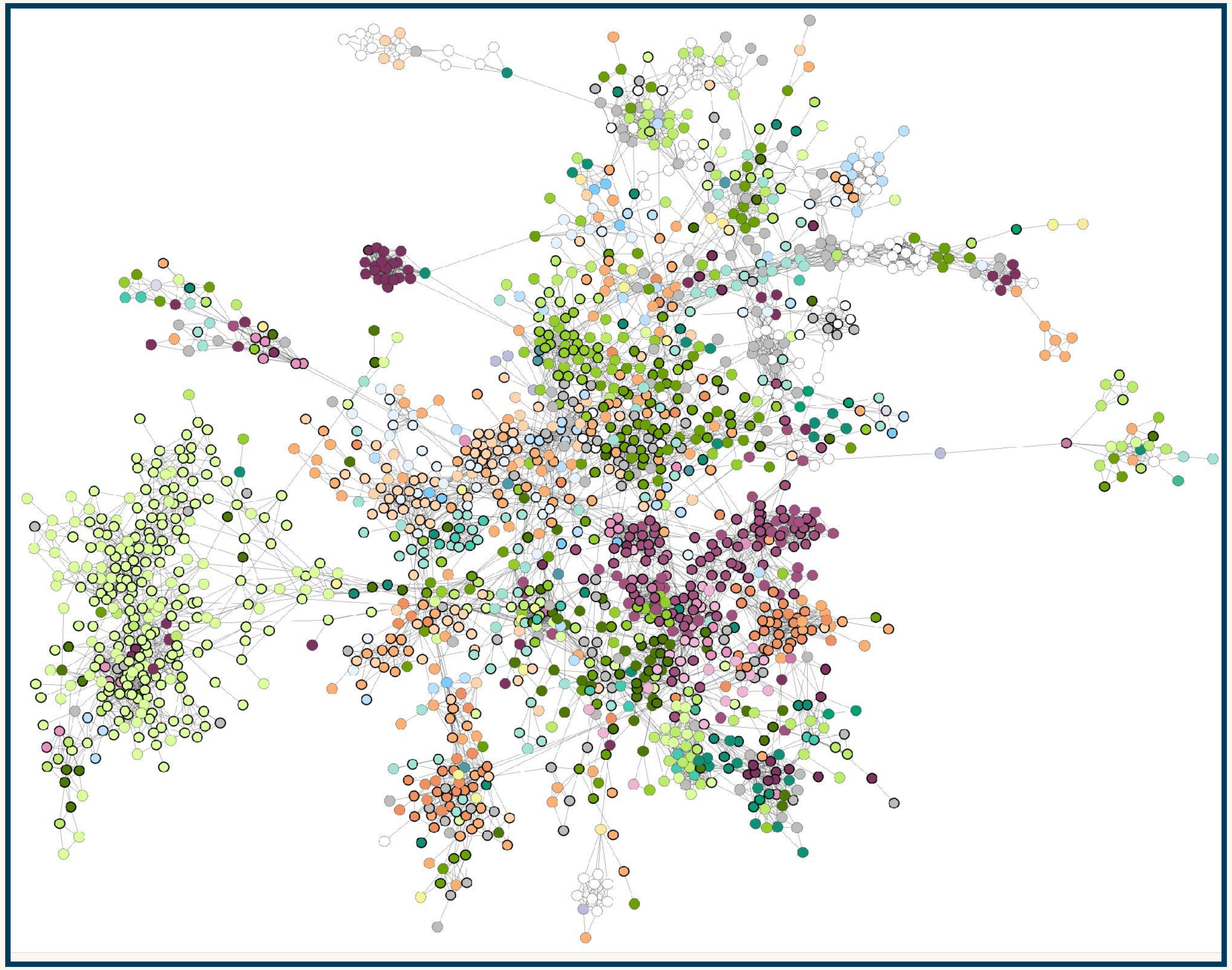

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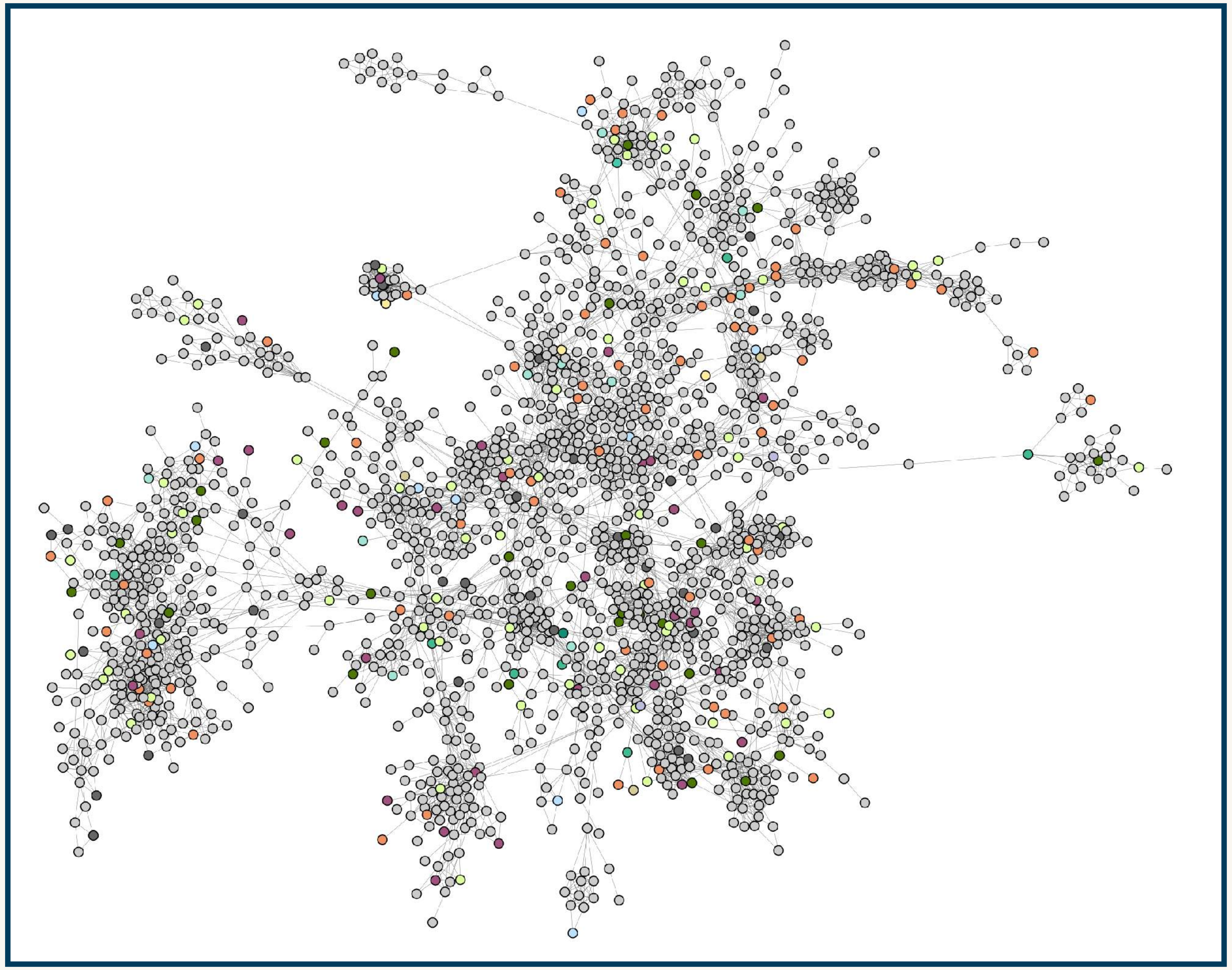
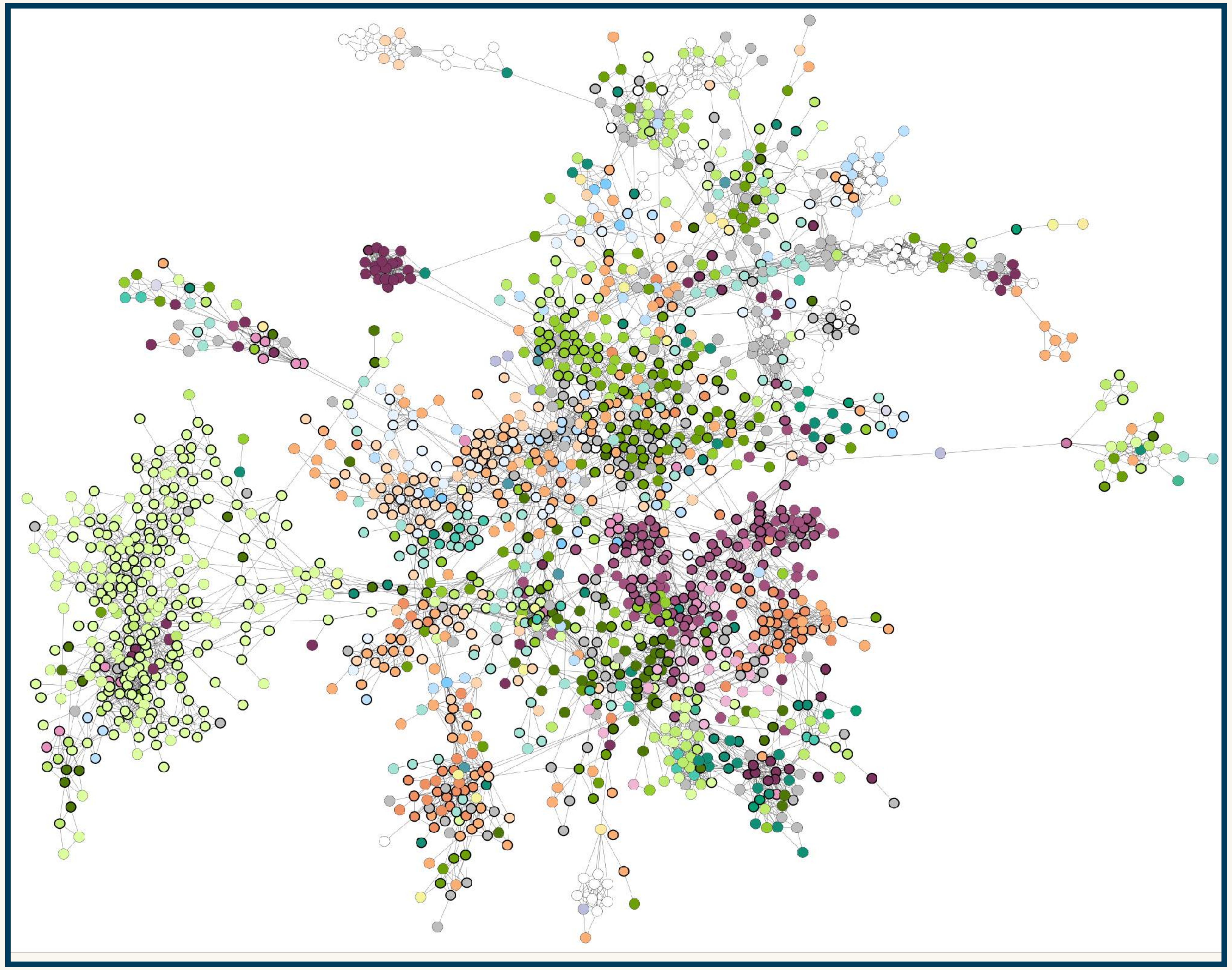


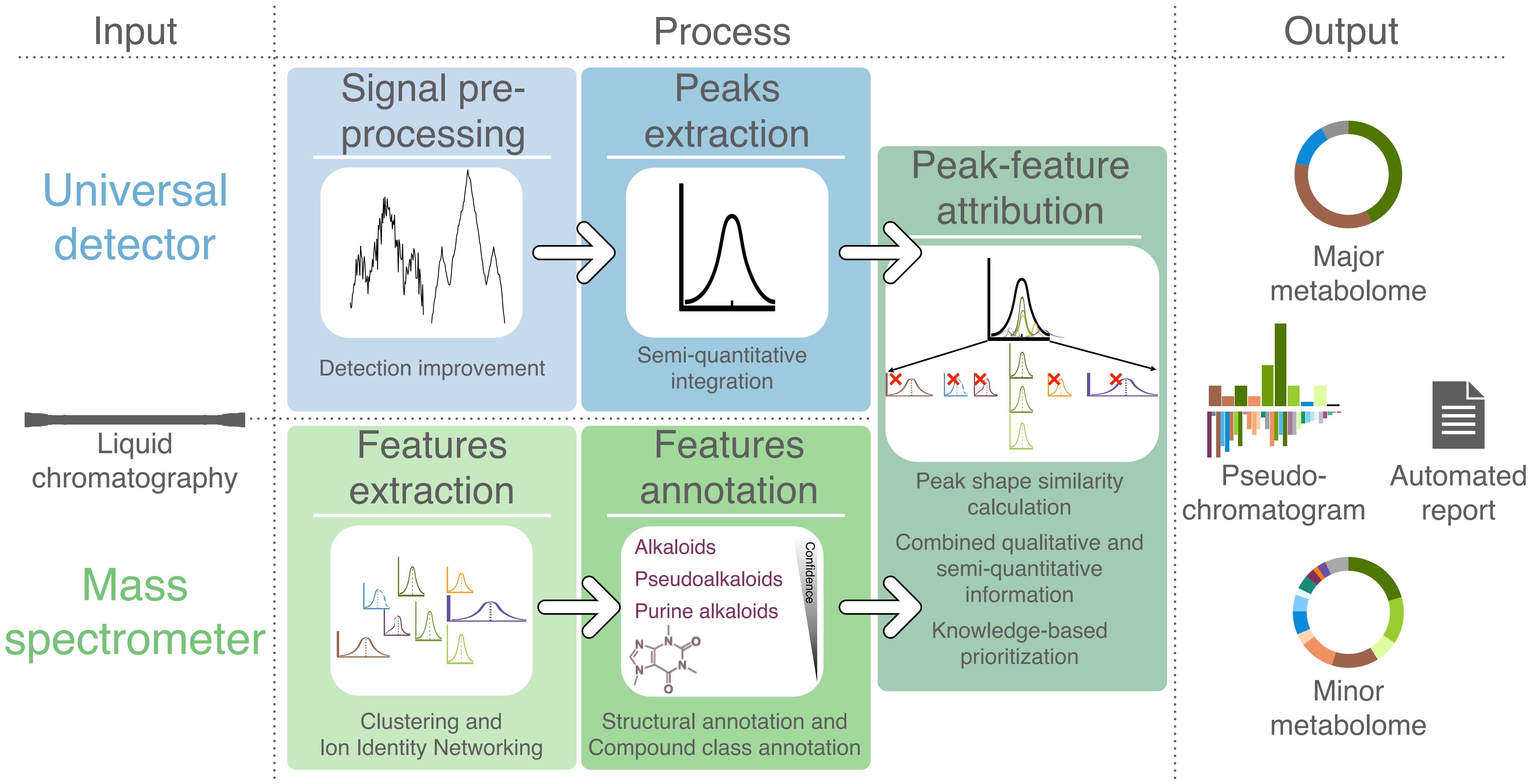


*Now that I am satisfied with the qualitative aspects, how could I include semi-quantitative data on the broadest range of signals?*

# Integrating Semiquantitative Detection

Because good annotations were not enough

# Automated Composition Assessment of Natural Extracts

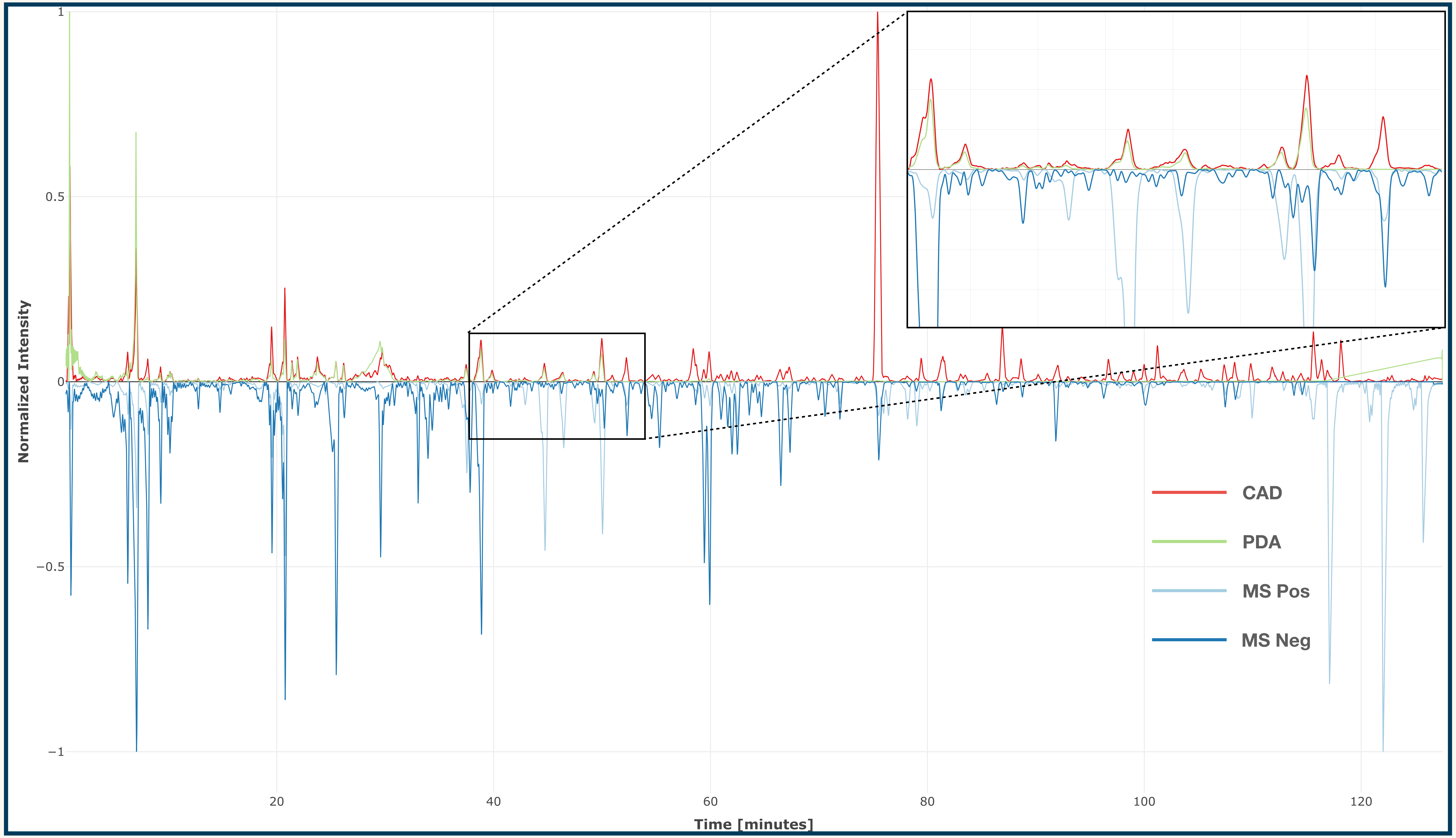



## Key message

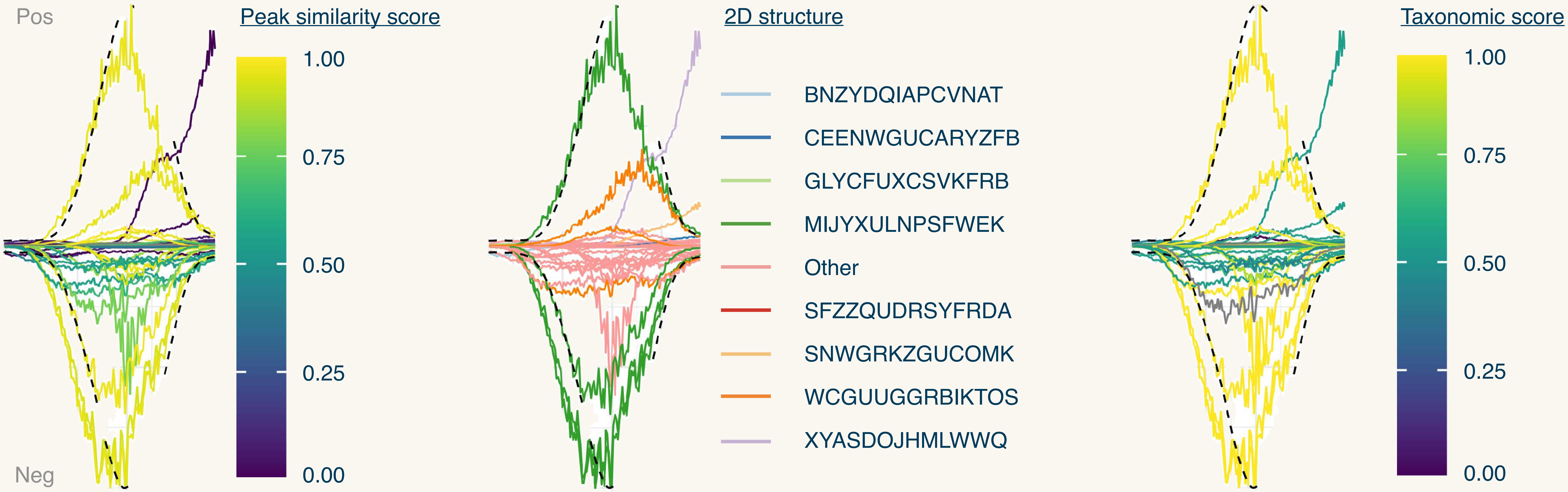
Some very minor components can be annotated in mass spectrometry with major ones ionizing poorly

- Automated reporting
- Metabolome split into “major” and “minor”

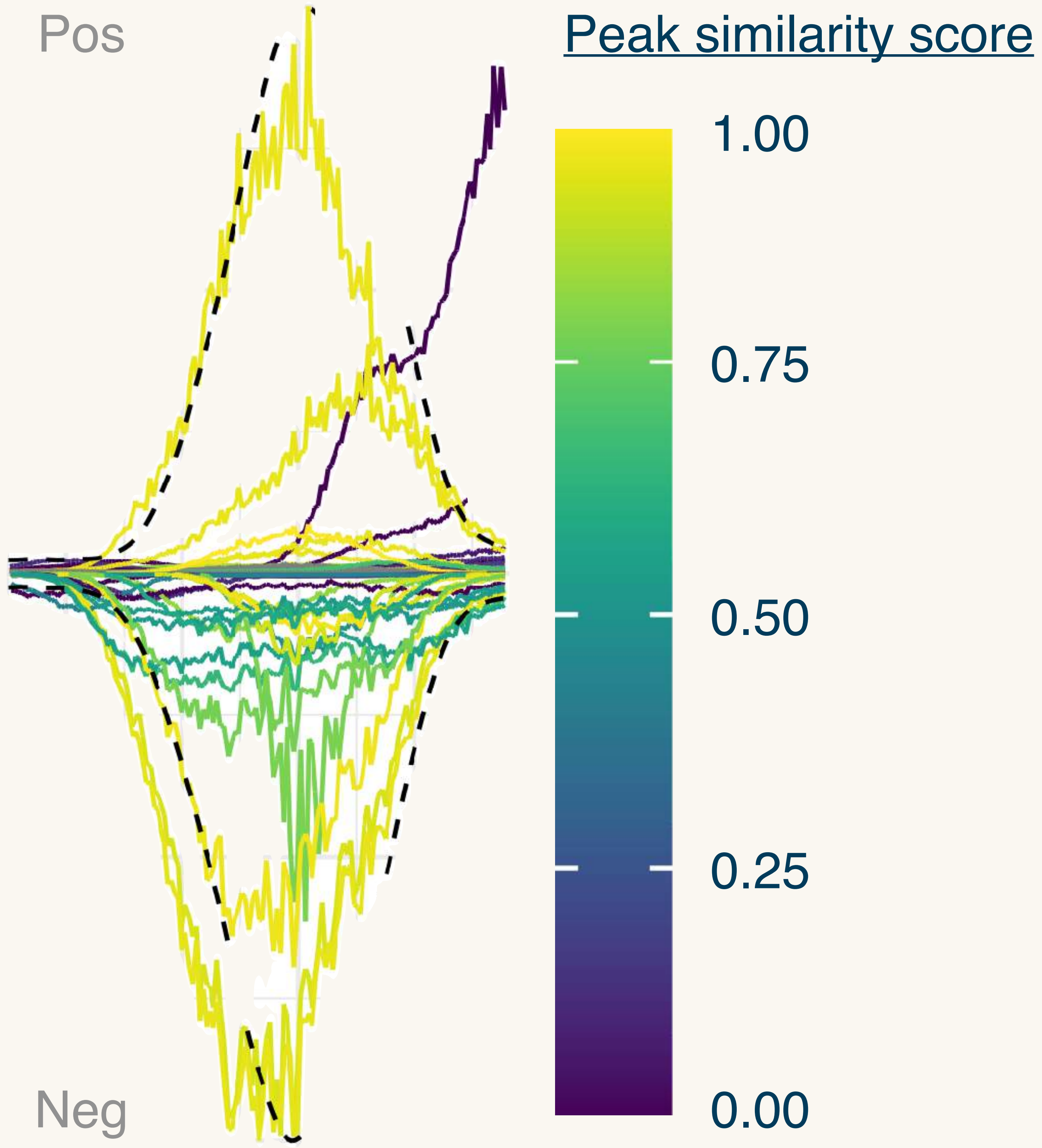
# Strong Requirement: Good Peak Shapes!

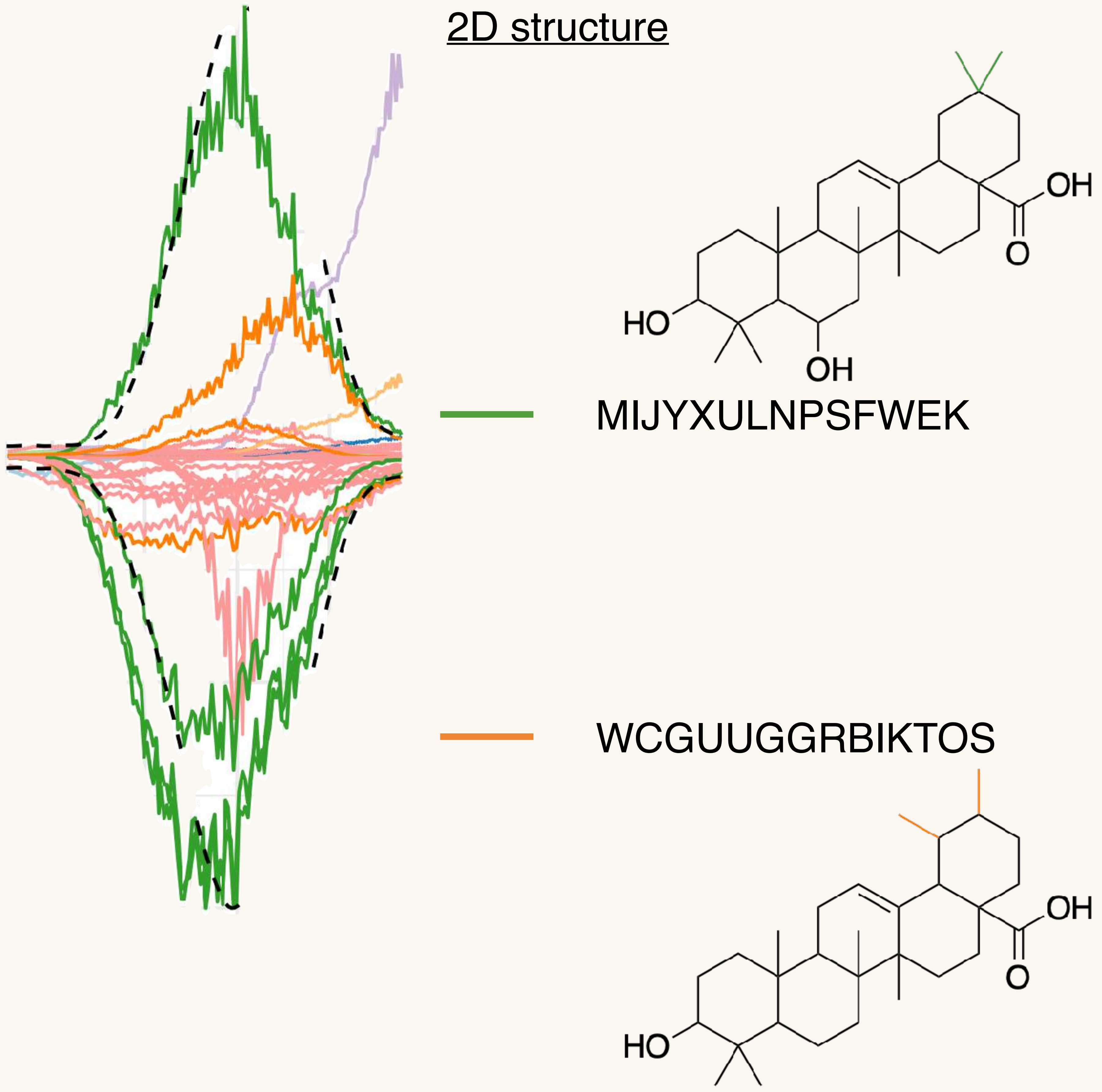


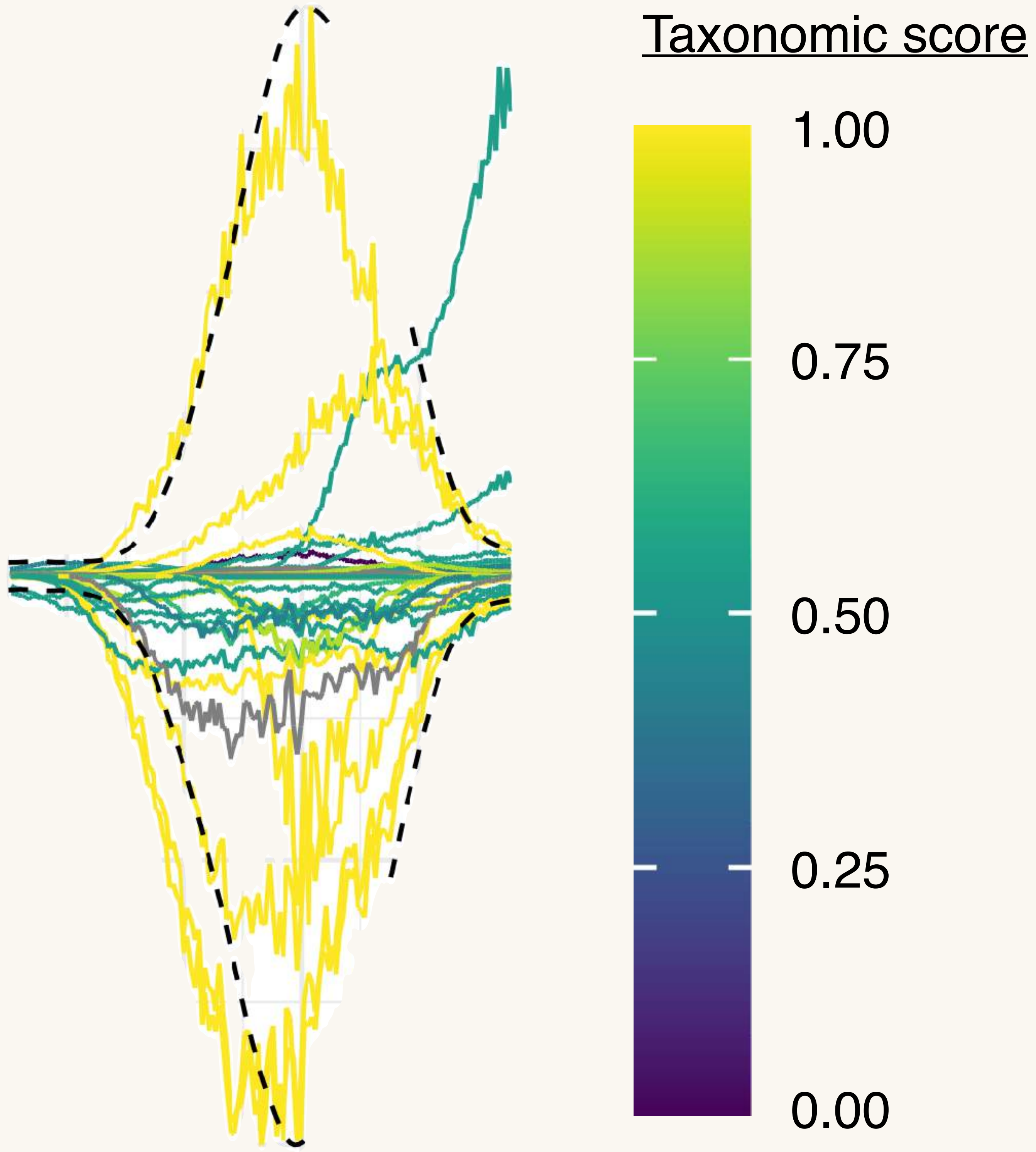
# Peak Shape Informed Decision



# Peak Shape Informed Decision

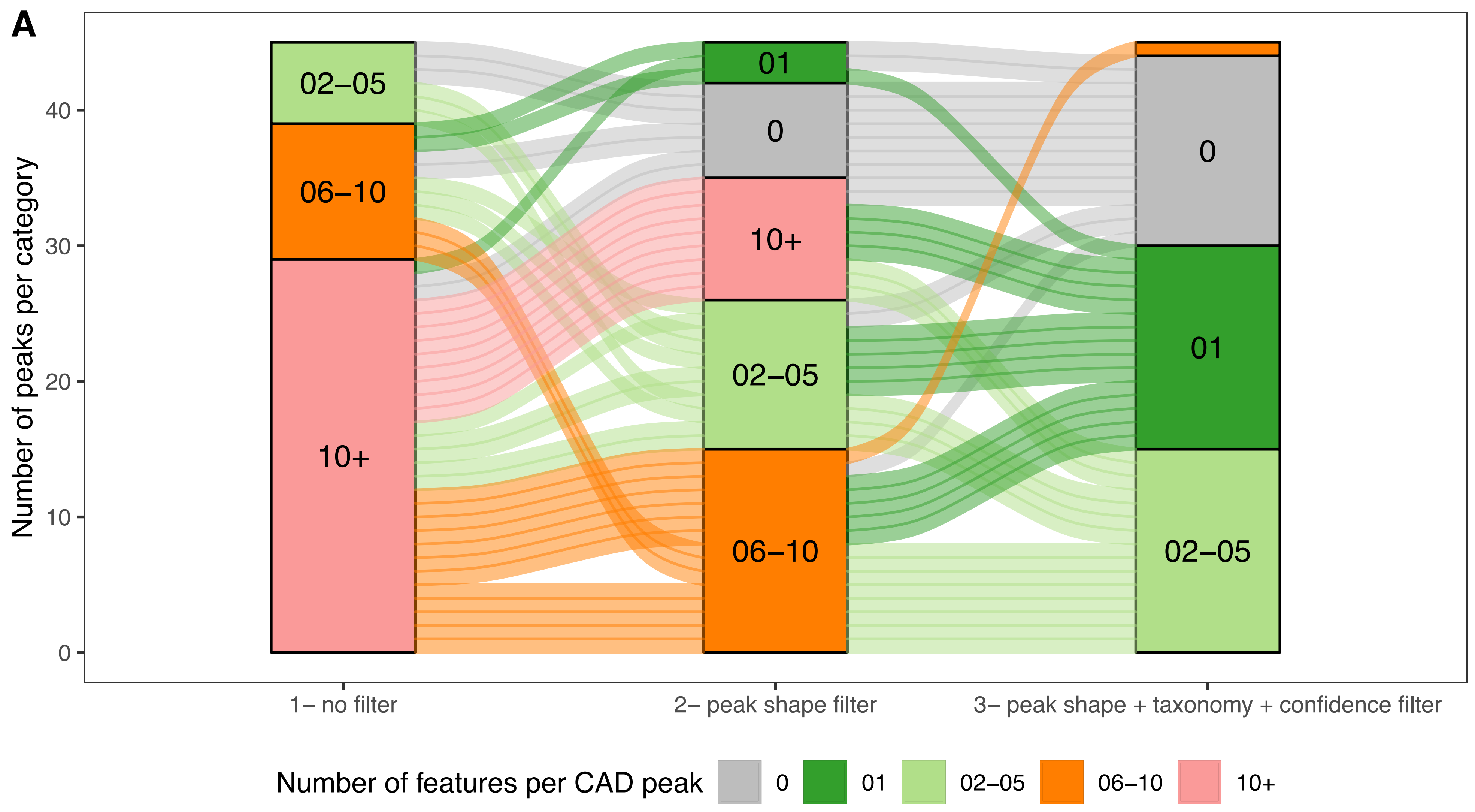


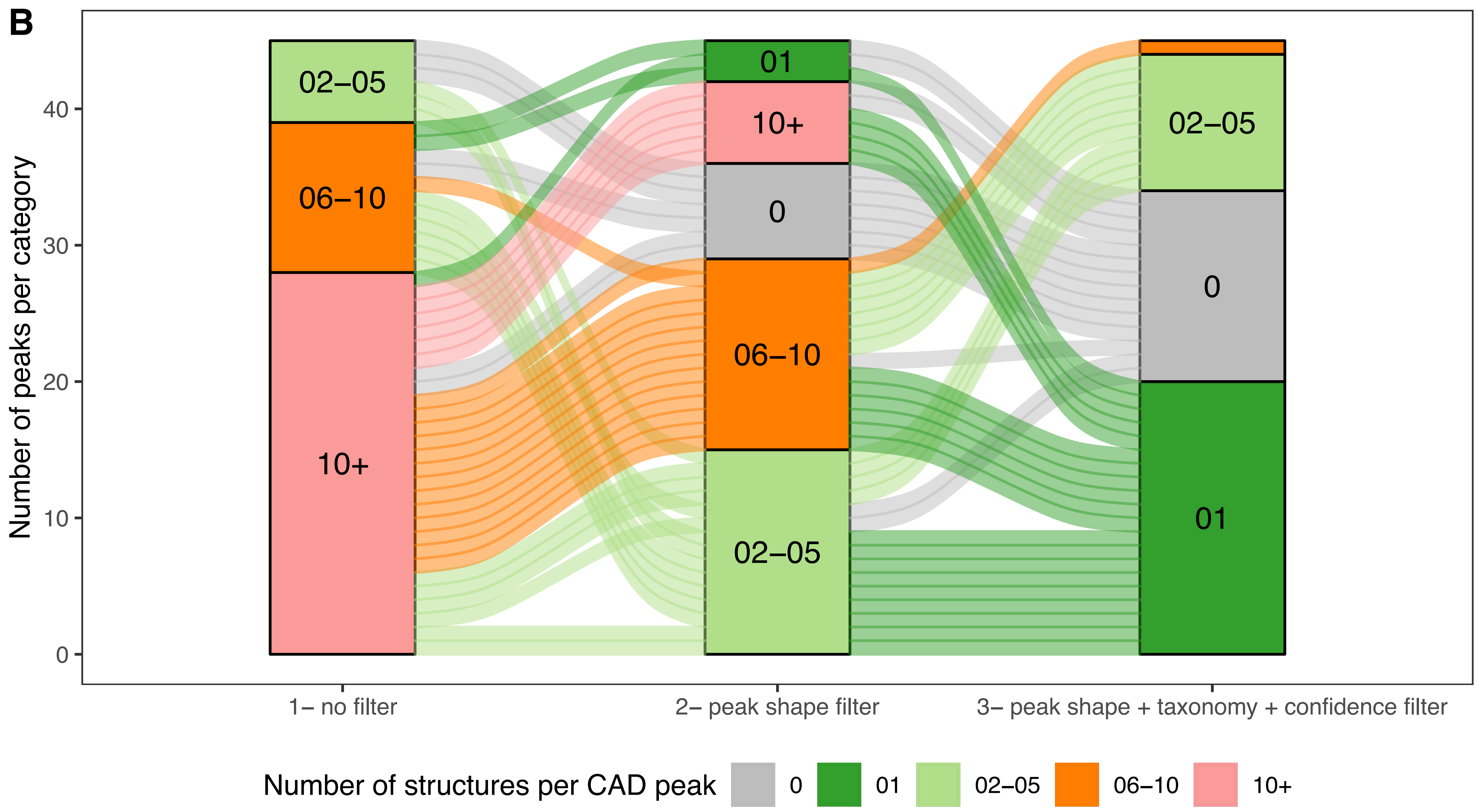



# Reduced Complexity

### Number of features per CAD peak

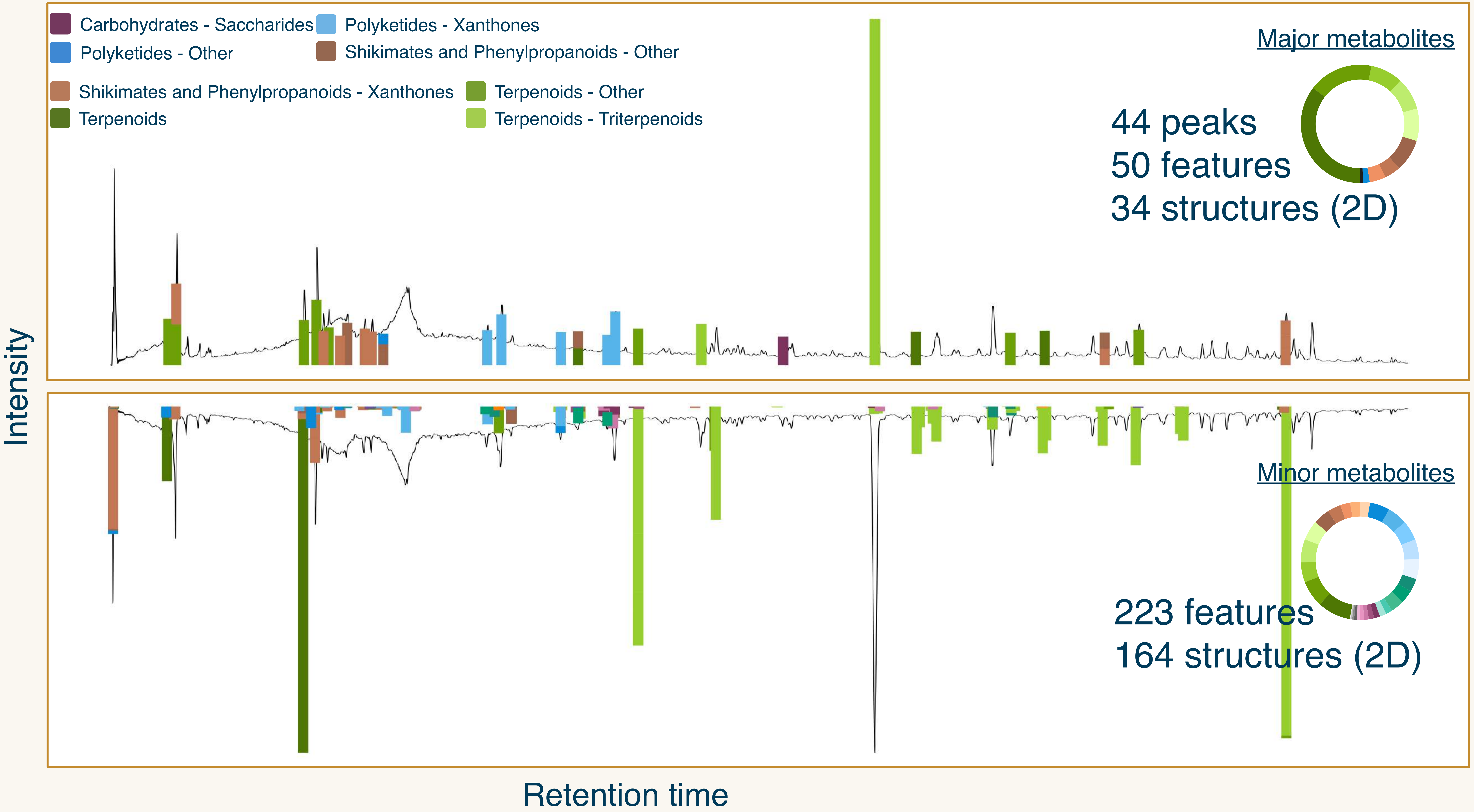



### Number of structures per CAD peak

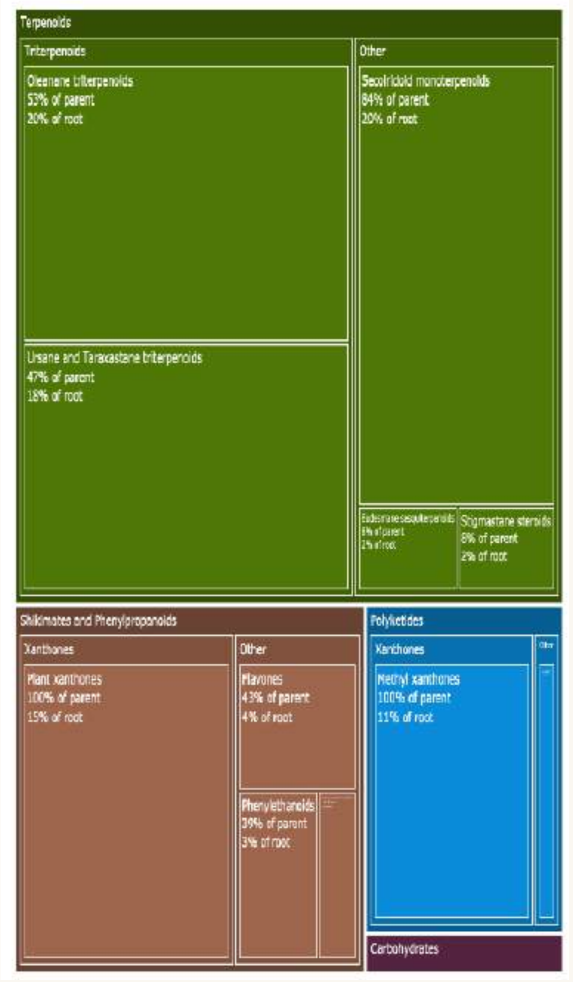
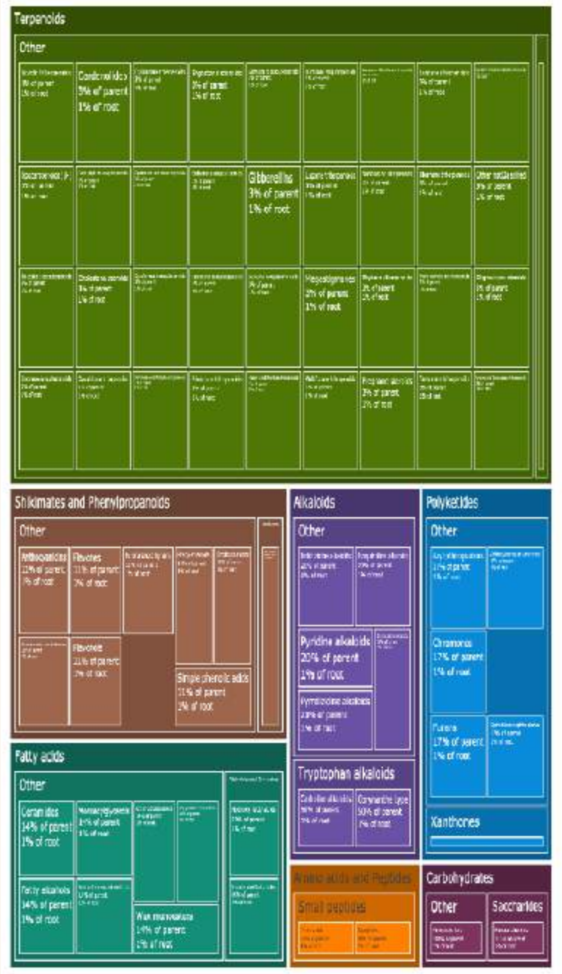



*Half of the peaks could be unambiguously annotated to a single structure!*

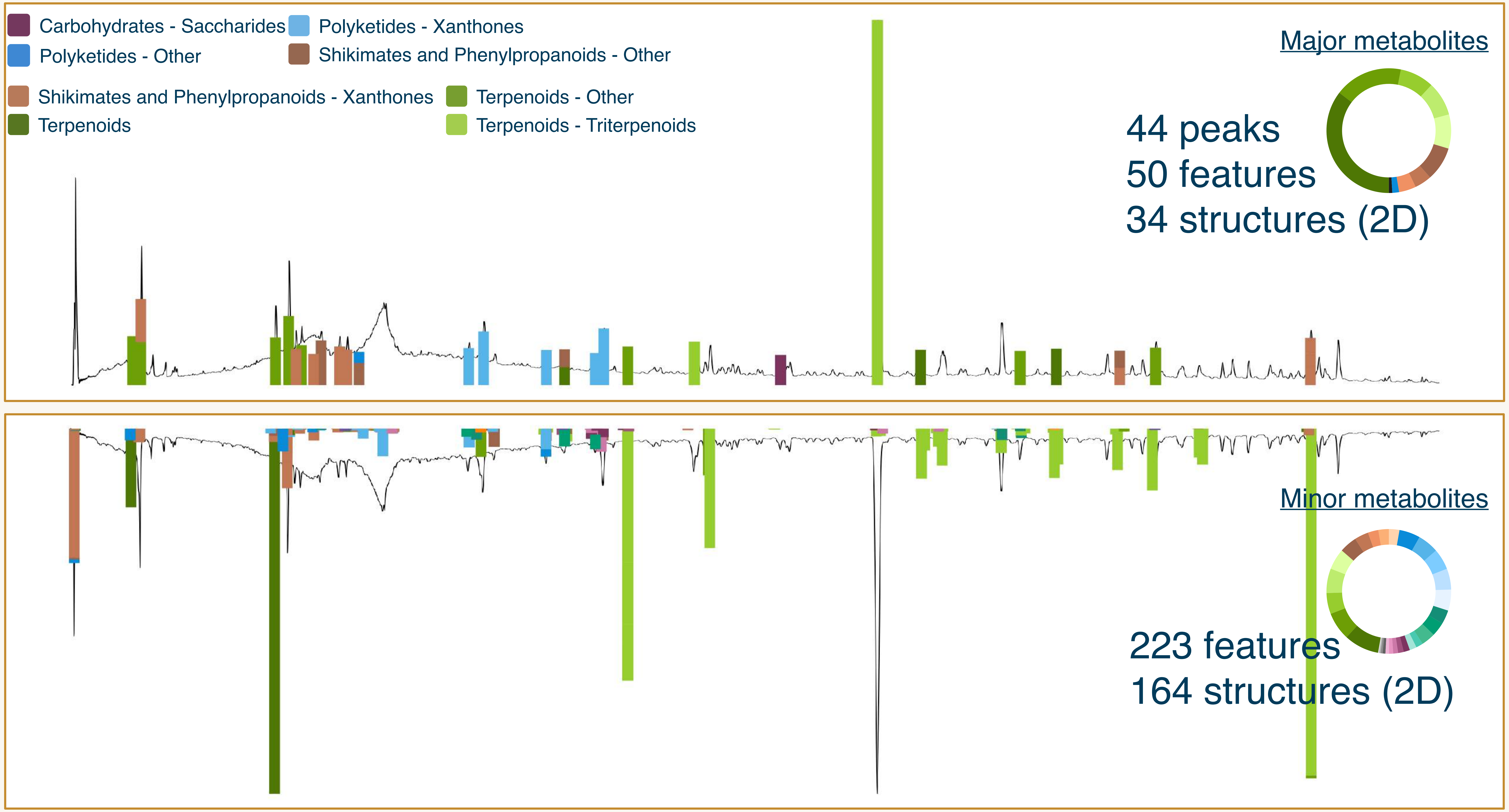
# Chemically Informed Pseudochromatogram



# Chemically Informed Pseudochromatogram

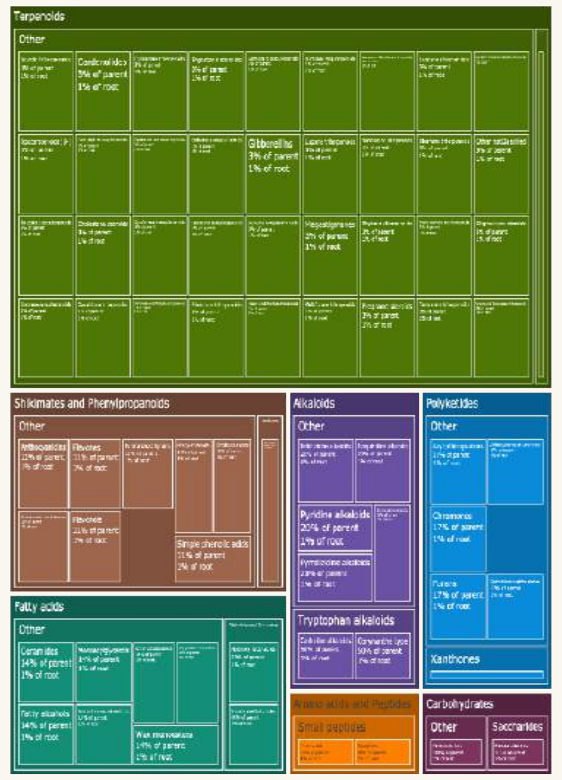
Intensity



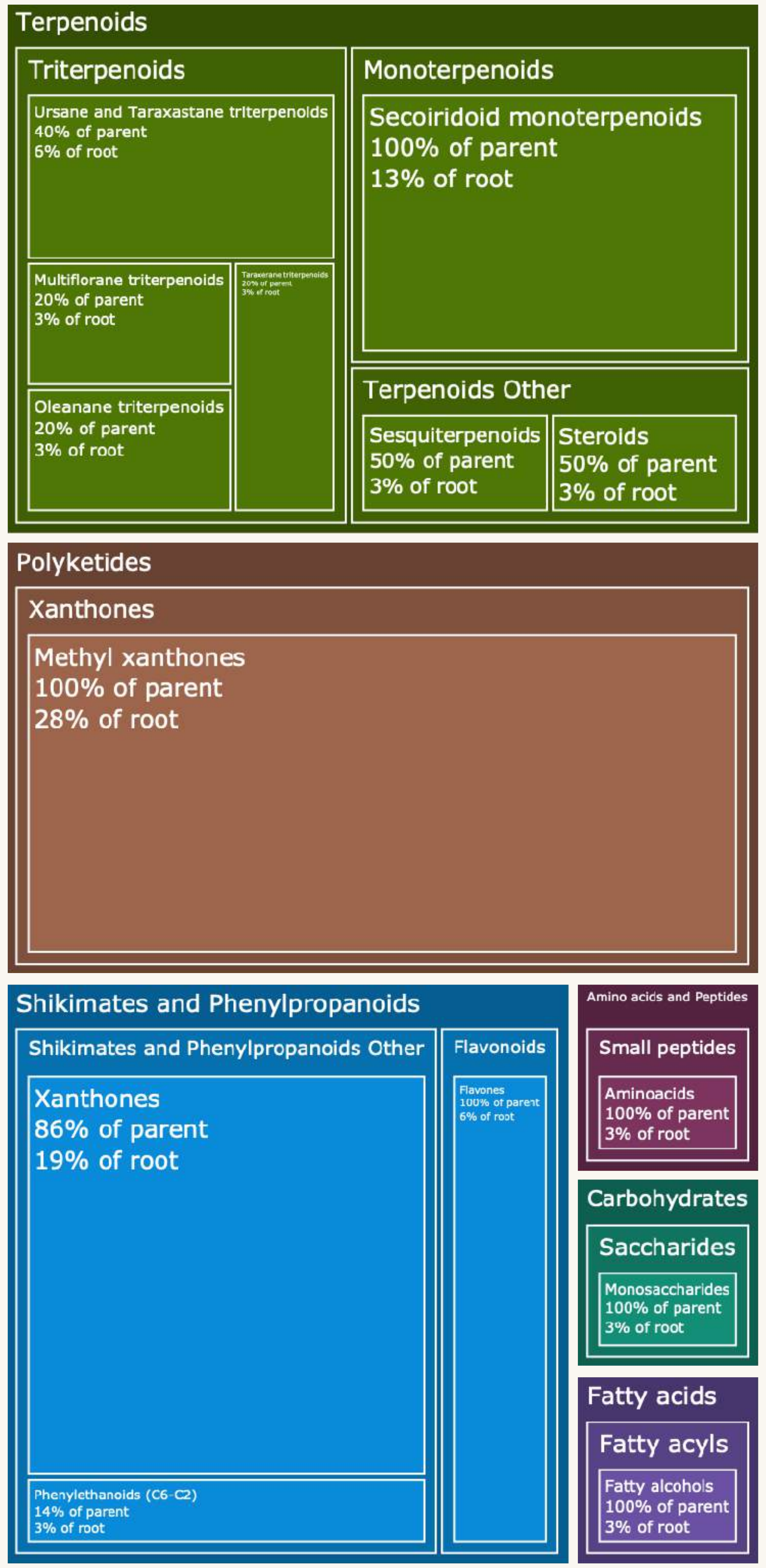

Retention time

# Comparison to Literature

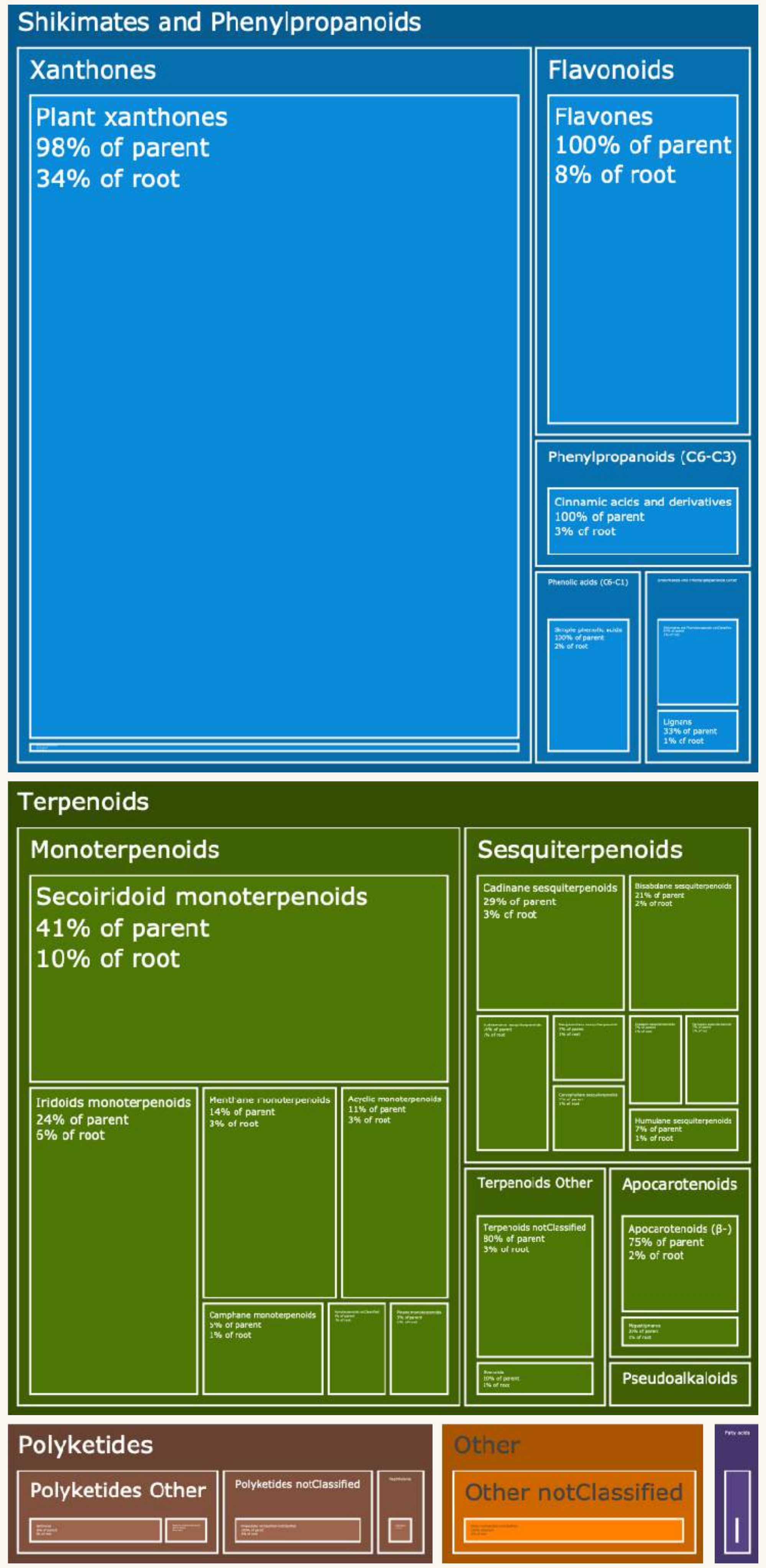
164 minor 2D structures



34 major 2D structures



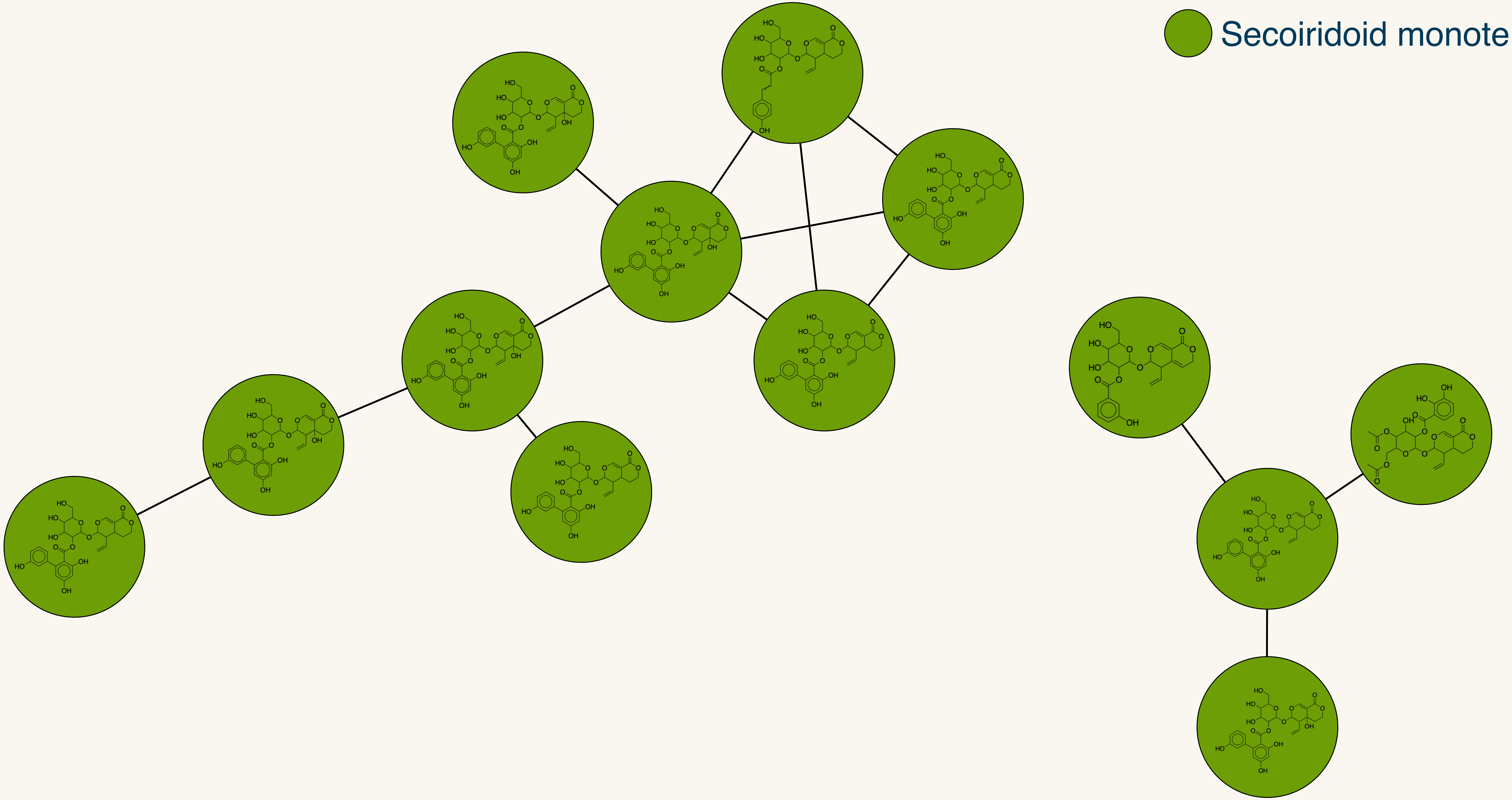

19 « detectable » 2D structures in *Swertia chirayita*



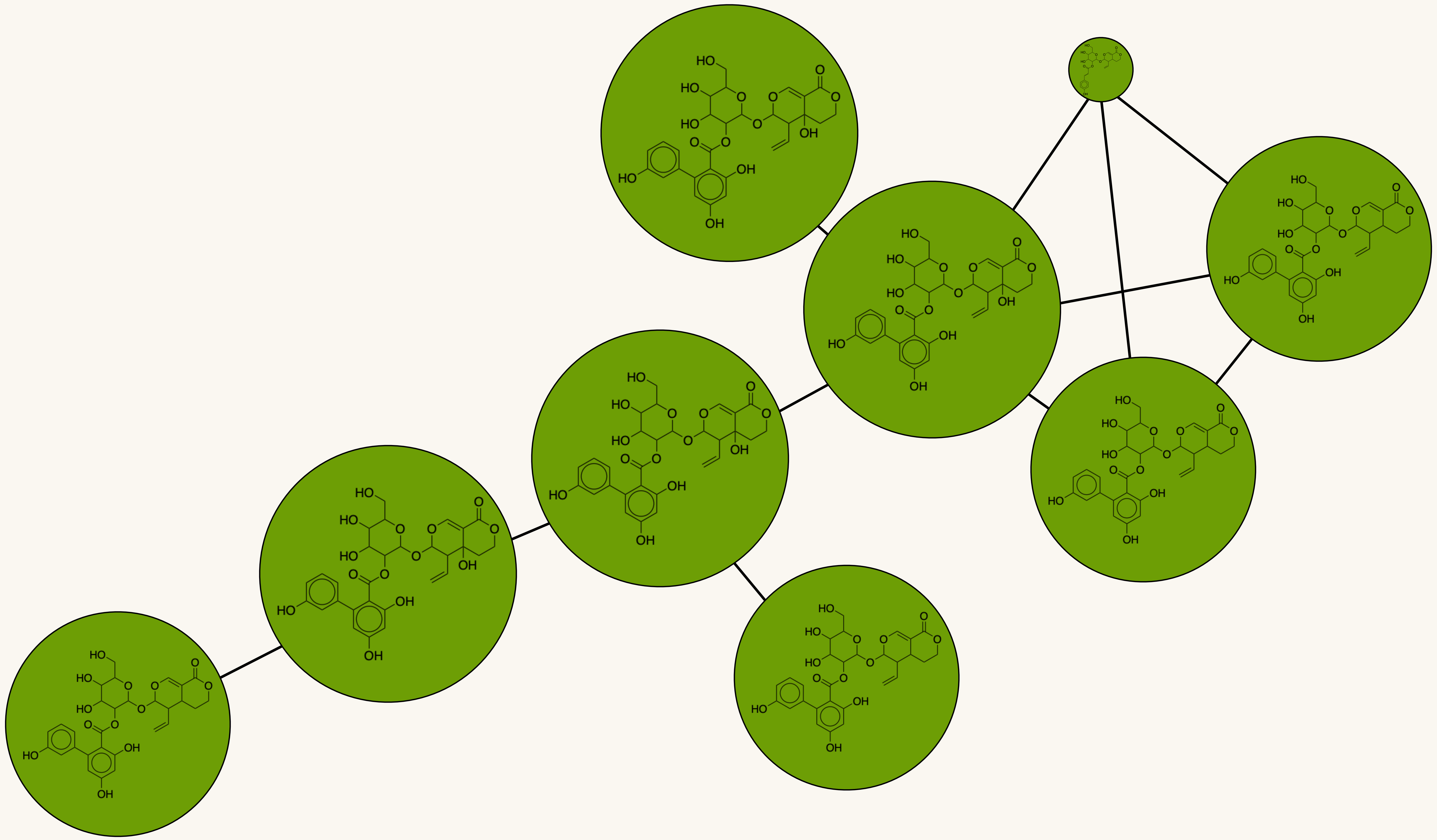

147 in *Swertia* genus

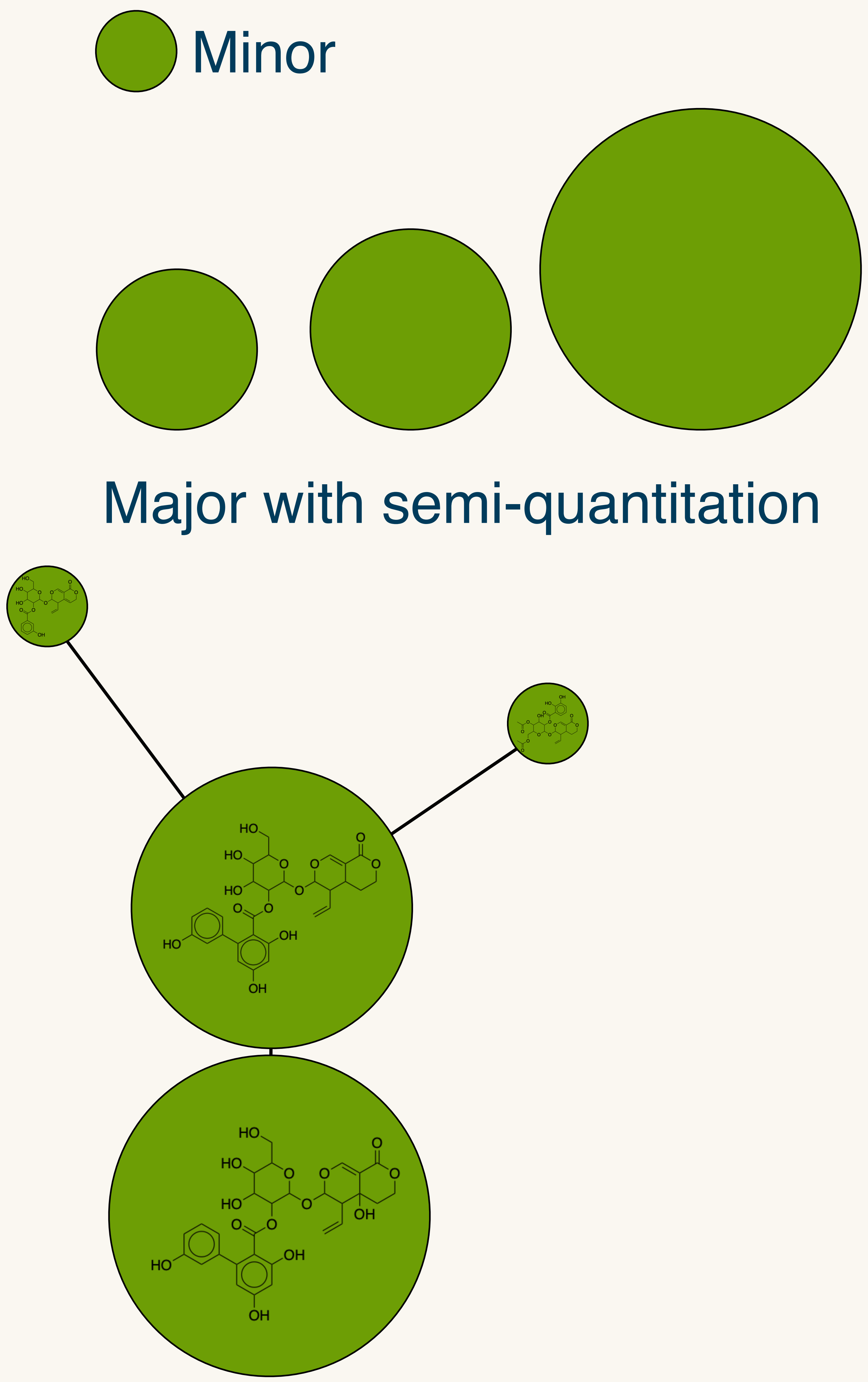
# Semi-Quantitative Molecular Networks

Secoiridoid monoterpenoids

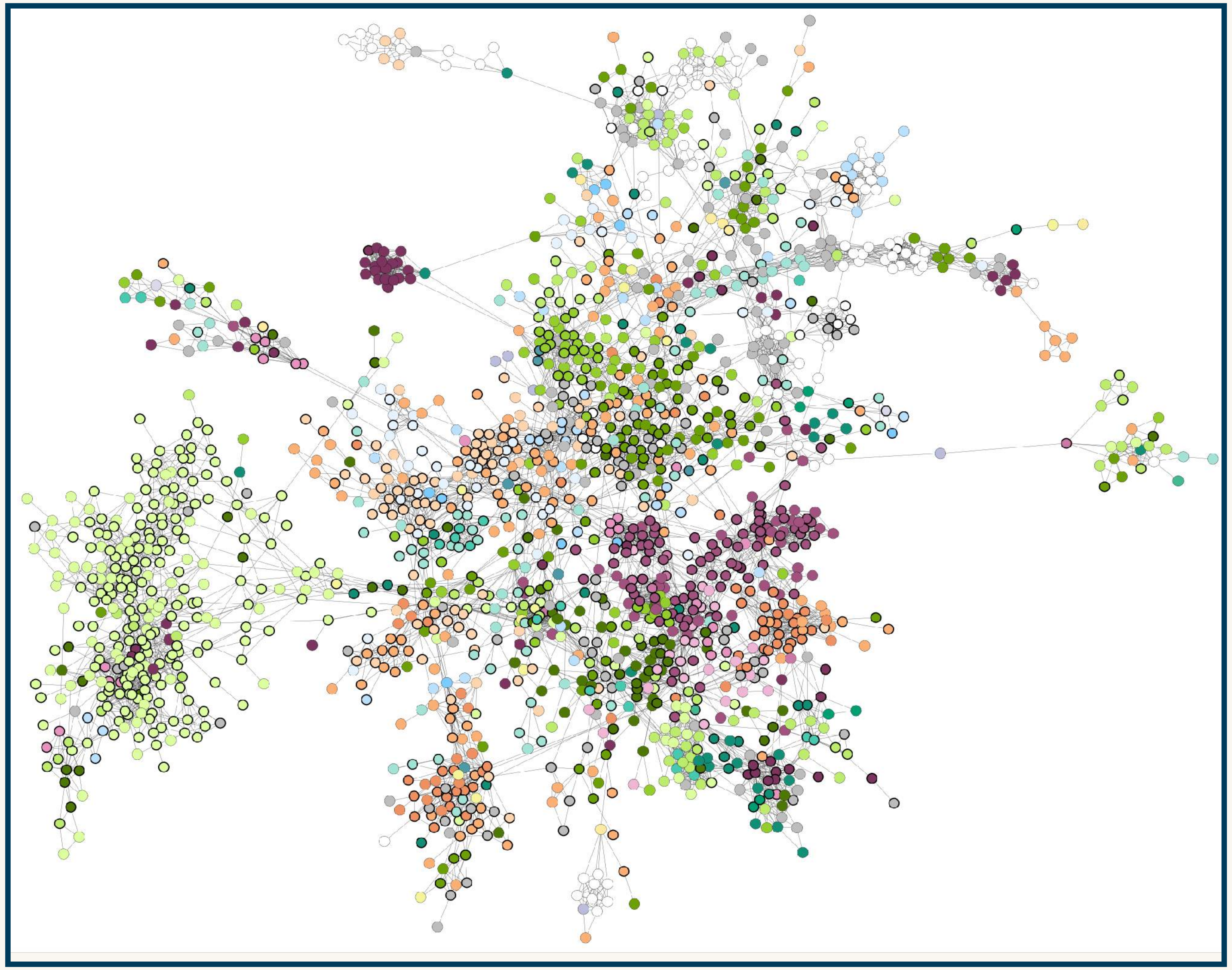


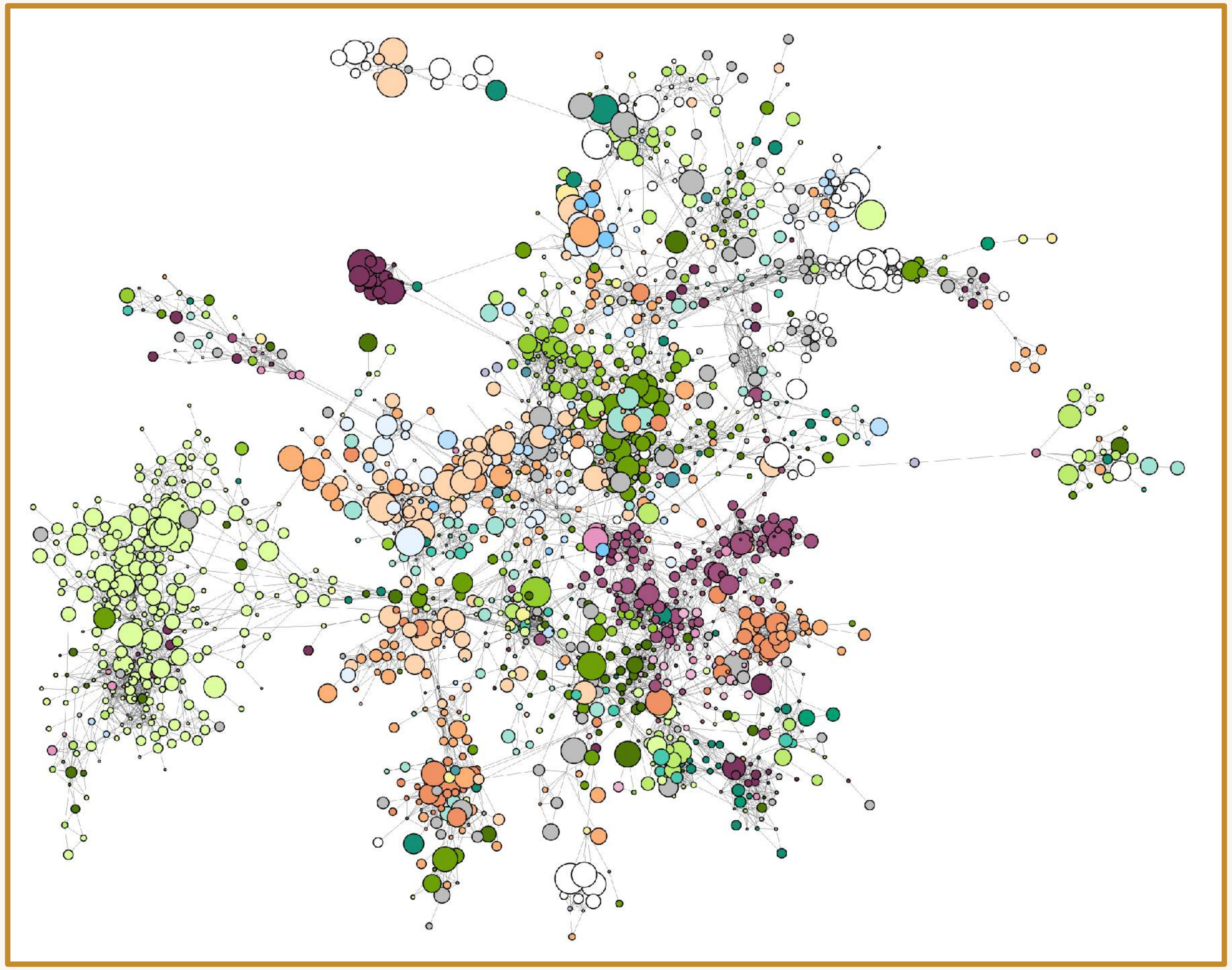
# Semi-Quantitative Molecular Networks





# Semi-Quantitative Molecular Networks





22-25 Giugno 2026

# Linking Comprehensive Compositional Assessment to Bioactivity

Because *bitter is better*

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Divisione di Spettrometria di Massa

# a Strategy to Analyze Plant extracts taste In Depth

MASSA 22-25 Giugno 2026


Selection

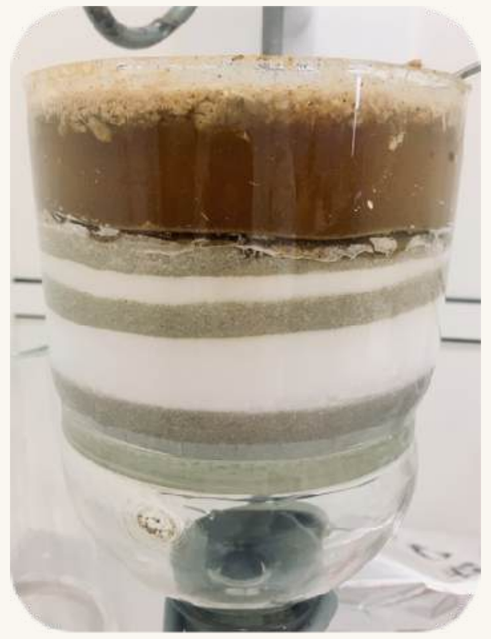
Extraction

11kg EtOH + 1.1kg *Swertia chirayita*

Lyophilisation

86.3 g

Pre-fractionation



30.0 g

Vanillin smell

16.1 g

Extremely bitter

10.4 g

10.0 g

Fractionation

82 fractions 20mg scale

Lyophilisation

Fractions for further study



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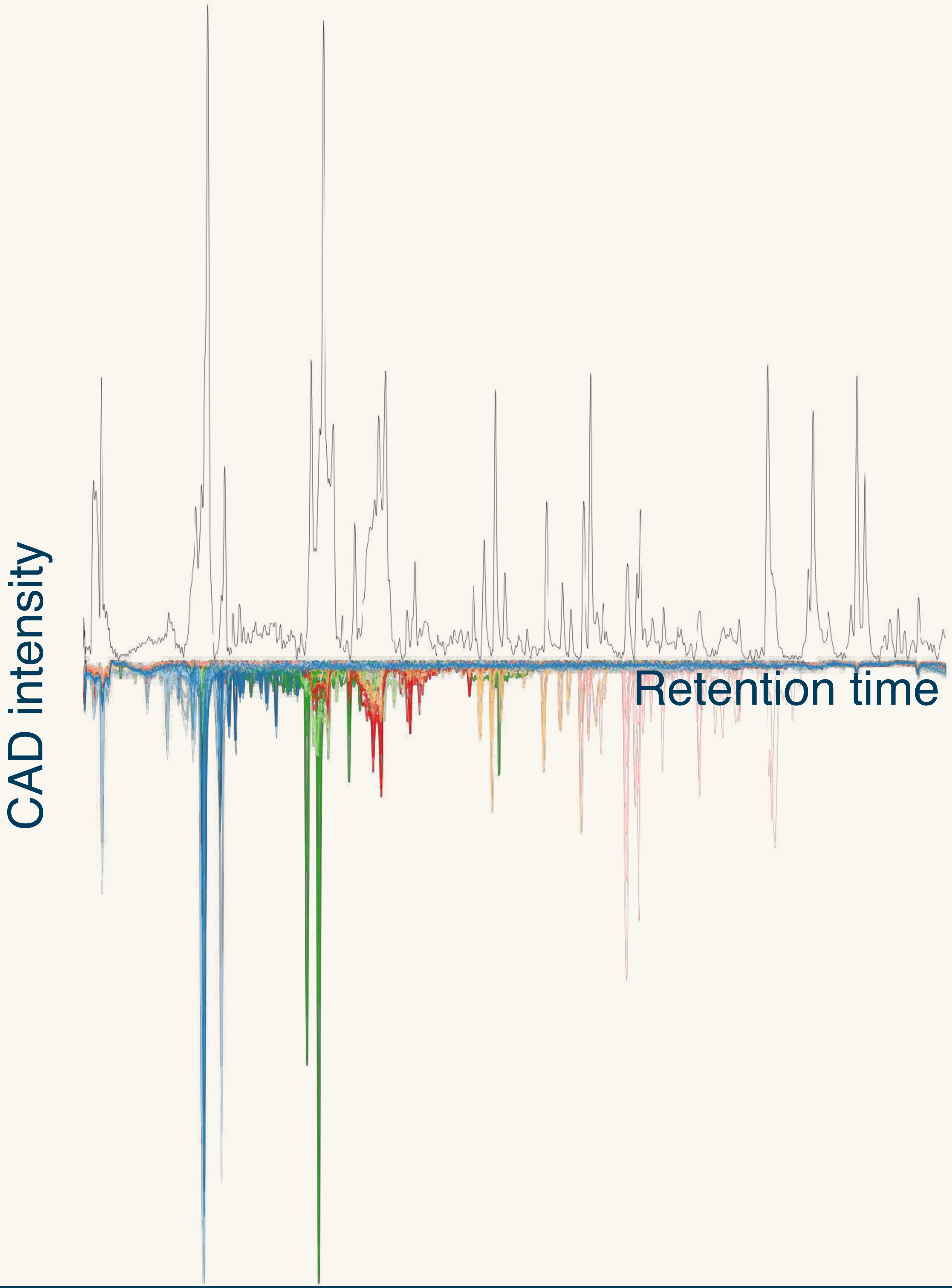
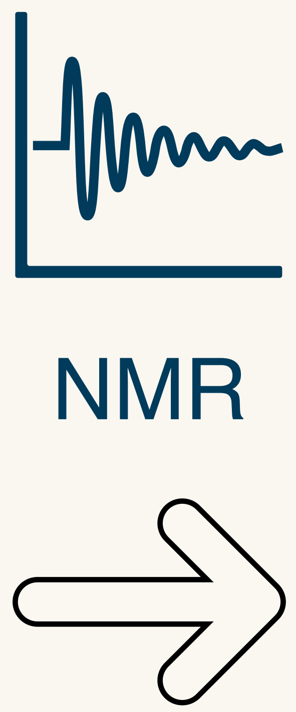
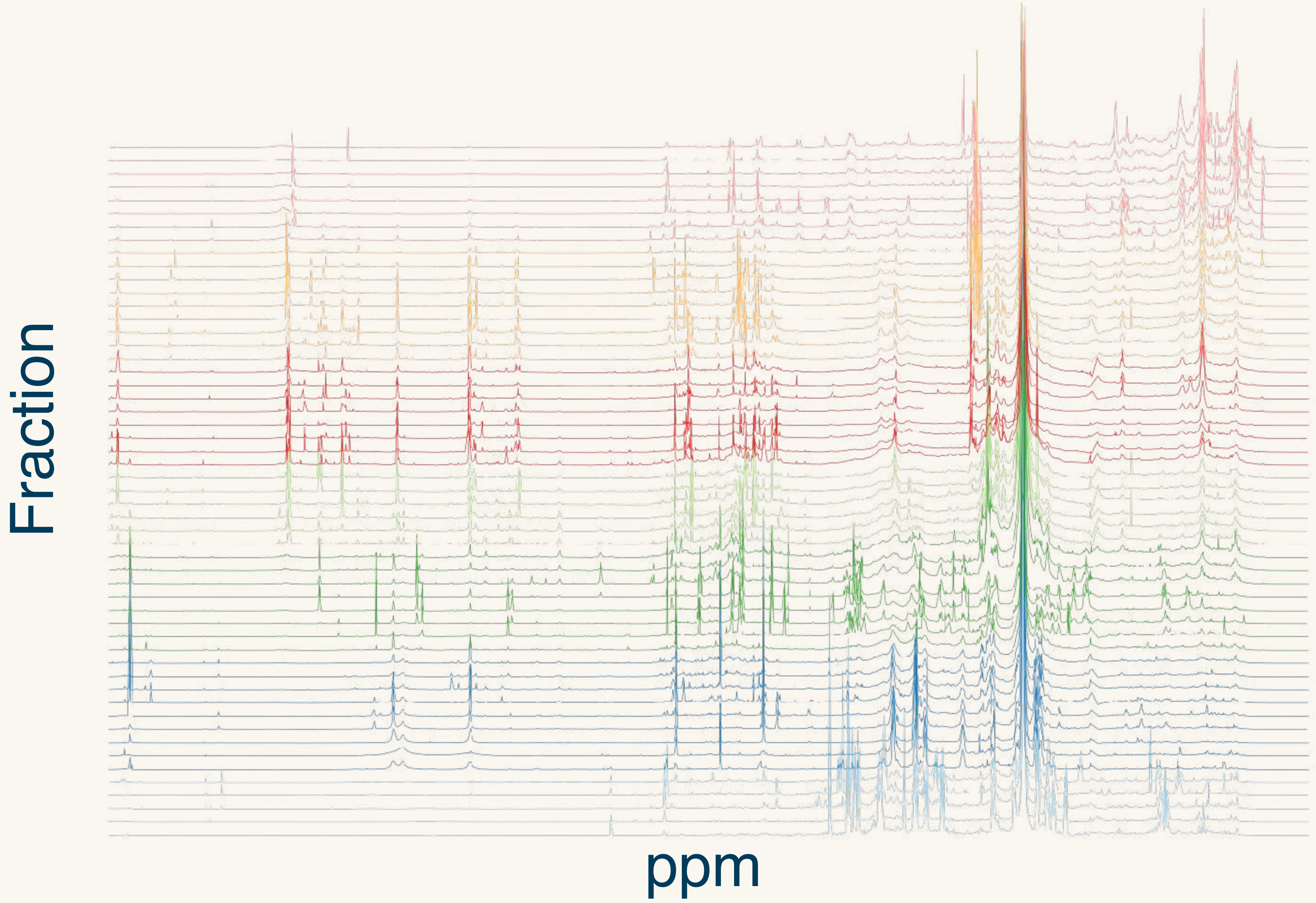
# a Strategy to Analyze Plant extracts taste In Depth



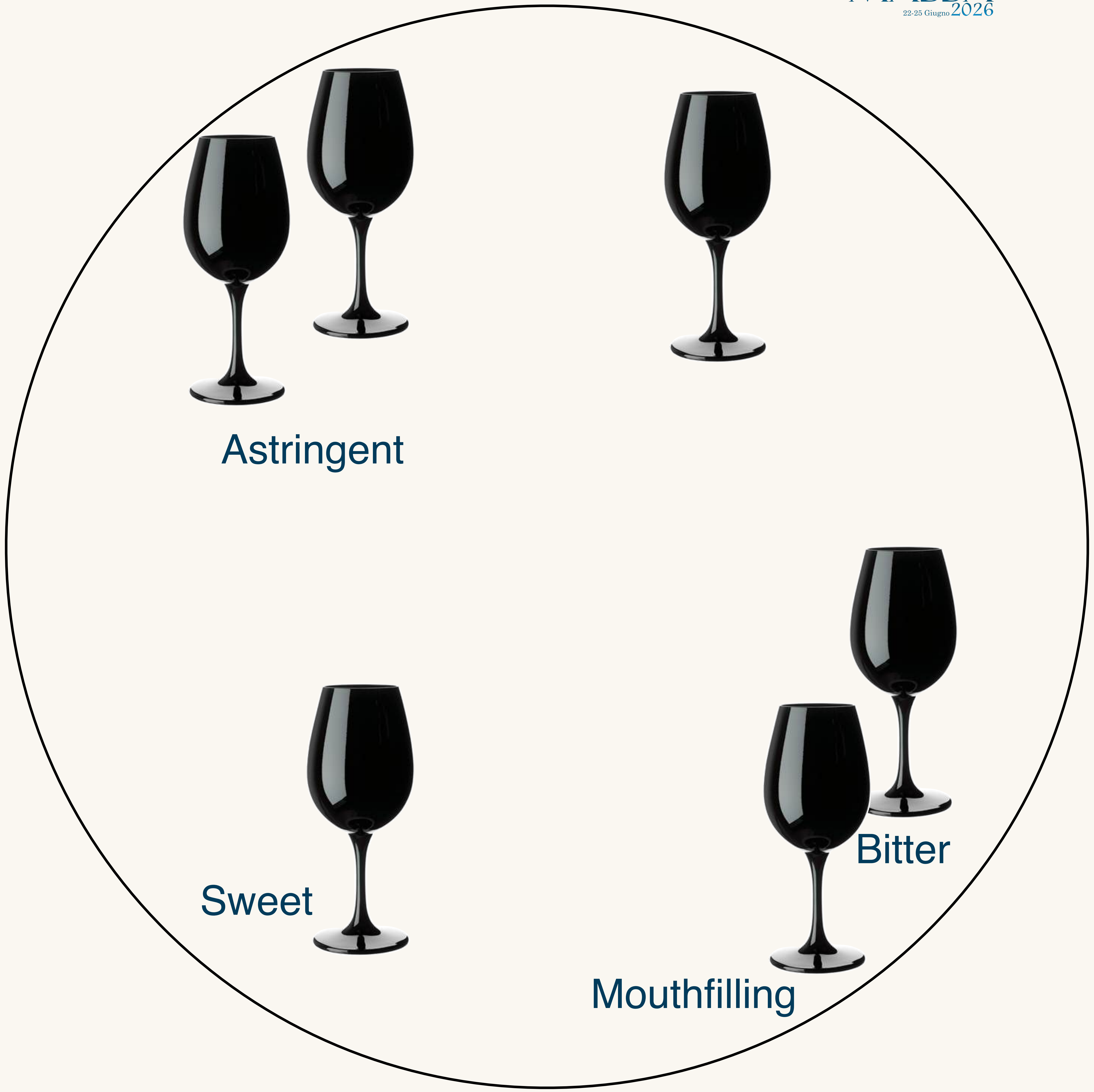




Chemically informed tasting

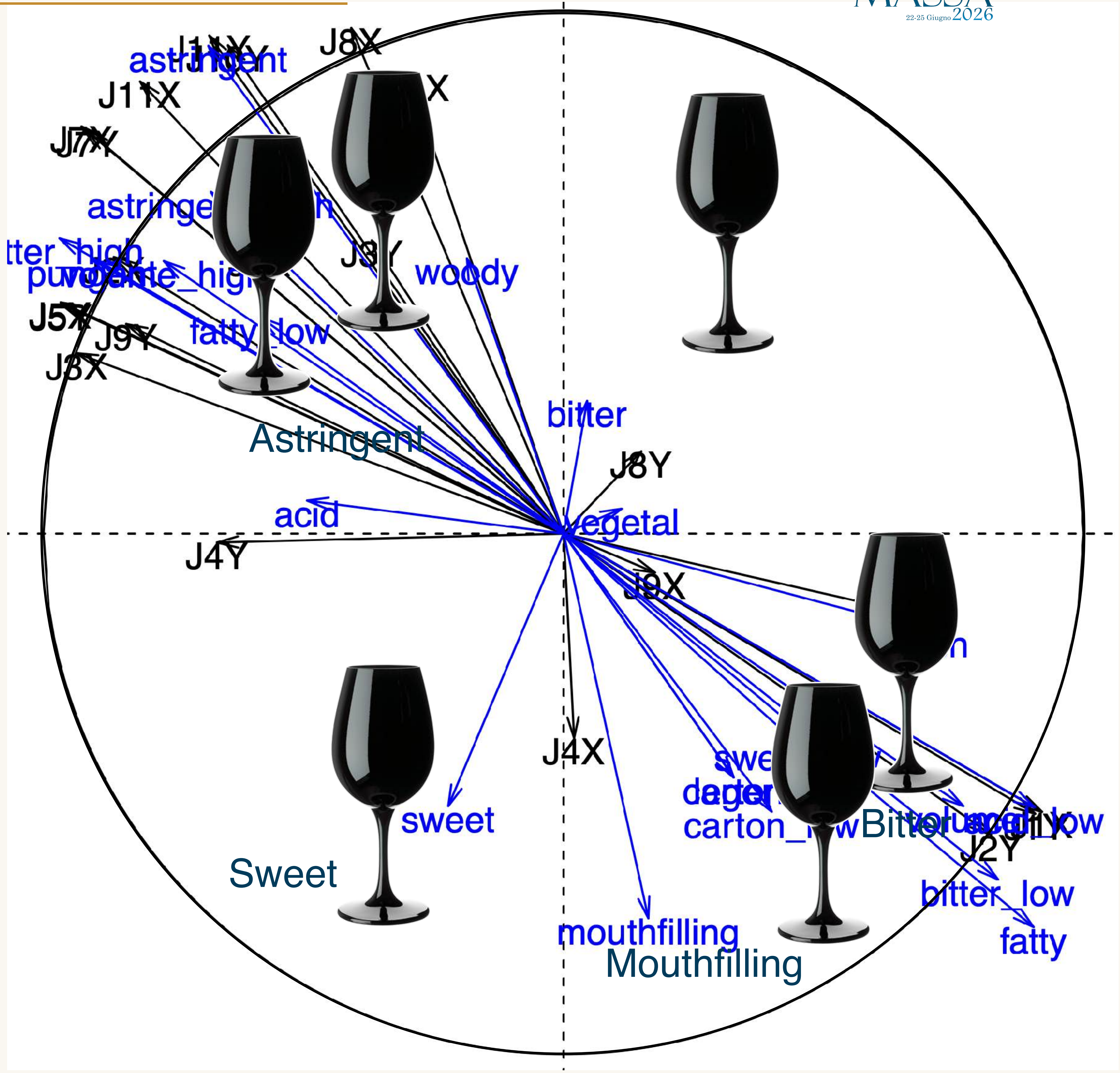
# a Strategy to Analyze Plant extracts taste In Depth

Group of chemically similar fractions

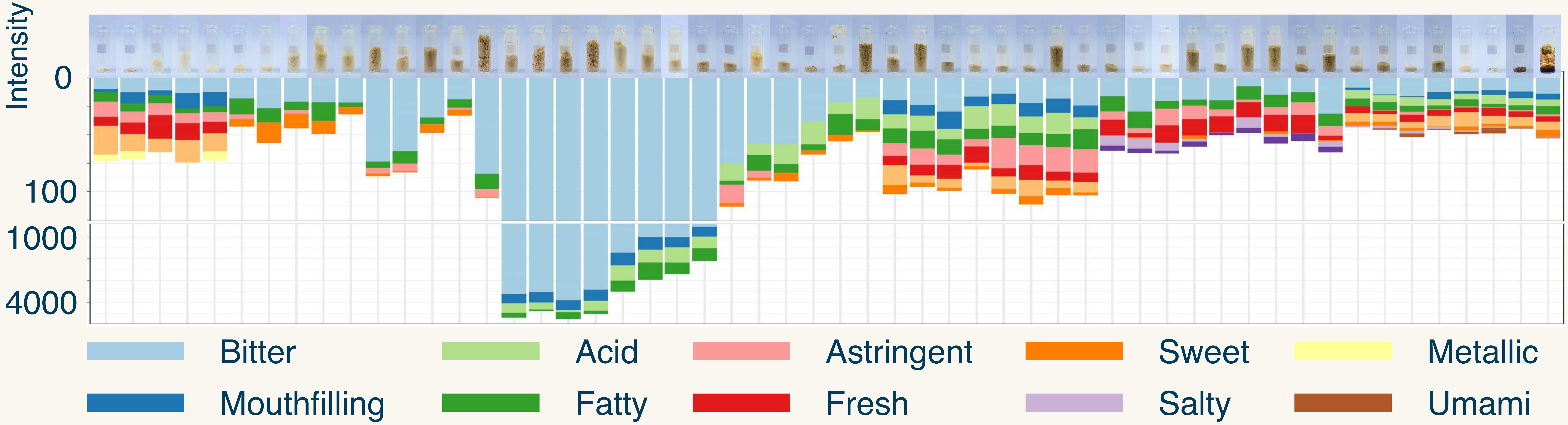


# a Strategy to Analyze Plant extracts taste In Depth

Group of chemically similar fractions



# Linking sensorial and chemical assessments



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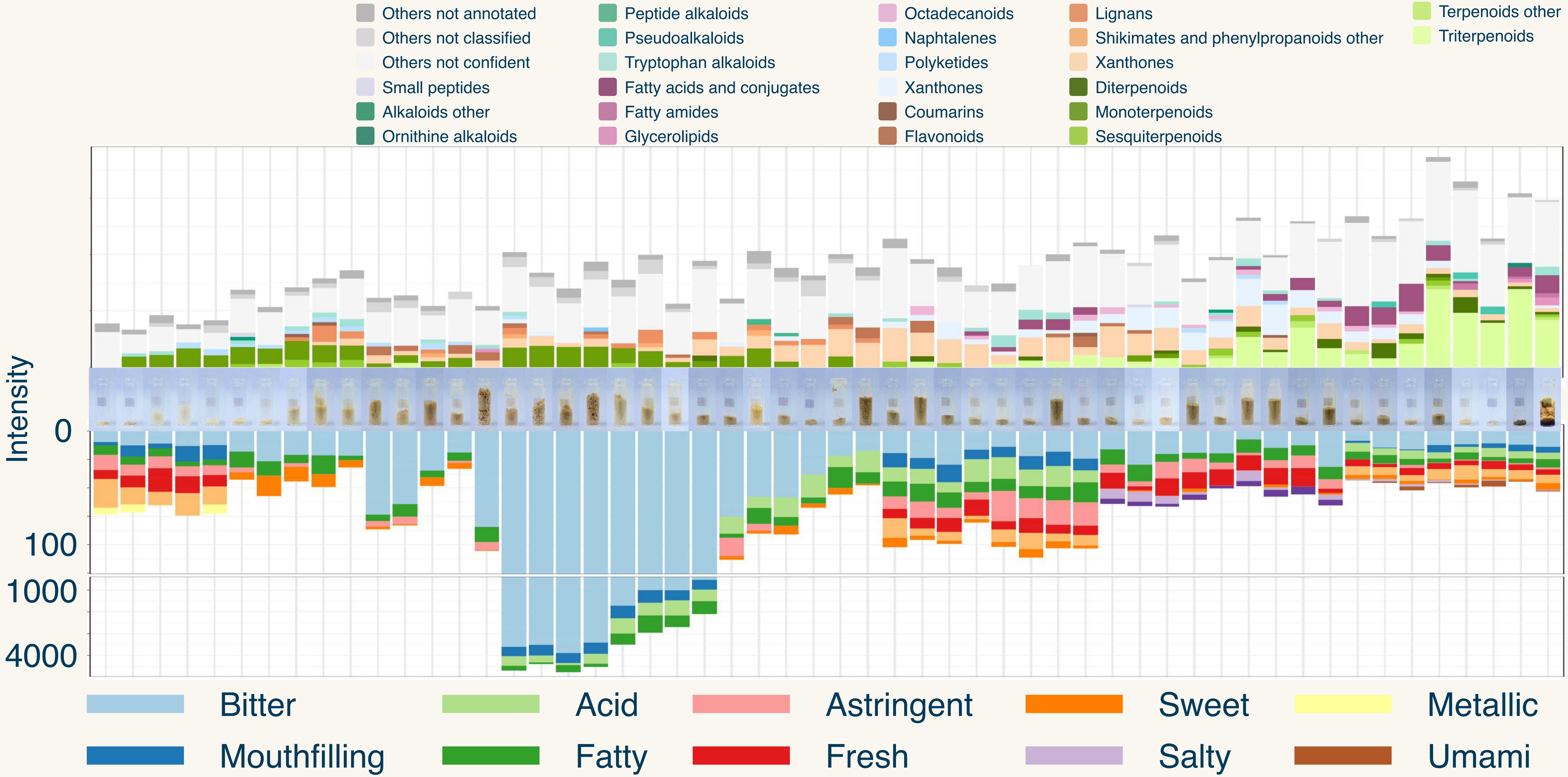

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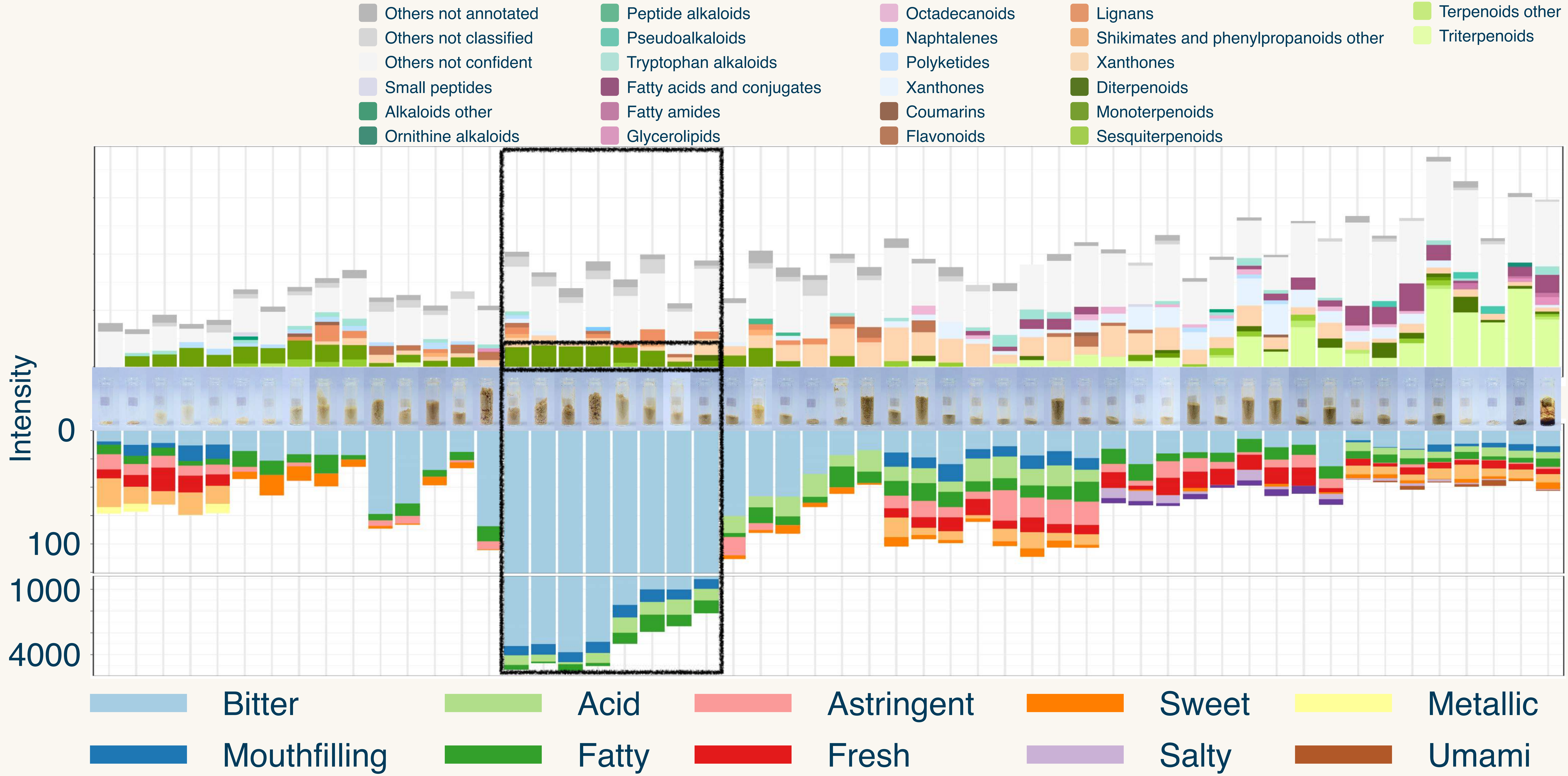

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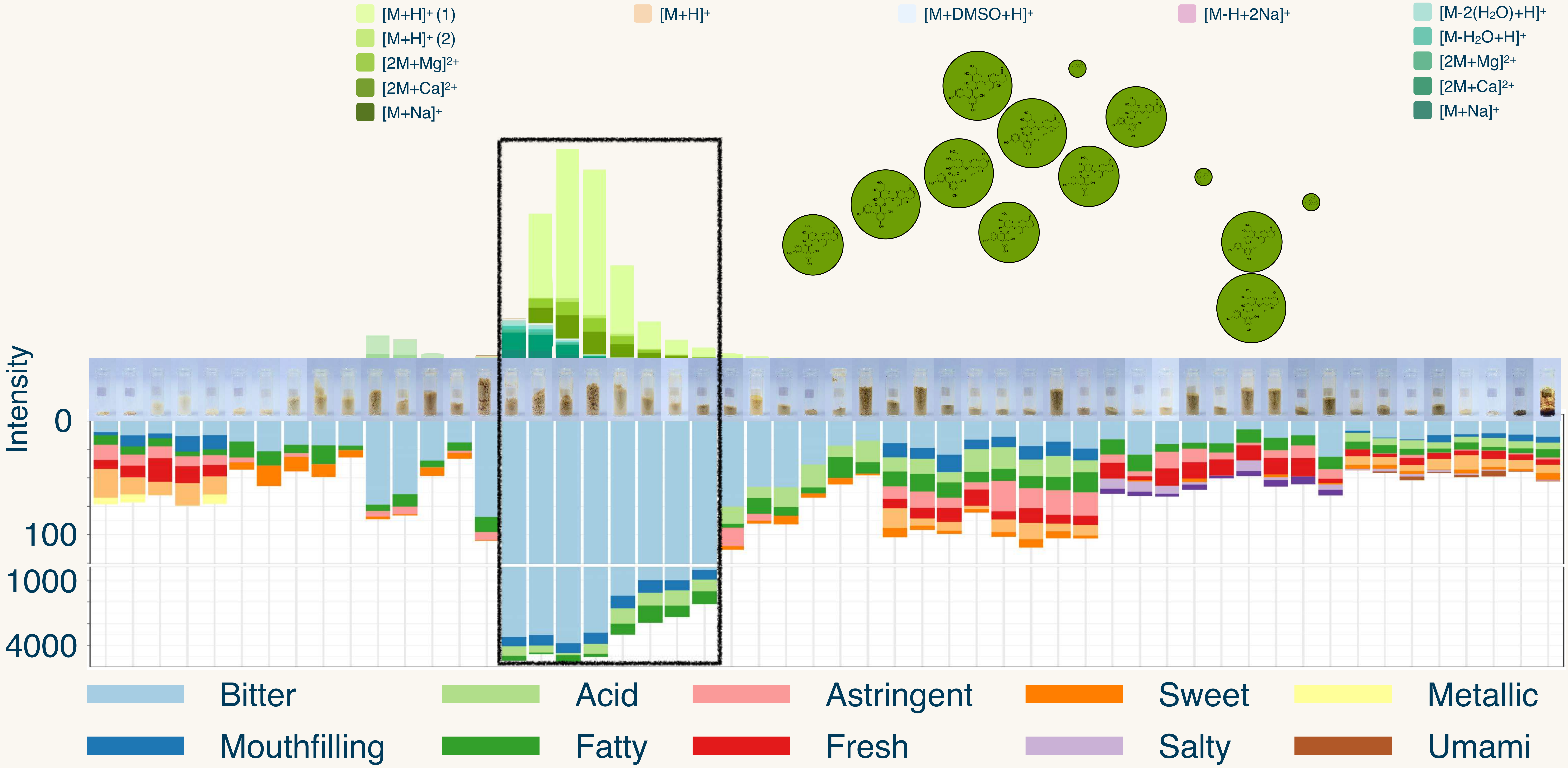

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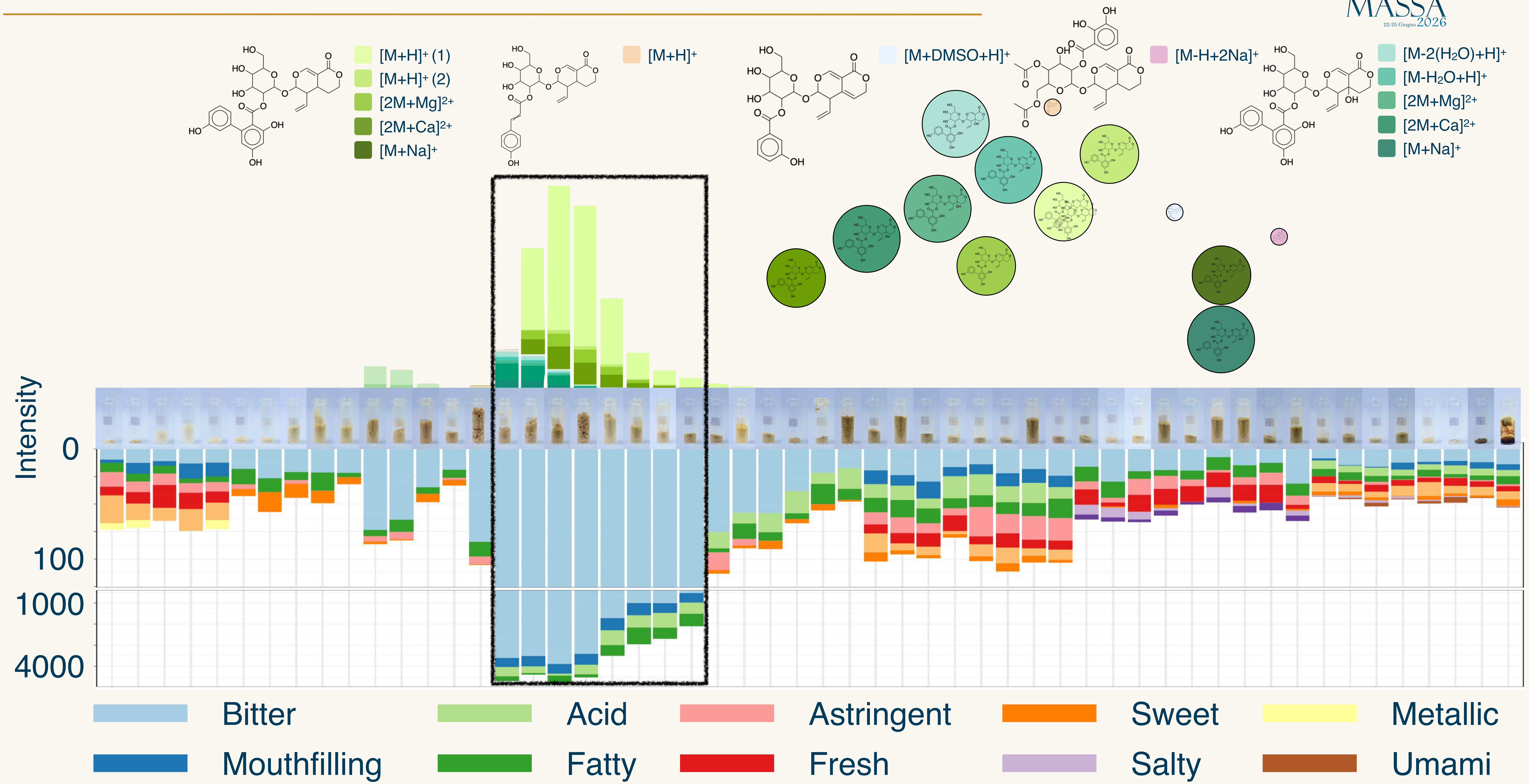

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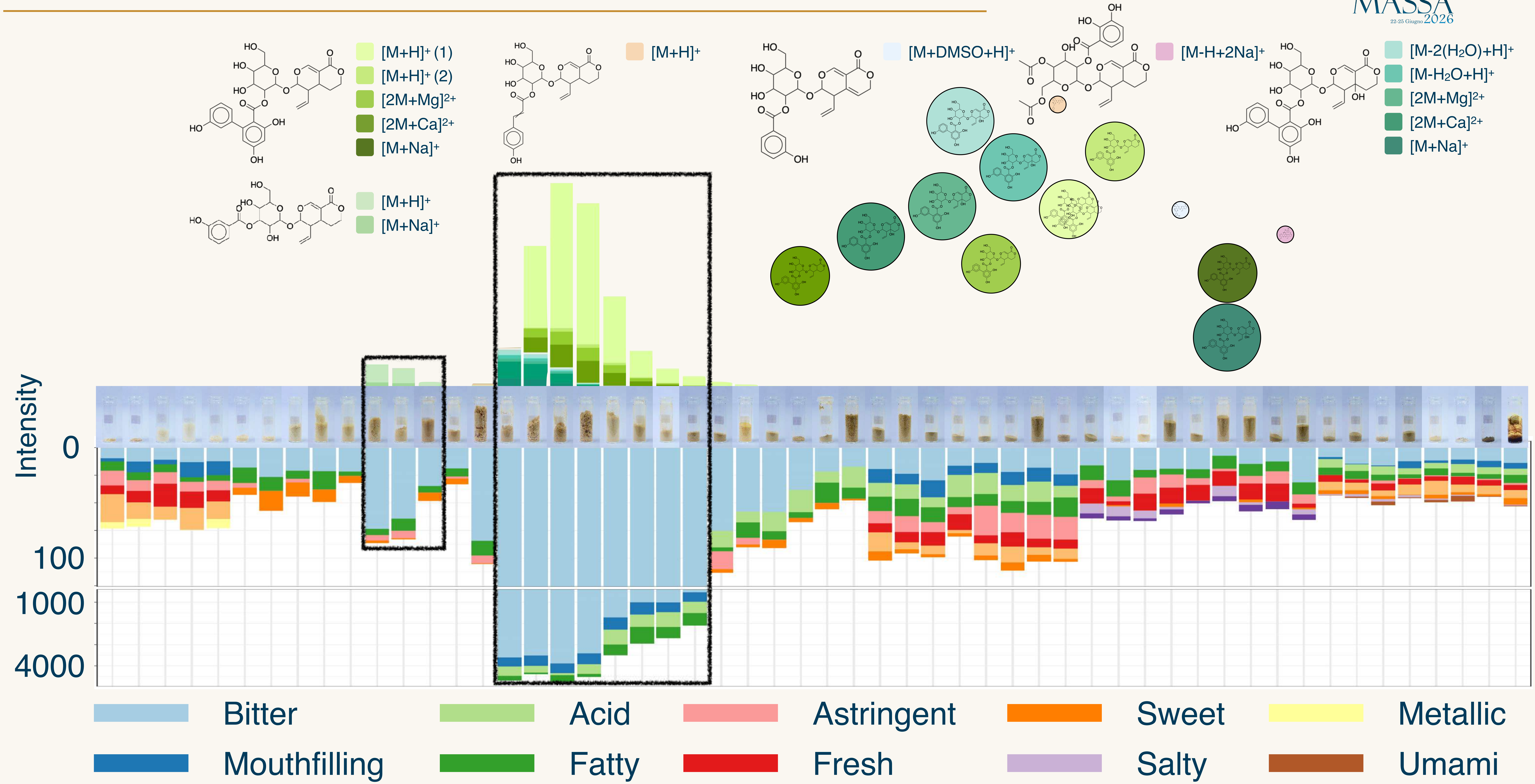
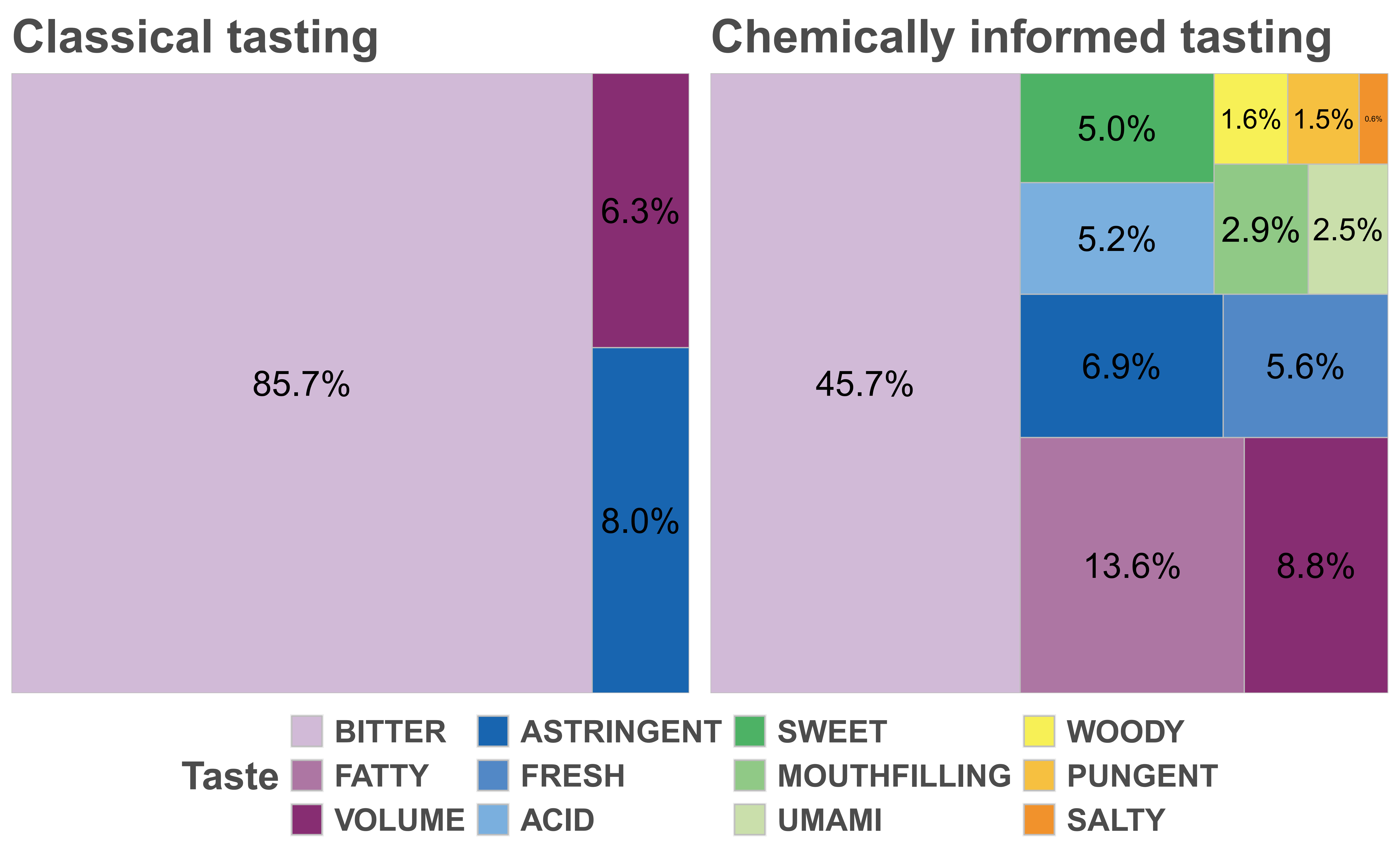


# a Strategy to Analyze Plant extracts taste In Depth




## Key finding

The complex taste and molecular composition of *Swertia chirayita* ethanolic extract could be deconvoluted

- Untargeted chemical and sensorial approaches
- Major and minor contributors were highlighted

MASSA

22-25 Giugno 2026

# Conclusions

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# Conclusion 1: Use open tools!

proteowizard

mzmine

OpenMS

Spectra

tima

WIKIDATA

*Open Science*

SIRIUS

LOTUS

MassBank
High Quality Mass Spectral Database

GNPS

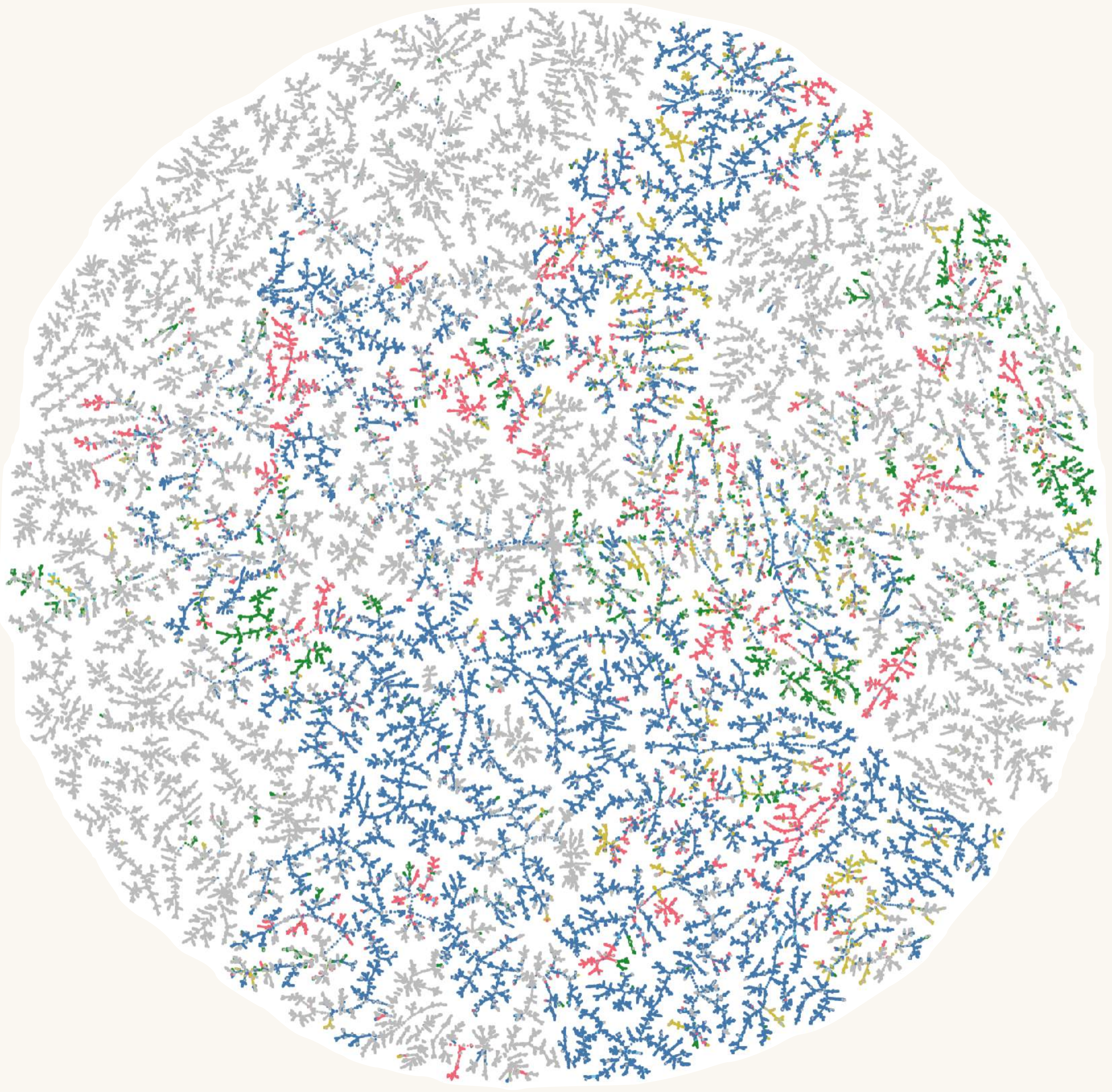
MASST

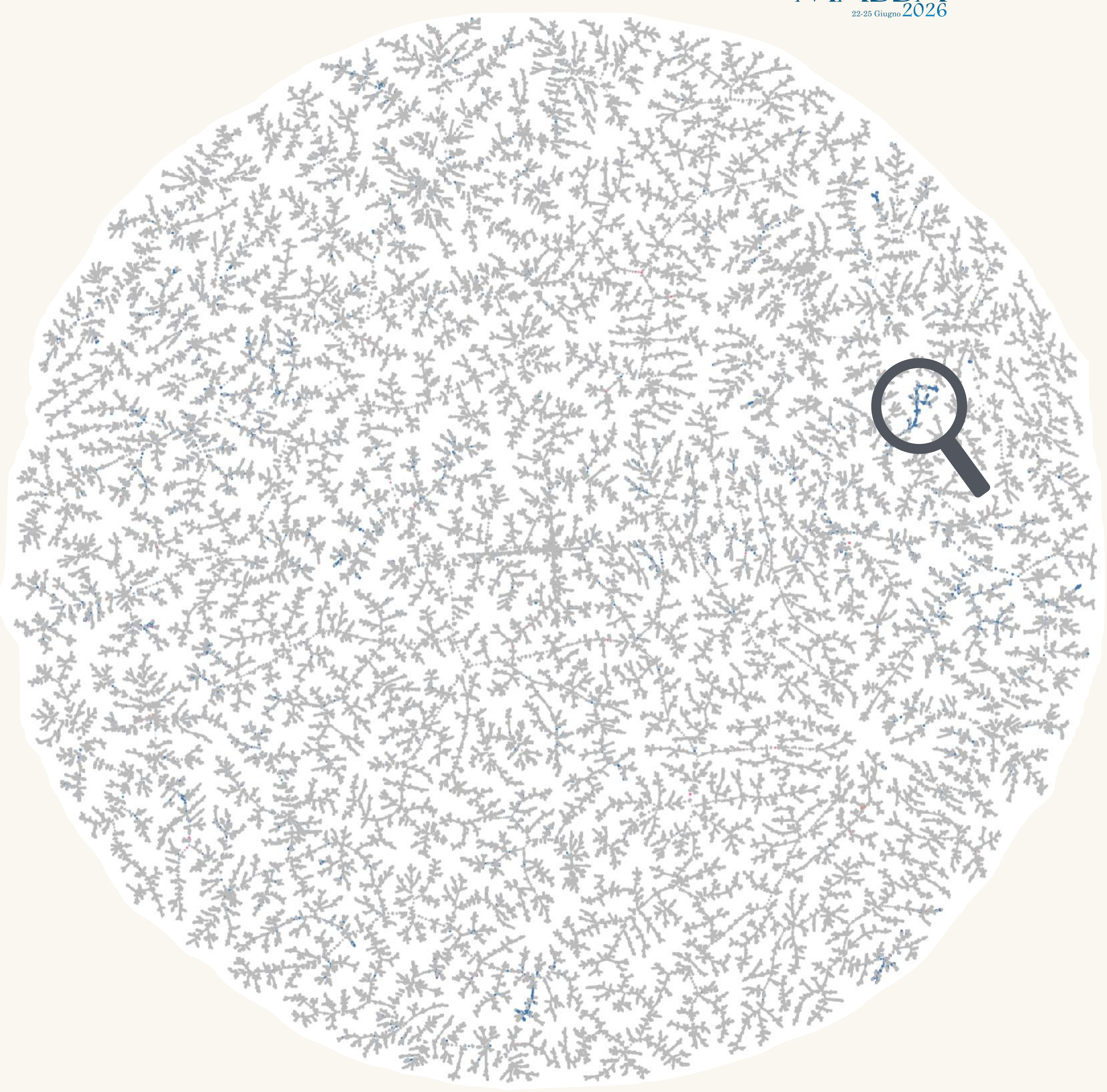
masst.ucsd.edu

CFM-ID
Competitive Fragmentation Modeling
for Metabolite Identification

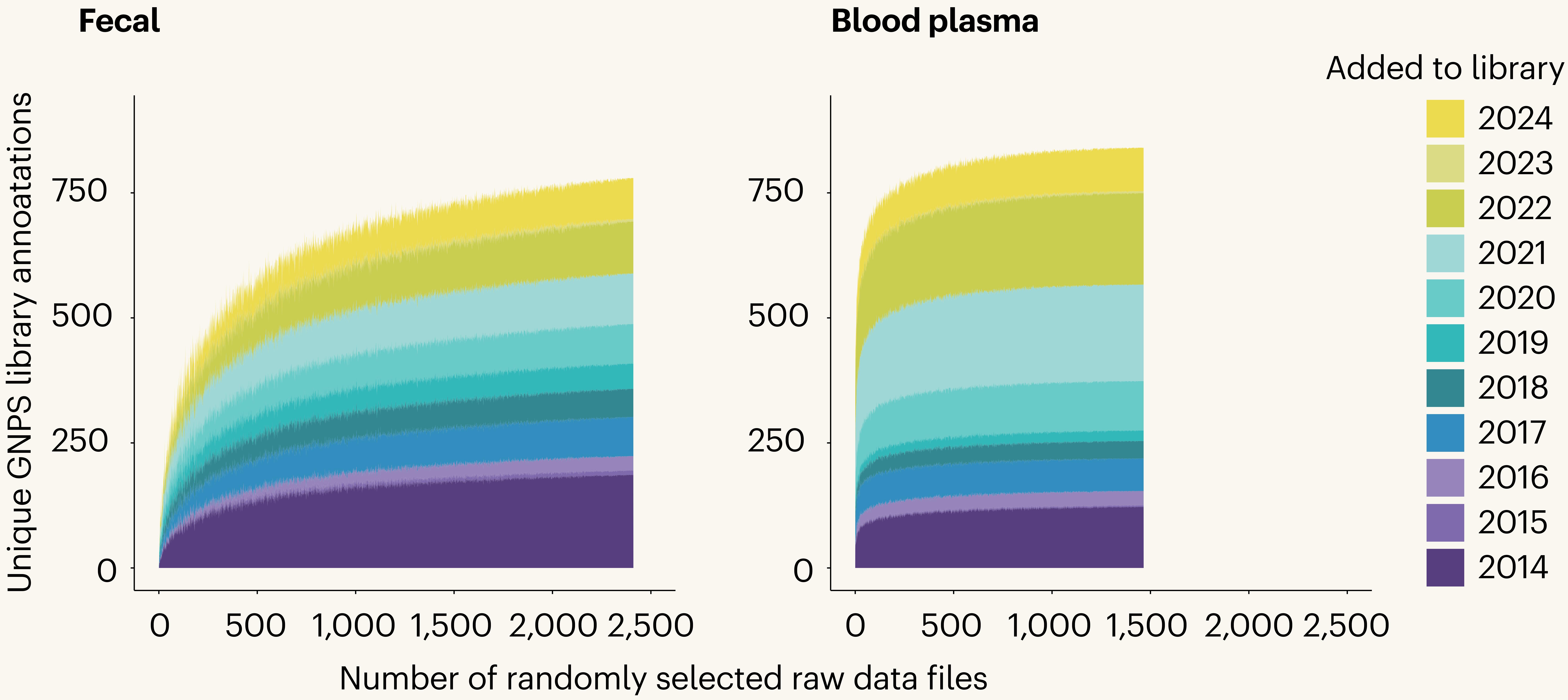
RDKit

# Conclusion 2: Contribute to the open world!



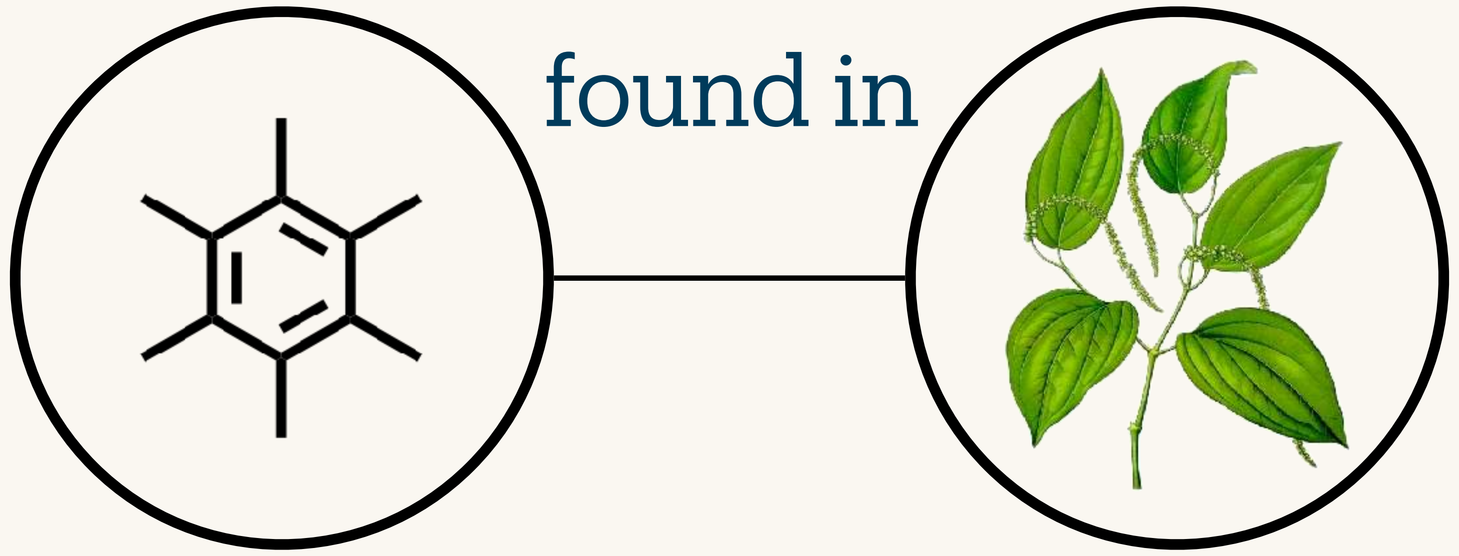
# Conclusion 3: Contribute to the open world!




El Abiead, Y., Rutz, A., Zuffa, S. *et al*. Discovery of metabolites prevails amid in-source fragmentation. *Nat Metab* **7**, 435–437 (2025). https://doi.org/10.1038/s42255-025-01239-4

# Conclusion 4: Contribute to the open world!

LOTUS




EARTH
METABOLOME
INITIATIVE

Prior Knowledge

Future Knowledge

https://lotus.nprod.net/

https://www.earthmetabolome.org/

Metabo
LinkAi

https://www.metabolinkai.net/

# Conclusions

1 **Open source is the only way forward**
Did someone try to open Vendor 1 raw files using Vendor 2 software?
Good, then go open source.

2 **Sharing is caring**
Your best dataset shouldn't retire on your hard drive.
Make it someone else's next discovery.

3 **Work with your peers**
Mass spectrometry is too rich to explore alone.
Let's build the knowledge together.

# Thank you

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